



THE UNIVERSITY
OF QUEENSLAND
A U S T R A L I A

Development of Low Compression Ratio Driver Conditions & General Facility Upgrades for X2

Student Name: Jaiden, HODSON

Course Code: ENGG7291

Supervisor: Dr. Chris James

Submission date: 3rd July 2025

Faculty of Engineering, Architecture and Information Technology

Jaiden Rhys Hodson
The University of Queensland
Queensland 4072

3rd July 2025

Professor Ross McAree
Head of School
School of Mechanical and Mining Engineering
The University of Queensland
Queensland 4072

Dear Sir,

I, Jaiden Rhys Hodson, am the author of the thesis entitled, Development of Low Compression Ratio Driver Conditions & General Facility Upgrades for X2.

I submit this thesis for consideration as partial fulfilment of the Master of Engineering/Bachelor of Engineering (Hons) at The University of Queensland.

I request that The University of Queensland make this thesis available in the open access collection of UQ eSpace, where I understand, it will be freely accessible on the Worldwide Web.

I warrant that no third-party copyrights or legal rights are infringed by my thesis.

All the work contained within this Thesis is my original work except where otherwise acknowledged.

Further, I declare that Appendix A accurately reflects my use of AI tools in this research. While AI assistance is acceptable, I affirm that the critical thinking, analysis, and conclusions presented in this thesis are my own work.

Signature:

A handwritten signature in black ink, consisting of several overlapping, stylized strokes that form a unique, cursive-like mark.

Student ID: 46410287

Executive Summary

X2 officials have expressed interest in providing validation data for the Modified Marrone-Treanor (MMT) thermochemical nonequilibrium air chemistry model which was recently developed at the University of Minnesota. In order to provide the required data, X2 must target low-density test flows with freestream flight equivalent velocities, U_e , in the range of 4-6 km/s. However, in its standard configuration, X2 is only capable of generating test flows with U_e in the range of 6-16 km/s.

The primary aim of this project was the design and modelling of low compression ratio driver conditions to enable standard X2 operation in the 3-6 km/s test flight regime; this range bridges the performance gap between X2's standard free-piston-driven configuration and the ultra-low-enthalpy cold-helium driver. The auxiliary aim of the project was the development of a series of general facility upgrades.

To accomplish these aims, the project began with the design of preliminary driver conditions using SPARK and PITOT3 ran in series. Once a promising set of low compression ratio driver conditions was established, the analysis transitioned to L1D4 for further refinement. In turn, the L1D4 launcher loss-factor was calibrated using data from X2 blanked-off experiments. Finally, with the model coarsely-calibrated, the set of preliminary driver conditions was refined further in preparation for a secondary round of detailed blanked-off experiments.

Ultimately, eight coarsely-refined CR 15 driver conditions were developed for X2. These low compression ratio driver conditions utilised a wide variety of helium-argon driver gas mixtures, ranging from pure-argon to almost pure-helium. The test flows generated by these conditions exhibited freestream U_e in the range of 3.1-5.8 km/s, thereby satisfying the goal of enabling standard X2 operation in the 3-6 km/s test flight regime. Unfortunately, this new envelope came with a caveat in the form of a performance gap which spans 3.7-4.7 km/s; bridging this gap will require four additional 'mid-enthalpy' CR 15 driver conditions with a U_e target of approximately 3.8-4.6 km/s. On a more positive note, as a result of the novel infrastructure-minimising philosophy applied throughout the design process, the eight new driver conditions require just two new primary diaphragms (rupturing at 1.8 & 5.0 MPa) to be fully integrated with X2's

existing 'long buffer' driver configuration. Additionally, several X2 facility upgrades were successfully implemented throughout the course of the project, specifically: a more accurate pressure monitoring system for the reservoir, an improved wear pad material and rear O-ring gland for the piston, and recommissioning of the X2 static rupture rig. Appreciable progress was also made towards a new laser-centric diaphragm scoring process and driver-length instrumentation.

The future work required for the completion of the CR 15 driver conditions includes: design of additional 'mid-enthalpy' CR 15 driver conditions, detailed blanked-off testing, detailed L1D4 loss-factor calibration, finalisation of the driver conditions, development of the required primary diaphragms, and finally diaphragm-rupturing validation tests. Future work towards the goal of laser-scored primary diaphragms should include: finalisation of the laser-scoring rig design, manufacturing of the laser rig, and iterative tuning of the laser-scoring process such that it can accurately capture the static rupture characteristics of X2's 1.2 mm, 2.0 mm and 2.5 mm steel diaphragms. Finally, there is a great deal of work required to complete the driver-length instrumentation, including: finalisation of the PCB housing design, FEA of the design installed in the tunnel wall, development of a custom magnetic drill mount (with an associated drilling procedure), and final installation.

Acknowledgements

I extend my deepest gratitude to my supervisors, Dr. Chris James and Toby van den Herik, whose supervision, advice and encouragement have steadied the course of this thesis and helped me to become a better engineer. Their meticulous attention to detail and shared enthusiasm for advancing the X2 facility have significantly shaped the outcomes of this thesis.

My appreciation also goes to Robert Hawken and Neil Duncan, whose patience and readiness to assist ensured the success of my brief X2 experimental campaign. Finally, to the broader XLabs team, thanks for the ‘pong’ – I shall miss it dearly.

Table of Contents

Executive Summary	i
Acknowledgements.....	iii
1 Introduction	1
1.1 A Brief History of The X2 Expansion Tube.....	1
1.2 Project Motivation.....	2
1.3 Project Aims	3
1.4 Project Scope	4
1.5 Project Deliverables	5
1.5.1 Project Outcomes	5
1.5.2 ENGG7291 Assessment.....	5
2 Literature Review.....	7
2.1 The X2 Expansion Tube.....	7
2.1.1 Test Time & Terminating Wave Processes.....	9
2.2 Tuned Operation of The Free-Piston Driver.....	9
2.2.1 Piston Overdrive	10
2.2.2 Piston Soft Landing Condition	11
2.3 Hornung Analysis of The Free-Piston Driver	12
2.4 PITOT3.....	14
2.5 L1D4	14
2.6 Low Enthalpy Driver Condition Design Principles	15
3 Project Plan	18
3.1 Project Methodology.....	18
3.2 Project Timeline	20
4 Development of Preliminary Driver Conditions	21
4.1 Definition of Key Driver Criteria	21
4.2 Exploration of The Driver Condition Design Space.....	22
4.3 Preliminary L1D4 Simulations	28
4.3.1 L1D4 Geometry	28
4.3.2 L1D4 Simulation Parameters	36

4.3.3 Preliminary L1D4 Results.....	37
5 Coarse Refinement of Driver Conditions.....	39
5.1 Preliminary L1D4 Blanked-Off Simulations	39
5.2 X2 Blanked-Off Experiments.....	40
5.3 Coarse L1D4 Calibration	43
5.4 Summary of Coarsely-Refined Driver Conditions	45
5.5 Proposal for Future Experimental Campaigns.....	49
6 Conclusions.....	50
7 Recommendations for Future Work	51
References.....	52
Appendices.....	55
A Documentation of Generative AI Usage in Research.....	56
B Project Gantt Chart	57
C Project Risk Assessment.....	58
C.1 Health & Safety Risks.....	59
C.2 Scheduling Risks	59
C.3 Technical Risks.....	60
C.4 Project Opportunities	61
D SPARK Control Driver Conditions	62
E SPARK Plots for Preliminary Driver Conditions	66
E.1 3 MPa Rupture - 85 mm Orifice Plate.....	66
E.2 3 MPa Rupture - 77 mm Orifice Plate.....	67
E.3 3 MPa Rupture - 72.1 mm Orifice Plate.....	68
E.4 3 MPa Rupture - 65 mm Orifice Plate.....	69
E.5 7 MPa Rupture - 85 mm Orifice Plate.....	70
E.6 7 MPa Rupture - 77 mm Orifice Plate.....	71
E.7 7 MPa Rupture - 72.1 mm Orifice Plate.....	72
E.8 7 MPa Rupture - 65 mm Orifice Plate.....	73

F	L1D4 Plots for Coarsely-Refined Driver Conditions	74
F.1	CR15-1.8MPa-85mm	74
F.2	CR15-1.8MPa-77mm	76
F.3	CR15-1.8MPa-72.1mm.....	78
F.4	CR15-1.8MPa-65mm	80
F.5	CR15-5MPa-85mm	82
F.6	CR15-5MPa-77mm	84
F.7	CR15-5MPa-72.1mm	86
F.8	CR15-5MPa-65mm	88
G	Buffer Stud Failure Analysis	90
H	X2 Facility Upgrades.....	93
H.1	Commissioning of Improved Reservoir Pressure Monitoring System	93
H.2	Commissioning of Improved Piston Wear Pad Material	93
H.3	Correction of The X2 Piston's Rear O-Ring Gland.....	94
H.4	Recommissioning of The X2 Static Rupture Rig.....	95
H.5	Development of Laser-Scoring Techniques for X2 Primary Diaphragms.....	96
H.6	Development of Driver-Length Instrumentation	98
I	Standard Operating Procedures.....	104
I.1	Updated X2 Blanked-Off Test SOP	104
I.2	Static Rupture Test SOP	109
J	Technical Drawings.....	111
J.1	Excerpt from Corrected X2 Piston Drawing	111
J.2	Laser-Scoring Rig Drawing	112
J.3	PCB Housing Drawing.....	113

List of Figures

Figure 1.1: Schematic of the X2 expansion tube	1
Figure 1.2: Performance map of current X2 driver conditions	2
Figure 2.1: Schematic of the X2 expansion tube and corresponding flow processes	7
Figure 2.2: Effect of piston overdrive on driver pressure	10
Figure 2.3: x-t diagram showing wave processes for a tuned free-piston driver	10
Figure 2.4: Characteristics of piston motion	11
Figure 2.5: Approximate X2 simulation times for various computational methods	14
Figure 2.6: Post-compression sound speed of nitrogen and argon	17
Figure 3.1: Project chapters	18
Figure 4.1: Simplified reservoir geometry	29
Figure 4.2: Simplified launcher geometry	30
Figure 4.3: Simplified cylindrical representation of the X2 driver	31
Figure 4.4: Internal volume conservation for a gradual area change	32
Figure 4.5: Basic L1D4 X2 driver configuration – without buffers or an orifice plate ..	35
Figure 5.1: The stages of a blanked-off experiment	40
Figure 5.2: Time-aligned blanked-off pressure traces comparing shots x2s5983 & x2s5984, and the calibrated ‘low-enthalpy’ L1D4 model	44
Figure 5.3: Time-aligned blanked-off pressure traces comparing shots x2s5987 & x2s5988, and the calibrated ‘high-enthalpy’ L1D4 model	44
Figure H.1: Newly manufactured X2 piston	94
Figure H.2: The X2 static rupture rig	95
Figure H.3: Schematic of a rupturing diaphragm	96
Figure H.4: Example of iterative laser engraving calibration	97
Figure H.5: UQ Innovate laser machine peak bed	97
Figure H.6: Driver-length instrumentation plots (35.7 MPa condition)	99
Figure H.7: Driver-length instrumentation plots (27.9 MPa condition)	100
Figure H.8: Driver-length instrumentation plots (15.5 MPa condition)	101
Figure H.9: Key events during X2 piston launch	102
Figure H.10: Preliminary PCB housing design	103

List of Tables

Table 1.1: Project scope definition	4
Table 2.1: Primary X2 driver conditions	15
Table 4.1: CR 15 nitrogen driver conditions	25
Table 4.2: Helium-maximising driver conditions	26
Table 4.3: CR 15 infrastructure-minimising driver conditions	27
Table 4.4: Simplified cylindrical representation of the X2 driver	31
Table 4.5: Basic L1D4 breakpoint representation of the X2 driver	34
Table 4.6: L1D4 simulation parameters	36
Table 4.7: Preliminary CR 15 driver conditions	38
Table 5.1: Preliminary L1D4 blanked-off results	39
Table 5.2: X2 blanked-off results	41
Table 5.3: L1D4 calibrated blanked-off results	43
Table 5.4: Coarsely-refined CR 15 driver conditions	45
Table C.1: Risk matrix	58
Table C.2: Health & safety risk management	59
Table C.3: Scheduling risk management	59
Table C.4: Technical risk management	60
Table D.1: CR 17.5 control driver conditions	62
Table D.2: CR 15 control driver conditions	63
Table D.3: CR 12.5 control driver conditions	64
Table D.4: CR 10 control driver conditions	65

1 Introduction

This introductory section begins with an overview of the project's context and motivation, before delving into more detailed definitions of the project's aims, scope and deliverables.

1.1 A Brief History of The X2 Expansion Tube

X2 is a free-piston-driven expansion tube commissioned by The University of Queensland in 1995. Designed by Morgan and Doolan [1], X2's primary purpose was to serve as a proof of concept for UQ's premiere expansion tube facility, X3. At the beginning of 2004, X2's original two stage free-piston driver was converted to a single stage driver by Scott [13] which provided both increased performance and reduced operational complexity. X2's driver was upgraded once more by Gildfind [3] in 2010 with the introduction of a light-weight piston and corresponding driver conditions which remain in use to this day. In the years since, X2 has enabled scaled flow testing of a broad range of hypersonic applications; particularly in the field of planetary entry vehicles. A full-facility diagram of X2 is provided in Figure 1.1.

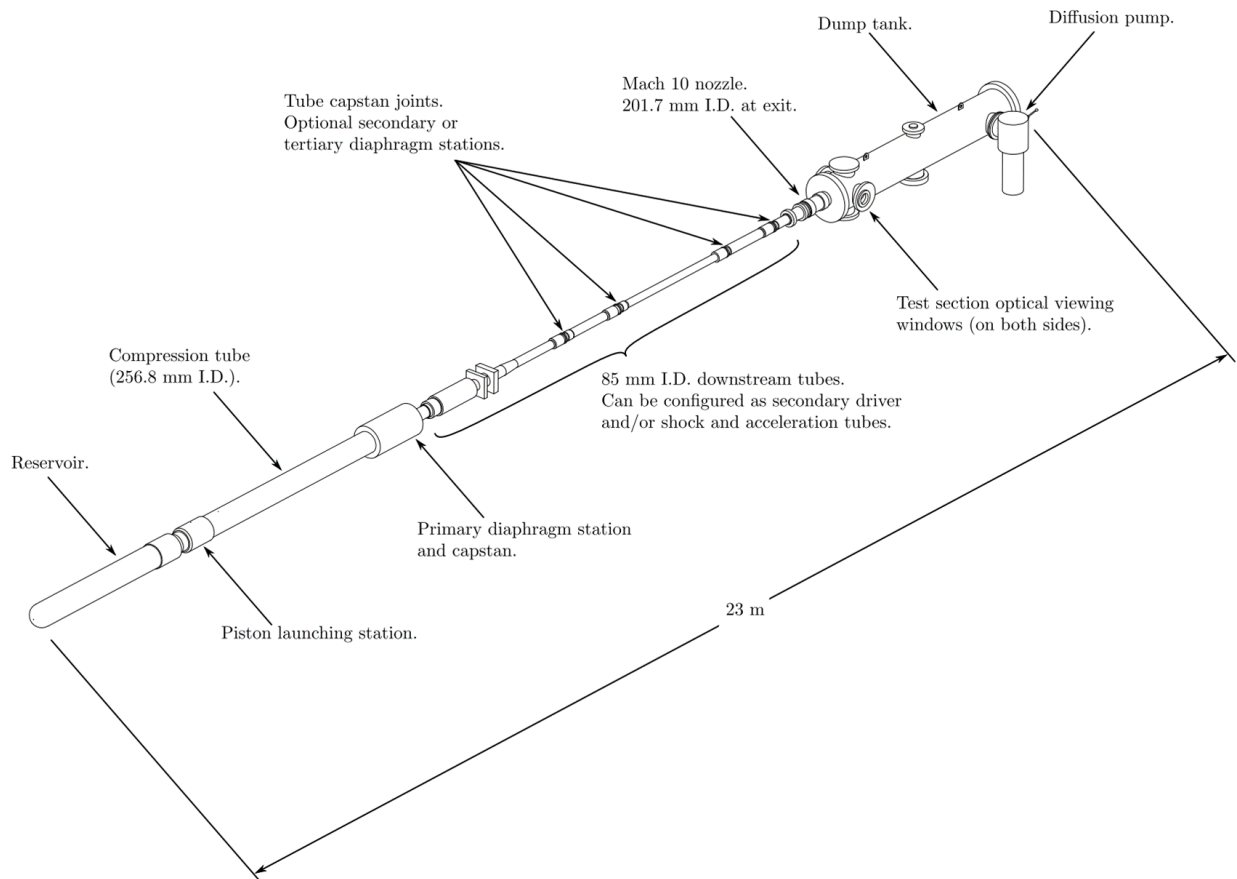


Figure 1.1: Schematic of the X2 expansion tube; drawn to scale [3].

1.2 Project Motivation

High-quality air chemistry models are fundamentally important for designing hypersonic vehicles, as post-shock thermochemical nonequilibrium chemistry can have large effects on heat transfer, forces and moments, vehicle control, wake signatures, and vehicle communications. The Modified Marrone-Treanor (MMT) thermochemical nonequilibrium air chemistry model [17,18], which utilises state-of-the-art quantum chemistry calculations in its derivations, was recently developed at the University of Minnesota.

Appreciating the magnitude of this modelling advancement, X2 officials have expressed interest in providing additional validation data for the MMT project. It is believed that X2 is uniquely equipped to deliver novel, ‘high-quality’ validation data, which cannot be reproduced in competing facilities such as the T5 reflected shock tunnel at Caltech [19-21]. In order to provide the required data, X2 must target low density test conditions with flight equivalent velocities, U_e , in the range of 4-6 km/s.

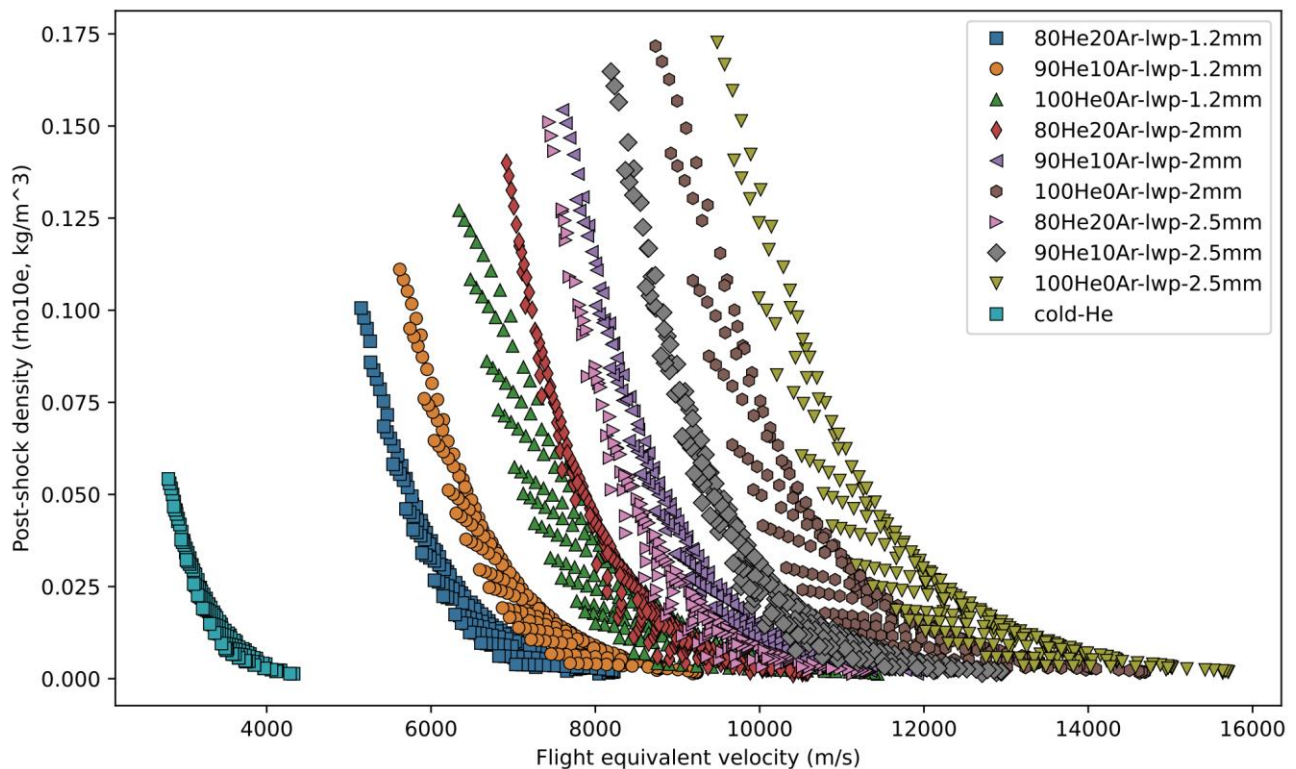


Figure 1.2: Performance map of current X2 driver conditions [7].

In its standard configuration (excluding the cold-helium driver), the X2 facility is currently capable of producing test conditions with freestream U_e in the range of 6-16 km/s (see Figure 1.2) [7]. This performance envelope has enabled the study of countless hypersonic phenomena in the past. However, the present demand for low-enthalpy, low-density validation data for the MMT model has highlighted the inherent value in reducing the lower bound of the standard envelope to 3 km/s – thereby bridging the performance gap to the cold-helium configuration. In recent years, researchers at X3R have demonstrated that low compression ratio drivers can successfully produce such ‘low-performance’ test conditions [14]. In hopes of replicating the success of X3R, and in doing so, bolstering the ‘low-performance’ capabilities of the X2 facility, officials have commissioned the development of new low compression ratio driver conditions for X2.

1.3 Project Aims

The objectives of the project were categorised into the primary aim and auxiliary aims:

- **Primary Aim:** Design and modelling of low compression ratio driver conditions to enable standard X2 operation in the 3-6 km/s test flight regime.
- **Auxiliary Aims:** Development of general X2 facility improvements pertaining to reservoir pressure monitoring, piston design, static rupture testing, primary diaphragm scoring techniques and driver-length instrumentation.

1.4 Project Scope

To ensure focused progression throughout the course of the project, it was necessary to first define what was considered within the scope of the project. Key inclusions were listed in Table 1.1 alongside several items which were deemed beyond the scope of the project.

Table 1.1: Project scope definition.

In Scope	Out of Scope
• Driver condition design via the manipulation of reservoir fill pressure, driver fill pressure and composition, piston mass, buffer length, primary diaphragm rupture pressure and orifice plate diameter. • Simulation of driver condition performance using SPARK, PITOT3 and L1D4. • Calibration of the L1D4 launcher loss factor using blanked-off test results. • Assisting in general, incremental improvements of the X2 facility as requested by supervisors. • Producing technical drawings for all designs.	• Physical alteration of major components of the X2 facility (compression tube, launcher, reservoir, etc.) for the purpose of driver condition design. • Quantification of the losses incurred across the buffer plate, primary diaphragm and secondary diaphragm (loss factors for these regions were simply adopted from previous X2 models). • Modelling complex flow phenomena not natively supported in L1D4 such as Mirels effect. • Optimisation of test flow conditions via manipulation of shock tube or acceleration tube fill conditions; air was used in both tubes at pressures of 10 kPa and 10 Pa respectively.

1.5 Project Deliverables

The project deliverables were most appropriately divided into the outcomes of the project and the ENGG7291 coursework requirements.

1.5.1 Project Outcomes

- **New Low Compression Ratio Driver Conditions:** A collection of eight new CR 15 driver conditions were designed. The theoretical performance envelope of the new conditions is $3.1 \text{ km/s} < U_e < 5.8 \text{ km/s}$.
- **Improved Reservoir Pressure Monitoring System:** A high-resolution digital pressure gauge was commissioned for the X2 reservoir fill station.
- **Improved X2 Piston:** A new X2 piston was commissioned with an upgraded wear ring material and a corrected rear O-ring gland.
- **Recommissioning of The X2 Static Rupture Rig:** A high-resolution digital pressure gauge, a new Enerpac and quick-release hydraulic fittings were commissioned to restore the X2 static rupture rig.
- **Development of Laser-Scoring Techniques for X2 Primary Diaphragms:** Progress was made towards a highly-automated laser-engraving process designed to supersede the existing mechanical scoring process.
- **Development of Driver-Length Instrumentation:** An optimised driver-length PCB distribution was proposed alongside a preliminary PCB housing design.

1.5.2 ENGG7291 Assessment

- **Project Proposal – 17/03/2025:** A brief report that focused on outlining the project context, aims and deliverables. This item also contained a preliminary literature review, project plan and risk assessment.
- **Progress Review Meeting – 06/05/2025:** An interim presentation of project progress designed such that the supervisory team could provide feedback and discuss plans for the remainder of the project. Evidence of final report drafting was also required at this checkpoint.
- **Project Presentation – 19/06/2025:** A formal presentation of the culmination of the project to peers, academic staff members and placement supervisors.
- **Final Report – 03/07/2025:** A comprehensive documentation of the progress and outcomes of the project.

- **Workplace Feedback Forms – 14/04/2025, 30/06/2025:** Feedback forms designed to represent an authentic workplace appraisal in which technical and professional development were discussed with the project supervisor.

2 Literature Review

The following section focuses on the fundamental principles of X2 operation and the philosophy behind driver condition design. To that end, the review begins with an overview of the X2 facility, before delving into the primary tools used for driver condition design and concluding with a summary of relevant driver condition design principles.

2.1 The X2 Expansion Tube

UQ's X2 expansion tube is a 23 m long, free-piston-driven expansion tube facility with an 85 mm driven tube bore and a 201.8 mm nozzle exit diameter. X2 is commonly used to conduct studies of blunt-body planetary entry radiation, simulating test flows for several celestial bodies including Earth, Mars, Titan, Venus, and Uranus [9]. X2 has also been successfully used for the study of scramjet combustion [11]. A partial cross section schematic of the X2 expansion tube is presented in Figure 2.1.

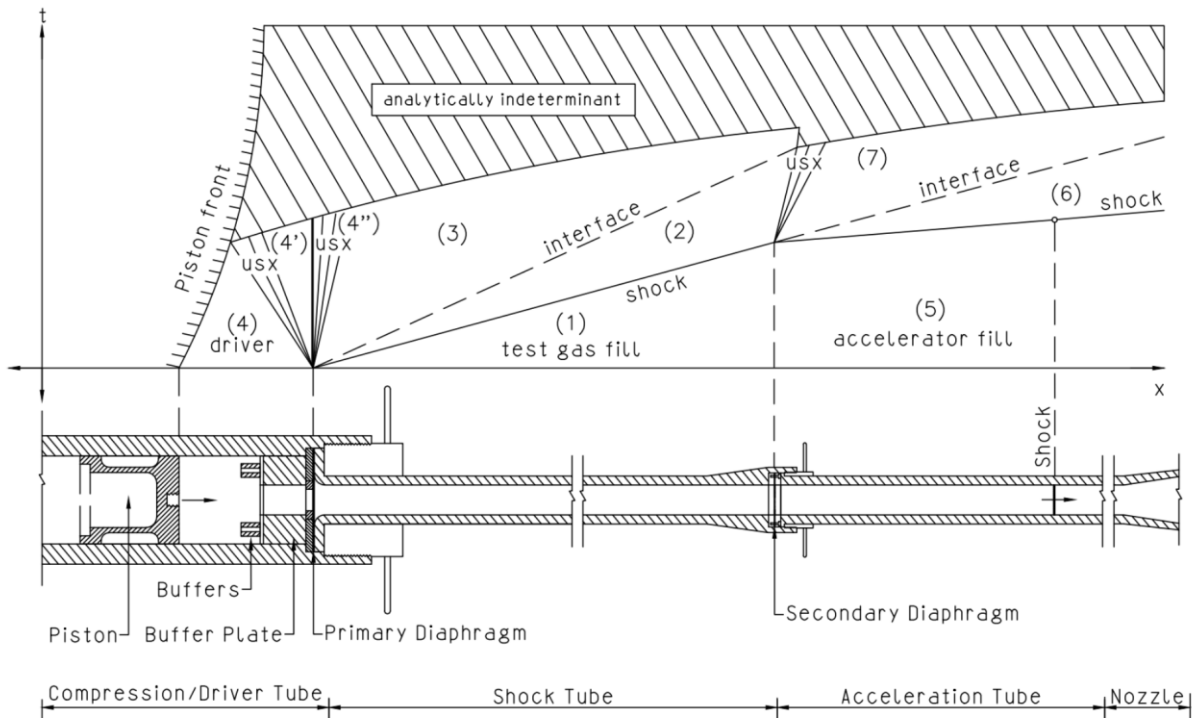


Figure 2.1: Schematic of the X2 expansion tube and corresponding flow processes [10].

In the configuration pertinent to this project, the X2 facility is comprised of the assembly of subsystems showcased in Figures 1.1 and 2.1. The first subsystem and the focus of this project is the driver, which consists of a high-pressure reservoir and a compression tube, separated by a free piston which is held at rest on a launcher mechanism prior to ‘firing’ of the facility. The driver is initially isolated from the subsequent subsystem, the

driven section, by a thin sheet (typically made of mild steel) referred to as the primary diaphragm. The driven section consists of a shock tube and an acceleration tube, isolated initially by a very thin mylar sheet referred to as the secondary diaphragm. Finally, a Mach 10 diverging nozzle [13] connects the acceleration tube to the test section and subsequently the dump tank.

Prior to ‘firing’, all sections of the X2 facility are evacuated and then filled with specified gas compositions, to specified fill pressures; air is used in the reservoir, a mixture of helium and argon, the so-called “driver gas”, is used in the compression tube, and the shock tube is filled with a “test gas” which typically corresponds to the atmosphere of the planet of interest. Upon ‘firing’, the piston is released from the launcher and the high-pressure reservoir gas rapidly accelerates the piston. As the piston travels down the compression tube, the driver gas is compressed from its initial fill pressure to the rupture pressure of the primary diaphragm. At the moment of diaphragm rupture, the pressure and temperature of the driver gas is in the order of 10’s of megapascals and 1000’s of Kelvin respectively.

Figure 2.1 includes an x-t diagram which illustrates the longitudinal wave processes that occur during an X2 experiment; the following description of these wave processes is adapted from [9]. Due to the cross-sectional area change between the compression tube and the primary diaphragm, the driver rupture condition (state 4) undergoes a steady expansion to a choked sonic state before it unsteadily expands into the shock tube (becoming state 3). In turn, this drives a shock wave through the shock tube gas (state 1), processing it to state 2. When the primary shock wave reaches the secondary diaphragm, the diaphragm immediately ruptures and the shock-processed test gas (state 2) begins to unsteadily expand into the acceleration tube (becoming state 7). The state 7 gas drives a secondary shock wave through the accelerator gas (state 5) and processes it to state 6. Finally, the nozzle at the end of the acceleration tube steadily expands the state 7 gas to the nozzle exit condition (state 8) which subsequently passes over the scaled test model and into the dump tank.

2.1.1 Test Time & Terminating Wave Processes

In an expansion tube such as X2, test time is defined as the period between the arrival of the state 8 gas at the test model, and the arrival of any waves which disturb the uniform test flow conditions. There are generally three key wave processes which can terminate the test period [1, 3, 12]:

1. The downstream edge of the unsteady expansion wave which originates from the secondary diaphragm. The arrival of this disturbance marks the end of the nominal test time.
2. An unsteady expansion reflected off the driver/test gas contact surface as a $(u + a)$ characteristic. This reflected disturbance travels very fast and can overtake other flow features to cause premature termination of the test time.
3. An unsteady expansion reflected off the front of the free-piston as a $(u + a)$ characteristic. In X2, the transmission of this disturbance is typically delayed through tuned piston operation.

Maximum theoretical test time corresponds to the ideal case in which disturbances 1 and 2 arrive at the test model simultaneously. Test time was a critical metric to consider throughout the driver condition design process as a new condition is pointless if it does not provide sufficient test time for establishment of key flow features over a given test model. The test times yielded by the new driver conditions designed during this project were primarily evaluated against existing X2 conditions, as accounting for the specific geometry of future test models was deemed beyond the scope of this project.

2.2 Tuned Operation of The Free-Piston Driver

There are effectively two tuning philosophies for a free-piston driver, constant-volume or constant-pressure operation. X2's driver utilises constant-pressure tuning, meaning that after primary diaphragm rupture, the piston maintains approximately constant gas properties in the driver (state 4) by balancing the mass loss from driver gas escaping into the shock tube, with further volumetric compression [9]. In order to achieve this, the piston must have excess velocity at the time of diaphragm rupture – this leads to the concept of piston overdrive.

2.2.1 Piston Overdrive

Piston ‘overdriving’ is formally defined by Itoh [5] as a driver configuration for which the piston overdrive parameter β is greater than one; β is the actual piston speed at the moment of diaphragm rupture divided by the reference speed. When overdriven, the piston briefly continues to increase the driver pressure (state 4) post diaphragm rupture, before pressure decreases again; this is illustrated in Figure 2.2. A deviation of $\pm 10\%$ from rupture pressure is arbitrarily permitted within the definition of constant-pressure operation. Thus driver ‘hold time’ is measured from the time of diaphragm rupture to the point at which the driver pressure falls below the 90% of the rupture pressure.

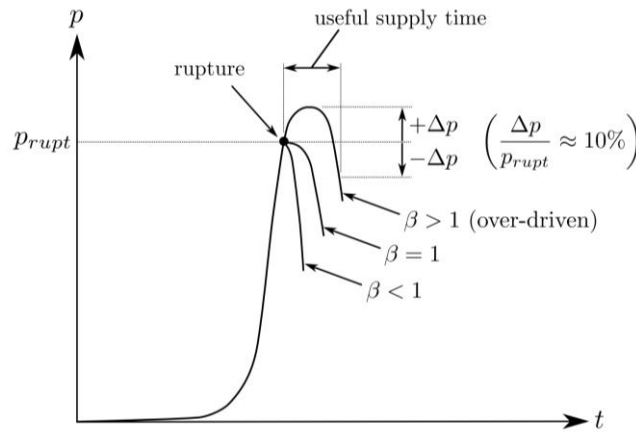


Figure 2.2: Effect of piston overdrive on driver pressure [3].

Overdriven operation has the additional benefit of delaying the transmission of the piston-reflected ($u + a$) wave, as allowing the piston to maintain its high speed later into the stroke ensures that the initial unsteady expansion is nullified by the strong compression waves produced by the piston, see Figure 2.3 [3].

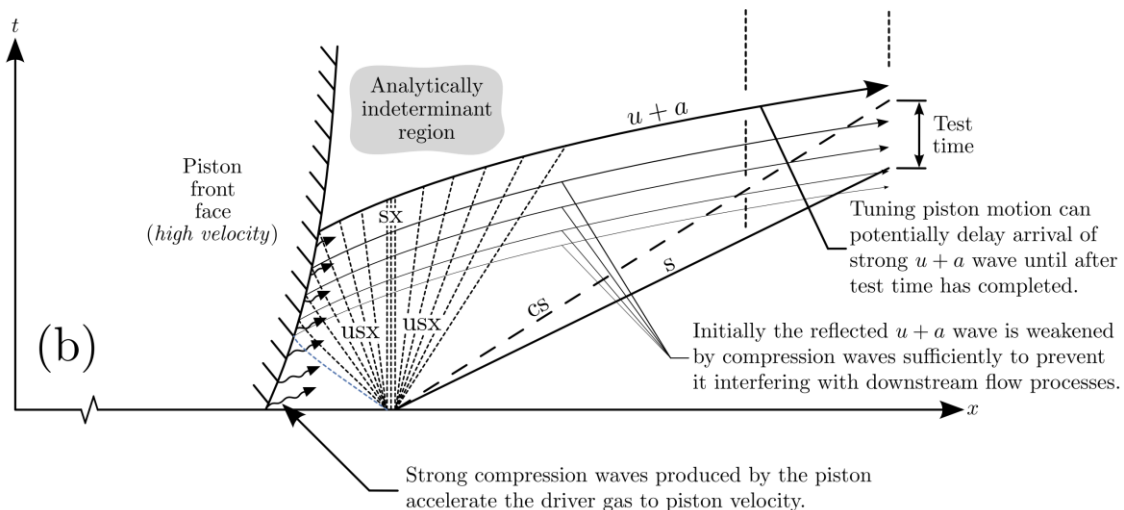


Figure 2.3: x-t diagram showing wave processes for a tuned free-piston driver [3].

2.2.2 Piston Soft Landing Condition

For tuned X2 operation, at the time of diaphragm rupture, the piston maintains a relatively high velocity, typically in the order of 100-300 m/s [3]. For the safety of the facility and its operators alike, the piston must undergo significant deceleration before it collides with the end of the compression tube. Depending on several driver parameters, the collision event can be categorised as either a ‘direct impact’, a ‘rebounding impact’, or a ‘soft landing’ where impact speed is effectively minimised, see Figure 2.4.

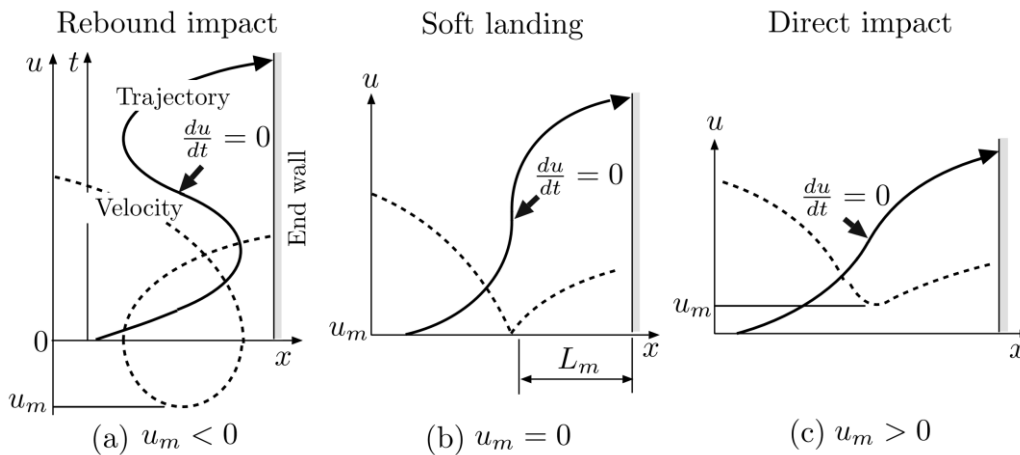


Figure 2.4: Characteristics of piston motion. Subscript m corresponds to the instant for which the piston acceleration is zero [3].

A direct impact occurs when the driver gas is insufficiently worked by the piston during its stroke and the piston reaches the end of the compression tube with appreciable velocity [14]; this characteristic is generally a consequence of the driver gas escaping too rapidly into the shock tube post diaphragm rupture. As there is no point in the trajectory where piston velocity equals zero, it can be challenging to design safe driver conditions with a directly impacting piston trajectory.

A rebounding impact occurs when the driver gas fails to escape quickly enough into the shock tube and the piston halts while the driver pressure is still very high. As a result, the pressure of the driver gas dominates the force balance acting on the piston, reversing its motion and driving it back up the compression tube [14]. As the driver gas continues to vent into the shock tube, the residual reservoir pressure ultimately dominates the piston force balance once again and the piston is accelerated downstream once more. In severe cases, this piston reacceleration can occur over an appreciable distance which can, in turn, lead to a dangerously high impact velocity. A rebounding impact can be

prevented by installing piston brakes which prevent reverse motion, however, X2 officials have decided to avoid this solution due to the increased operational complexity.

A soft piston landing can be induced by first generating an inflecting piston trajectory, where the piston is stationary for an instant near the end of its stroke before subsequent reacceleration. Next, a buffer system is strategically placed immediately after the zero-velocity, zero-acceleration ‘inflection’ point to catch the piston, thereby preventing the high velocity impact that would otherwise occur due to the reacceleration of the piston as the driver pressure continues to decay after diaphragm rupture. A soft piston landing is an essential aspect of safe driver condition design.

2.3 Hornung Analysis of The Free-Piston Driver

A rapidly solvable analytical driver model is an important tool for the early phase of driver condition design. Such models are based on the free-piston driver equations of motion derived in 1988 by Hornung [4]. Note: for brevity, only the conclusions of Hornung’s original analysis are provided below – for the full derivation and associated assumptions please refer to [4].

Nomenclature,

x	:	position, m
t	:	time, s
L	:	piston position from driver area change, m
M	:	piston mass, kg
D	:	piston diameter, m
P	:	pressure, Pa
A	:	speed of sound, m/s
γ	:	ratio of specific heats
m	:	driver gas mass, kg
d	:	throat diameter, m

Subscripts,

- A : air
- D : driver
- 0 : initial state
- r : rupture state

The piston is initially at rest, held in place on the launcher. Upon firing of X2, the high reservoir pressure accelerates the piston, in turn initiating the compression of the driver gas. Prior to diaphragm rupture, the piston motion is governed by the following equation of motion,

$$\frac{d^2x}{dt^2} = -\frac{\pi D^2}{4M} \left\{ p_{A,0} \left[1 + \frac{1}{a_{A,0}} \left(\frac{\gamma_A - 1}{2} \right) \frac{dx}{dt} \right]^{\frac{2\gamma_A}{\gamma_A - 1}} - p_{D,0} \left(\frac{L}{x} \right)^{\gamma_D} \right\} \quad (2.1)$$

The piston trajectory can be solved numerically with the following initial conditions,

$$x(0) = L, \quad \frac{dx}{dt}(0) = 0$$

Upon diaphragm rupture, the driver gas begins to flow downstream – the resulting mass flow rate can be modelled using Equation 2.2.

$$\frac{dm}{dt} = -\frac{\sqrt{\pi\gamma_D}}{2} \left(\frac{2}{\gamma_D + 1} \right)^{\frac{\gamma_D + 1}{2(\gamma_D - 1)}} \frac{d^2}{D^2} \sqrt{p_{D,r}} \left(\frac{x_r}{m_r} \right)^{\frac{\gamma_D}{2}} \left(\frac{m}{x} \right)^{\frac{\gamma_D + 1}{2}} \quad (2.2)$$

The introduction of mass loss to the shock tube necessitates a new governing equation of motion for the piston post diaphragm rupture,

$$\frac{d^2x}{dt^2} = \frac{\pi D^2}{4M} \left[p_{D,r} \left(\frac{x_r}{m_r} \right)^{\gamma_D} \left(\frac{m}{x} \right)^{\gamma_D} - p_{A,R} \right] \quad (2.3)$$

Equations 2.2 and 2.3 can be solved numerically to yield an approximation of piston trajectory post diaphragm rupture. The initial conditions for the solution are the piston's position and velocity at the time of diaphragm rupture. This analytical model forms the foundation of SPARK, a code currently under development at X2. SPARK models the dynamics of free-piston compression using Hornung's equations of motion with some additional features; the code integrates optional advanced equilibrium gas modelling,

along with piston friction and primary diaphragm opening effects. SPARK is also capable of optimising reservoir pressure to produce a perfectly inflecting piston trajectory for given input conditions. This functionality, which is offered in the order of seconds, encourages rapid development of ‘safe’ preliminary driver conditions.

2.4 PITOT3

UQ’s basic impulse facility simulation code, PITOT3 [9], is a ‘0-D’ state-to-state solver capable of efficiently estimating central flow processes through the X2 facility using established compressible and isentropic flow relations, and equilibrium and frozen chemistry. For the purpose of driver condition design, the user can input key parameters such as facility fill pressures, fill composition and driver compression ratio and the code will calculate an estimate of X2 performance.

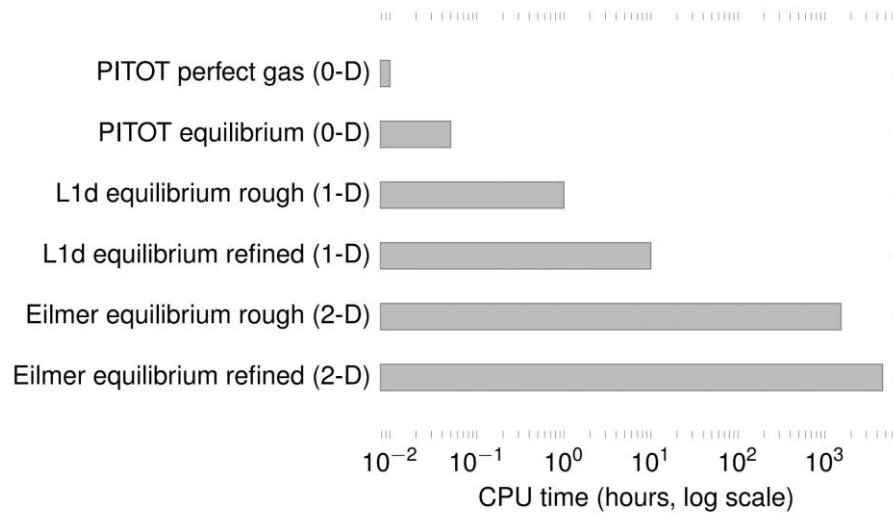


Figure 2.5: Approximate X2 simulation times for various computational methods [8].

As showcased in Figure 2.5 above, when compared to other UQ CFD codes, PITOT3 can deliver results significantly faster (typically in the order of minutes), thus enabling efficient evaluation of preliminary driver conditions during the early design phase.

2.5 L1D4

UQ’s quasi-one-dimensional CFD code, L1D4 [6], is a Lagrangian gas dynamics solver capable of simulating phenomena such as free-piston-driven compression, equilibrium chemistry, various diaphragm rupture phenomena, and longitudinal wave processes; this renders L1D4 particularly apt for modelling the dynamics of free-piston-driven expansion tubes such as X2. L1D4 operates on the basis of defining a series of end-to-end coupled objects, categorised as a GasSlug or a Piston, which are confined within a

tube of varying, user-defined cross-sectional area [6]. Each of the GasSlug objects are assigned initial flow state conditions which, over small time steps, are updated according to the constraints of mass, momentum and energy. By farming the time dependant gas state results across an array of locations, the performance of the physical facility can be estimated. Depending on the fidelity of the simulation, L1D4 can run a full-facility X2 simulation in the order of hours, making it more appropriate for driver condition design than more comprehensive CFD codes such as Eilmer [2, 9].

2.6 Low Enthalpy Driver Condition Design Principles

X2 is relatively simple by hypersonic ground testing facility standards. However, there are still several key parameters which directly impact facility performance. Fortunately, some of these parameters are fixed due to the construction of X2. This renders a far more manageable set of variables to manipulate in the pursuit of new 'low performance' driver conditions, these are: piston mass, reservoir fill pressure, driver gas composition, driver fill pressure, diaphragm rupture pressure, buffer length, and orifice plate diameter [7].

Compression ratio is a key metric within the scope of this project. Assuming isentropic compression, it follows from the selection of driver composition, driver fill pressure and primary diaphragm rupture pressure:

$$CR = \lambda = \left(\frac{p_{rup}}{p_{4i}} \right)^{\frac{1}{\gamma_D}} \quad (2.4)$$

X2 primarily utilises driver conditions with compression ratios of 20, 30 and 40. These conditions, designed by Gildfind [3], are summarised in Table 2.1. The target compression ratio for the new 'low performance' driver conditions was set at CR 15.

Table 2.1: Primary X2 driver conditions.

CR (-)	p_{4i} (kPa)	p_{rup} (MPa)	p_{res} (MPa)	He / Ar Mole Fractions (%)
20	110.3	35.7	4.85	100 / 0
30	92.8	27.9	6.85	100 / 0
40	77.2	15.5	6.00	100 / 0

Lowering compression ratio can raise many challenges, the most prominent of which is the necessity for longer buffers, as the low volumetric compression means the piston inflects further from the end of the compression tube and therefore needs to be caught earlier to maintain a soft landing. In the case of Stennett's [14] implementation of low CR conditions for X3R, a new enhanced buffer extension system was required as the previous design was found to be susceptible to buckling when elongated.

Ultimately, X2 performance is innately linked to the driver gas sound speed, a_4 , at the time of diaphragm rupture [3]. Decreasing a_4 reduces the velocity of the driven shocks which in turn decreases the test flow enthalpy; lowering the driver compression ratio is one method for achieving this [14]. However, as Equations 2.5 and 2.6 suggest, there are other means by which to decrease driver sound speed, such as low driver fill temperatures, high molecular-weight driver gas (e.g. pure Ar), or driver gas with a low ratio of specific heats (e.g. pure N₂).

$$T_4 = T_{4i}(\lambda)^{\gamma_D-1} \quad (2.5)$$

$$a_4 = \sqrt{\gamma_D R_D T_4} \quad (2.6)$$

In fact, in 2019, a nitrogen driver was studied in the T4 facility, and was later proposed as a suitable driver gas for low enthalpy free-piston-driven impulse facilities [16]. The efficacy of nitrogen as a driver gas is illustrated in Figure 2.6, where the sound speed of nitrogen and argon, post adiabatic compression, are compared for a range of compression ratios. As is evident, nitrogen yields lower sound speeds across all useful compression ratios.

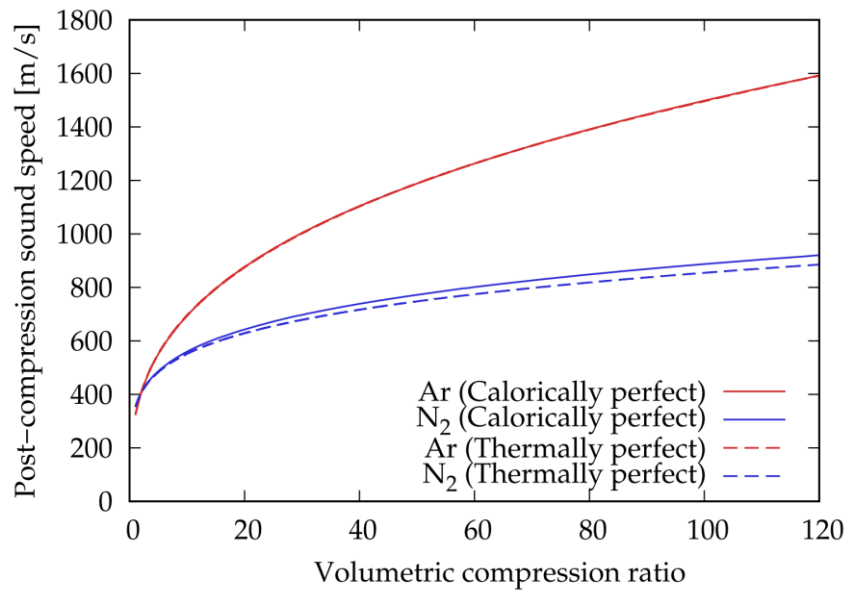


Figure 2.6: Post-compression sound speed of nitrogen and argon [16].

However, it is important to note that the ‘low performance’ benefit of nitrogen becomes less pronounced at low compression ratios. Another advantage of using nitrogen as a driver gas is its improved cost efficiency compared to helium. Conversely, the T4 researchers also highlighted that as nitrogen is a diatomic gas, there are two main disadvantages relevant to free-piston-driven facilities. Firstly, the ratio of specific heats deviates at significantly lower temperatures than that for monatomic gases, such as helium and argon, which complicates driver condition analysis. Secondly, calculating fill pressures is more tedious for mixtures of monatomic and diatomic gases, which only exacerbates the first disadvantage. The use of nitrogen as a ‘low performance’ X2 driver gas was explored briefly in Section 4.2.

3 Project Plan

The project was divided into a series of chapters, shown in Figure 3.1, some of which formed the remaining body of this thesis. An overview of each chapter and its subtasks are presented in the following methodology. This section concludes with a review of the project timeline.

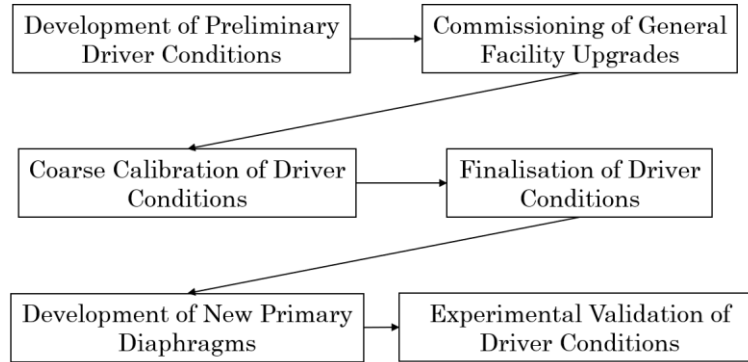


Figure 3.1: Project chapters.

3.1 Project Methodology

Development of Preliminary Driver Conditions: The first step towards the new driver conditions was the definition of key performance criteria by which to evaluate the efficacy of the conditions. Next, the driver condition design space was rapidly explored using SPARK to analytically model the free-piston dynamics of the X2 driver; PITOT3 was run in series with SPARK to estimate the primary shock speeds generated by the preliminary driver conditions. A set of new driver conditions was then iteratively tuned to produce a linearly spaced range of primary shock speeds from 1.5-3 km/s. Next, an L1D4 model of the X2 facility was established with great care taken to ensure the internal volumes of the facility were preserved in the transition to quasi-one-dimensional space. Subsequently, the driver conditions developed in SPARK were modelled in L1D4 and iteratively tuned until the piston trajectories observed in SPARK were replicated in L1D4. The test flows generated by each condition was analysed and iterative edits were made to the conditions back at the SPARK phase where applicable. Finally, once a freestream U_e range of approximately 3-6 km/s was observed, the conditions were documented for future calibration.

Commissioning of General Facility Upgrades: The process of commissioning the X2 facility upgrades is detailed at length in Appendix H, however a brief overview has been included here for convenience. Firstly, for the selection of both the new digital pressure

gauges and the new wear pad material, the process broadly consisted of researching the available options and selecting the product which best met X2's requirements. Correction of the O-ring gland was as simple as referencing the relevant handbooks and implementing the alteration in the CAD model. Restoration of the X2 static rig was equally straight forward, as assembly of the new hydraulic system was outsourced to specialists at Rhodes Hydraulics. The early development of laser-scoring techniques for the primary diaphragms was more complicated, starting with calibration of the UQ Innovate laser machine to enable engraving, before transitioning into the design of a dedicated laser-scoring rig to address repeatability concerns. Finally, the development of driver-length instrumentation consisted of optimisation of the PCB distribution along the X2 compression tube and preliminary design of a flush-mounted PCB housing.

Coarse Refinement of Driver Conditions: Coarse refinement began with the simulation of pure-helium blanked-off tests in L1D4. Subsequently, identical experiments were conducted in X2, and the driver pressure trace and maximum piston travel were recorded. The experimental results were then compared to the L1D4 results and the launcher loss factor was tuned such that the two data sets aligned. In turn, the newly calibrated L1D4 model was used to update the preliminary driver conditions. The coarsely-refined driver conditions were documented for future finalisation. This was the last step of the overall methodology completed during this thesis – the following stages of the methodology are representative of the future work required to complete the low compression ratio driver conditions.

Finalisation of Driver Conditions: The final L1D4 calibration will once again begin with simulating blanked-off tests, however this time, simulations will be conducted for each individual driver condition in order to capture the effects of driver gas composition on the pre-rupture piston mechanics. Corresponding blanked-off experiments will be conducted in X2, the results of which will hopefully allow for highly accurate, driver-composition-specific calibration of the L1D4 model. In turn, the final driver conditions will be dialled in and a grid independence study will be conducted to verify that the L1D4 solutions have converged.

Development of New Primary Diaphragms: With the driver conditions finalised, the only remaining step before validation testing is the physical manufacturing of the new primary diaphragms. To achieve this, a thickness of aluminium sheet metal must first be selected. This sheet will be manufactured into a small batch of diaphragms which will be statically ruptured using the X2 rupture rig. The rupture pressure determined from these experiments will be correlated to an estimated dynamic rupture pressure and the process will be repeated until the diaphragms align with the rupture pressures specified in the finalised driver conditions.

Experimental Validation of Driver Conditions: The final step in the development of the low compression ratio driver conditions will be a series of diaphragm-rupturing validation experiments for each of the candidate driver conditions. The results of these experiments will be used by X2 officials to evaluate the success of this project.

3.2 Project Timeline

Throughout the course of the project, progress generally followed the path laid out in the project methodology. January was primarily spent building foundational knowledge and forming an early project plan with the supervisory team. February was focused in equal parts on exploration of the driver condition design space and commissioning of facility upgrades. March centred around setting up the L1D4 code, which was utilised heavily in April for iterative design of preliminary driver conditions; early development of the driver instrumentation also began in April. May was dominated by the first round of blanked-off tests and a shift in focus following novel results. Finally, June saw progress made towards laser-scored diaphragms as well as coarse refinement of the driver conditions. While a secondary round of detailed blanked-off experiments was planned to take place mid-June, an unfortunate launcher failure derailed this campaign. A detailed timeline, including the background progression of ENGG7291 assessment items, is provided in Appendix B. Furthermore, several risks to the project timeline were identified at the beginning of the project, the details of which can be found in Appendix C alongside the full risk assessment created during the project proposal phase. While, overall, the scheduling risks were managed reasonably effectively, the combination of experimental misfortune and new learnings late in the project meant that several deliverables were not completed by the end of the ENGG7291 placement.

4 Development of Preliminary Driver Conditions

The first step towards the new, low compression ratio driver conditions was the development of preliminary conditions. This section begins with the definition of key driver criteria before delving into the driver condition design process. A summary of the initial set of conditions is provided at the end.

4.1 Definition of Key Driver Criteria

The development of the driver conditions began with defining the criteria by which to evaluate the conditions. The first key metric is pressure overshoot, described by Equation 4.1 below, which serves as a convenient indication of the degree of piston overdrive. Increasing piston overdrive increases driver hold time, however, the definition of constant-pressure operation only permits a deviation of up to 10% from the rupture pressure.

$$p_{over} = \left(\frac{p_{4,max}}{p_{rup}} - 1 \right) \cdot 100\% \quad (4.1)$$

Hold time, by the same definition, is the duration for which the driver is assumed to operate at constant pressure. This occurs between the time of primary diaphragm rupture and the time at which driver pressure falls below 90% of the rupture pressure.

$$t_{hold} = t_{90\%} - t_{rup} \quad (4.2)$$

Another key metric used throughout the initial design process is the primary shock speed, v_{s1} , calculated using PITOT3. The speed of the primary shock is a useful estimation of net facility performance as experience shows that test flow flight equivalent velocity is approximately double v_{s1} [8]. Primary shock speed was eventually superseded as the main performance metric once the test flow conditions were able to be modelled directly in L1D4. Within PITOT3, primary shock speed is calculated according to Equation 4.3.

$$v_{s1} = a_1 \sqrt{\frac{\gamma_1 + 1}{2\gamma_1} \left(\frac{p_2}{p_1} - 1 \right) + 1} \quad (4.3)$$

The driver hold time required for a useful X2 experiment follows from the primary shock speed, as X2 officials postulate that hold time must be at least long enough to drive the shock tube from the time of primary diaphragm rupture to the time of secondary diaphragm rupture. The rationale underpinning this idea is that once the primary shock reaches the acceleration tube, the shock front velocity increases by such a large margin that subsequent driver pressure decay is rendered inconsequential. Thus, an estimate of the required driver hold time can be defined as follows.

$$t_{required\ hold} = \frac{L_{shock\ tube}}{v_{s1}} \quad (4.4)$$

Ultimately, the most important criterion in the pursuit of ‘low performance’ driver conditions is the predicted stagnation enthalpy, H_t , of the test flow. Within the scope of the project, a slightly more physical term, the flight equivalent velocity, U_e , was the used to characterise the performance of the driver conditions. The relation between stagnation enthalpy and flight equivalent velocity is provided in Equation 4.5.

$$U_e = \sqrt{2H_t} \quad (4.5)$$

The final key design metric, which pertains to the safety of a driver condition, is the impact velocity of the free piston, v_{imp} . This is defined as the greatest velocity at which the piston impacts the buffer studs during its stroke; note that for severe rebounding trajectories, the piston can strike the buffers harder upon secondary contact. To ensure the safety of the facility and operators alike, the maximum permissible impact velocity was set to 10 m/s.

4.2 Exploration of The Driver Condition Design Space

As previously stated, the set of driver variables available for manipulation in the pursuit of new ‘low performance’ driver conditions includes: piston mass, reservoir fill pressure, driver fill composition, driver fill pressure, diaphragm rupture pressure, buffer length, and orifice plate diameter. While this set is only a fraction of the total performance-dictating facility variables, it still represents a daunting design space.

In order to obtain a basic intuition regarding how each of these variables impact net performance, a series of ‘control’ driver conditions were designed using SPARK and PITOT3. It should be noted that for this initial exploration of the design space, rupture

pressure was maintained at 15.5 MPa, maximum driver pressure was limited to +10% of the rupture pressure, and buffer lengths were selected to ensure a soft piston landing with an impact velocity less than 10 m/s. The control conditions are based on Gildfind's [3] CR 20 condition, however four low compression ratios (17.5, 15, 12.5 & 10) were trialled in parallel with three piston masses, the standard X2 10.5kg piston and two alternative heavy pistons (20 kg & 35 kg). The control driver conditions, and their associated results, are presented in Appendix D, Tables D.1-D.4.

Upon analysis of the control driver conditions, several design trends can be observed. Firstly, as the compression ratio was lowered, reservoir fill pressure, driver fill pressure, buffer length, hold time and primary shock speed all increased. The increase in driver fill pressure and required buffer length was a direct, expected consequence of lowering the volumetric compression ratio – increased reservoir pressure was in turn a consequence of the increased driver fill pressure. The increase in hold time was also expected, as for a given rupture pressure, the lower compression ratio conditions required higher fill pressures and therefore contained more moles of driver gas; The mass flow rate into the shock tube was therefore a smaller fraction of the total driver gas mass, so the driver pressure was less sensitive post-rupture and hence remained approximately constant for longer.

Conversely, the increase in primary shock speed was not initially predicted, as lowering compression ratio was previously shown to lower driver sound speed and thereby decrease driver performance. However, the original analysis failed to account for the observation that, all other things being equal, lower compression ratios required higher mole fractions of helium driver gas to maintain the soft-landing criterion (the highly-coupled nature of the driver variables highlights the benefit of a preliminary, brute-force design space exploration). The need for higher percentages of helium ultimately rendered several 10.5 kg piston conditions infeasible, denoted by a dash in Tables D.2-D.4. This occurred because even after the driver composition reached pure helium, the post-rupture driver gas drainage remained over-choked, so the piston inevitably rebounded, and the soft-landing condition was invalidated.

Secondly, as the piston mass was increased, reservoir fill pressure, driver gas helium content, buffer length and primary shock speed all decreased while driver hold time was extended. At first glance, these results implied that a heavy piston would be extremely desirable for low performance drivers. However, the old X2 35 kg piston, designed by Scott [13], requires special considerations due to its mass which increases the risks and complexity of facility turnaround. The theoretical 20 kg piston, on the other hand, would retain the turnaround procedure of the standard X2 piston, but was infeasible to develop within the limited timeline of the project. Moreover, it was later discovered that similar results could be achieved by decreasing the rupture pressure of the primary diaphragm – a far more convenient solution. As such, subsequent driver conditions exclusively utilised the standard X2 10.5 kg piston.

It is also worth noting that orifice plates were implemented such that the free-piston mechanics could be preserved across four different driver compositions and therefore four different primary shock speeds. The inclusion of an orifice plate with a throat diameter less than the driven tube diameter (85 mm) resulted in greater choking of the driver gas mass flow rate into the shock tube. In order to balance this restriction, the driver gas needed to be made lighter by increasing the helium content. The increased helium content, in turn, raised the driver gas sound speed and thereby increased the strength of the driven shock. Finally, as none of the control conditions satisfied the lower portion of the 1.5-3 km/s primary shock speed target, it was concluded that no sensible manipulation of Gildfind's CR 20 condition could meet the objectives of the project – therefore, entirely new driver conditions were required.

The first new driver concept trialled was the inclusion of nitrogen driver gas. The nitrogen driver conditions featured a compression ratio of 15, with a rupture pressure of 8 MPa. Once again, maximum driver pressure was limited to +10% of the rupture pressure, and buffer lengths were selected to ensure a soft piston landing with an impact velocity less than 10 m/s. Presented in Table 4.1, the results of this trial were not as promising as implied in the literature [16].

Table 4.1: CR 15 nitrogen driver conditions.

m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{di} (kPa)	He / N ₂ Mole Frac. (%)
10.5	85	1.40	97.545	60 / 40
	77	1.42	93.222	75 / 25
	72.1	1.45	91.439	83 / 17
	65	1.48	89.571	93 / 7

Table 4.1: Continued.

L_{buffer} (mm)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{s1} (m/s)
98	1.590	1.491	2293
110	1.626	1.349	2534
120	1.651	1.266	2700
135	1.674	1.150	2973

While these conditions yielded sufficient hold time and appropriate primary shock speeds, the disadvantages were numerous. The main disadvantage stemmed from nitrogen's ratio of specific heats ($\gamma = 1.4$) which is different to that of helium and argon ($\gamma = 1.667$). This resulted in variance of reservoir fill pressure, driver fill pressure and buffer length between different driver compositions; hence the free-piston mechanics were no longer preserved between various orifice plates. In particular, the range of required buffer lengths (98-135 mm) was too large to maintain a soft piston landing for all conditions using a single buffer length. This meant that a whole collection of nylon buffer studs would be required to implement nitrogen driver gas mixtures; this is operationally infeasible for X2. Chan et al [16] utilised pure nitrogen driver gas and therefore did not encounter this challenge. However, the exclusive use of pure nitrogen driver gas was deemed impracticable for this project as the broad performance target necessitates a wide range of driver compositions.

With nitrogen ruled out, there remained two leading design philosophies for consideration: helium-maximisation and infrastructure-minimisation. Showcased in Table 4.2, the helium-maximisation philosophy centred around the goal of producing steadier driver supply properties by minimising the driver pressure perturbations

believed to be caused by the addition of argon driver gas [3]. To achieve this, two compression ratios (CR 10 & CR 18), three theoretical orifice plates (50, 55 & 60 mm), two theoretical buffer lengths (108 & 220 mm) and two theoretical primary diaphragms (2.72 & 5.57 MPa rupture) were proposed.

Table 4.2: Helium-maximising driver conditions.

CR (-)	m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{4i} (kPa)	He / Ar Mole Frac. (%)
10	10.5	85	0.65	58.556	18 / 82
		77			48.5 / 51.5
		72.1			63 / 37
		65			79.3 / 20.7
		60			88.1 / 11.9
		55			94.9 / 5.1
		50			100 / 0
18	10.5	65	0.83	45.011	79.4 / 20.6
		60			88.1 / 11.9
		55			94.9 / 5.1
		50			100 / 0

Table 4.2: Continued.

L_{buffer} (mm)	v_{imp} (m/s)	p_{rup} (MPa)	p_{over} (%)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{s1} (m/s)
220	0.440	2.72	10.00	4.331	2.230	1533
	1.032		9.85	4.319	2.044	1672
	1.190		9.89	4.310	1.938	1764
	0.706		9.96	4.328	1.790	1909
	1.800		9.67	4.289	1.691	2021
	2.156		9.60	4.274	1.599	2137
	1.013		9.85	4.319	1.516	2254
108	1.710	5.57	9.96	2.026	1.408	2428
	1.752		9.95	2.027	1.325	2579
	2.128		9.77	2.021	1.247	2741
	1.384		9.98	2.030	1.177	2905

Presented in Table 4.3, the infrastructure-minimisation philosophy instead focused on making as few changes to the existing X2 driver infrastructure as possible, whilst meeting to target performance envelope. After a lengthy iteration process, it was found that only one theoretical buffer length (130 mm) and two theoretical primary diaphragms (3 & 7 MPa rupture) were required to produce a set of eight new CR15 driver conditions that were capable of covering the entire primary shock speed target.

Table 4.3: CR 15 infrastructure-minimising driver conditions.

CR (-)	m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{4i} (kPa)	He / Ar Mole Frac. (%)
15	10.5	85	0.49	32.853	0 / 100
		77			19.3 / 80.7
		72.1			40.5 / 59.5
		65			64.5 / 35.5
15	10.5	85	1.32	76.657	51.4 / 48.6
		77			70.9 / 29.1
		72.1			80.2 / 19.8
		65			90.7 / 9.3

Table 4.3: Continued.

L_{buffer} (mm)	v_{imp} (m/s)	p_{rup} (MPa)	p_{over} (%)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{s1} (m/s)
130	2.117	3.0	9.96	3.152	2.136	1501
	2.125		9.97	3.152	2.085	1639
	2.106		9.90	3.153	1.972	1733
	2.124		9.97	3.152	1.816	1882
130	8.549	7.0	9.89	2.031	1.556	2196
	8.544		9.90	2.030	1.420	2407
	8.496		9.91	2.036	1.342	2547
	8.495		9.91	2.036	1.235	2767

Comparing the results of the two sets of conditions, it is clear that either design philosophy is capable of populating the primary shock speed target range of 1.5-3 km/s. By virtue of the additional orifice plates, the helium-maximising set produced a slightly denser distribution; however, it is important to note that the net performance of any

driver condition can be manipulated precisely through selection of shock tube and acceleration tube fill parameters. Both sets of conditions provide ample hold time, safe piston impact speeds and feature feasible reservoir and driver fill pressures. One major concern for the helium maximising set regarded the stiffness of the 220 mm buffers used for the CR10 conditions. The longest buffers used for standard X2 operation are 100 mm in length, and for the existing stud design, 220 mm neared the limit of requiring a buffer extension system such as the one designed by Stennett for X3R [14]; conversely, the CR15 buffers, at just 130 mm in length, were deemed insusceptible to buckling failure. Ultimately, because the infrastructure-minimising set offered superior operational simplicity, it was selected for further refinement in L1D4. Furthermore, the potential for argon induced driver noise was acknowledged as an acceptable project risk, with X2 officials agreeing that gathering data on the topic might be of benefit to the broader scientific community.

4.3 Preliminary L1D4 Simulations

The final stage of the preliminary driver condition design process was the simulation of the SPARK conditions in L1D4, UQ's quasi-one-dimensional Lagrangian gas dynamics solver. But before any simulations could be conducted, the L1D4 input code needed to be constructed.

4.3.1 L1D4 Geometry

In the transition from the real, three-dimensional X2 driver geometry, to the quasi-one-dimensional L1D4 space (which is defined by lengths and associated cross section diameters), it can be challenging to accurately capture the essence of the real driver. In fact, particularly complex geometries such as the piston launcher and the buffer plate are virtually impossible to model directly in one-dimension. Nonetheless, experience suggests that L1D4 can simulate free-piston-driven impulse facilities with reasonable accuracy provided internal volumes are conserved. However, area changes in L1D4 cannot be discontinuous and therefore must be modelled gradually; thus, internal volume conservation is a non-trivial task. The following section details the process of defining an L1D4 representation of the X2 driver.

The first step in this process was to redefine the X2 driver as a series of cylindrical volumes stacked end-to-end. In order to do this however, the reservoir and the piston launcher needed to be simplified. The relevant region of the real reservoir is illustrated in Figure 4.1, alongside the simplified version.

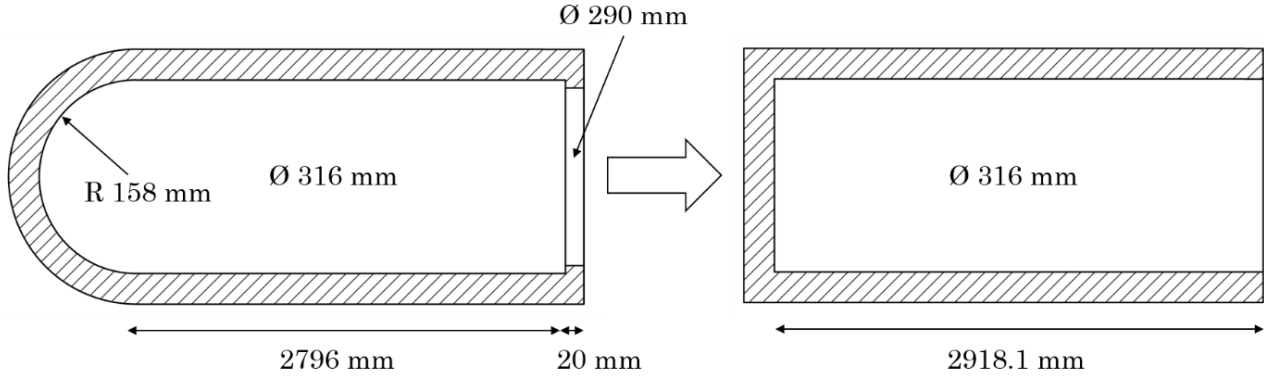


Figure 4.1: Simplified reservoir geometry (not drawn to scale).

The internal volume of the real reservoir section was calculated as the addition of the two cylindrical volumes and the hemispherical end.

$$V_{total} = \pi(0.02)(0.145)^2 + \pi(2.796)(0.158)^2 + \frac{2}{3}\pi(0.158)^3 = 0.229 \text{ m}^3 \quad (4.6)$$

This volume was then equated to that of a cylinder of identical diameter, and an effective length was determined for the simplified model.

$$L_{eff} = \frac{0.229}{\pi(0.158)^2} = 2.918 \text{ m} \quad (4.7)$$

Simplification of the launcher was more complicated as the rear edge of the piston is required to rest initially at the downstream edge of the launcher region. Therefore, the launcher region needed to proactively include a gradual area change to the diameter of the compression tube, otherwise the rear of the piston would be placed within an area change if standard volume conservation were to be applied directly at the start of the compression tube. The application of this observation is illustrated in Figure 4.2 – first the real launcher geometry was converted to an equivalent cylindrical representation, then a gradual area expansion was incorporated.

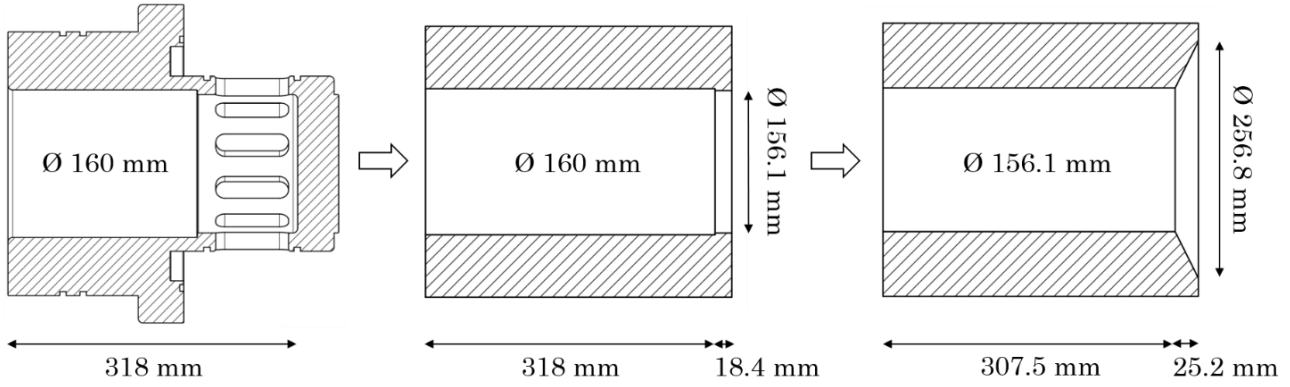


Figure 4.2: Simplified launcher geometry (drawn to scale).

To achieve this conversion, the internal volume of the launcher was approximated using the following equation (note: the launcher features ten slots which are 18.4 mm deep, with a combined cross-sectional area of 0.019 m^2 which is equivalent to a circular cross section with $\text{Ø } 156.1 \text{ mm}$).

$$V_{total} = \pi(0.318)(0.08)^2 + (0.0184)(0.019) = 6.746 \times 10^{-3} \text{ m}^3 \quad (4.8)$$

Subsequently, it was decided that the $\text{Ø } 160 \text{ mm}$ internal section, featured in the real launcher, would be converted to $\text{Ø } 156.1 \text{ mm}$, as to capture the chokepoint caused by the launcher slots which would otherwise be lost due to the inclusion of the area expansion on the downstream side of the launcher (see Figure 4.2); conveniently, applying the slots-area to the full length of the launcher ensured that the flow-choking characteristics of the launcher were better resolved across all mesh resolutions. Next, to implement this modification, the area expansion was assigned a radius-length aspect ratio of two, which enforced a length of 25.2 mm for the 50.4 mm radial expansion. In turn, the length of the $\text{Ø } 156.1 \text{ mm}$ section required for volume preservation was determined using the following equation.

$$L_{eff} = \frac{V_{total} - V_{expansion}}{\pi(0.07805)^2} = \frac{6.746 \times 10^{-3} - 8.603 \times 10^{-4}}{\pi(0.07805)^2} = 0.3075 \text{ m} \quad (4.9)$$

It is worth noting that while the area of the fully exposed slots represents the long-term launcher chokepoint, this approach does not capture the transient nature of a real piston launch; rather, as the piston begins to move off the launcher, the slots are gradually exposed, therefore the mass flow rate through the launcher is subject to a ramping effect. Unfortunately, this phenomenon cannot be modelled directly in L1D4; though an

L1D4 valve object could be incorporated within the launcher volume to provide an approximation of the ramping effect. Alternatively, the effects of this complex process can be empirically replicated with a loss region spanning the length of the launcher.

With the complex driver geometries converted to equivalent cylindrical forms, the complete cylindrical driver model was assembled based on measurements from the X2 facility CAD model. The full details of this cylindrical representation are provided in Table 4.4 and Figure 4.3.

Table 4.4: Simplified cylindrical representation of the X2 driver.

ID	x-location (m)	Diameter (m)	Comment	Volume (m³)	
A	-8.7188	0.3160	Start of Reservoir	Segment	Total
B	-5.8007	0.2440	-	0.2289	-
C	-5.1817	0.1800	-	0.0289	-
D	-5.1407	0.1561	-	0.0010	-
E	-4.8332	0.1561	-	0.0059	Reservoir:
F	-4.808	0.2568	End of Reservoir / Piston Rear	0.0009	$\Sigma = 0.2656$
G	-4.587	0.2568	Piston Front / Start of Compression Tube	-	-
H	-0.1120	0.0850	Buffer Plate	0.2318	Compression Tube:
I	0.0000	0.0850	Primary Diaphragm	0.0006	$\Sigma = 0.2324$

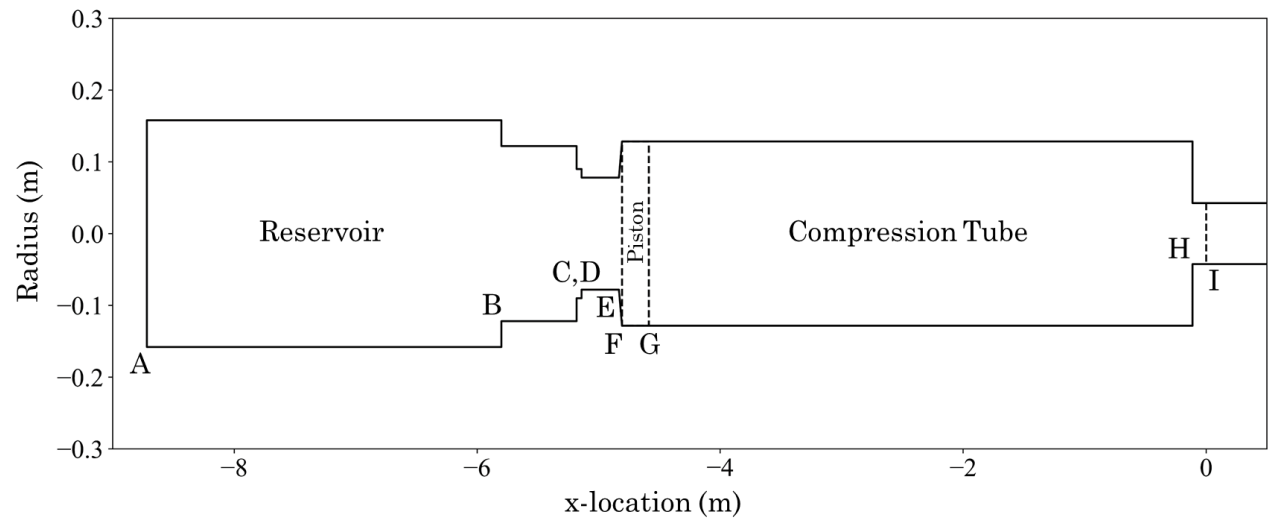


Figure 4.3: Simplified cylindrical representation of the X2 driver.

Finally, the cross-sectional area discontinuities (which occur at the interfaces between each cylindrical section) needed to be remodelled as gradual area changes, similar to the area expansion previously incorporated into the simplified launcher geometry. Past L1D4 representations of the X2 driver implemented these area gradients somewhat arbitrarily. However, in the pursuit of improved L1D4 accuracy, a mathematical model was derived to ensure the area gradients were implemented without changing the internal volumes of the driver.

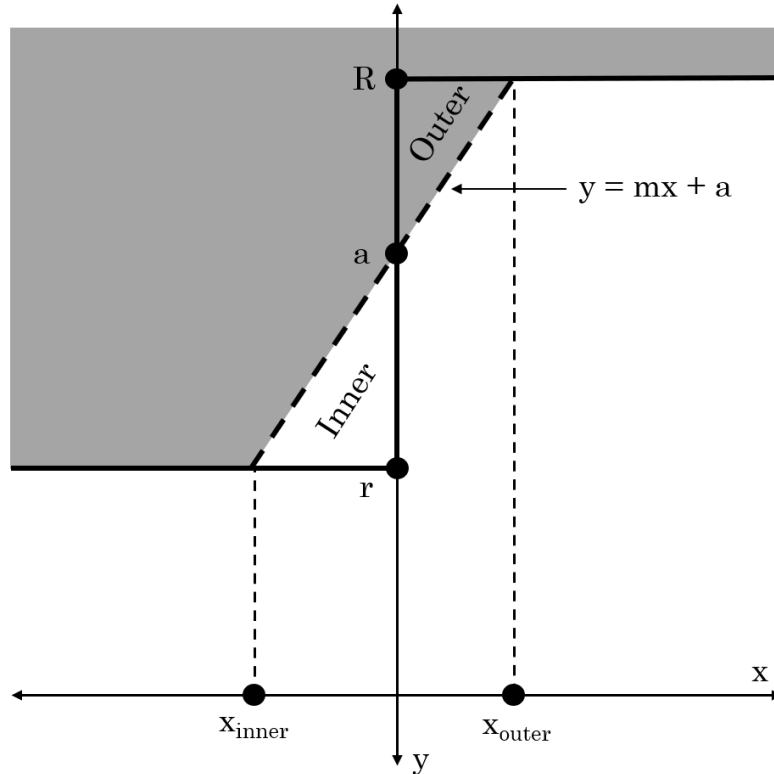


Figure 4.4: Internal volume conservation for a gradual area change.

Illustrated in Figure 4.4, first consider two triangular cross sections revolved about a central axis; for volume conservation, these revolutions must have equal volumes. The volume of the outer revolution was determined using the washer method of integration.

$$\begin{aligned}
 V_{outer} &= \pi \int_0^{\frac{R-a}{m}} (R^2 - (mx + a)^2) dx \\
 &= \pi \left[\frac{-m^2}{3} x^3 - amx^2 - a^2x + R^2x \right]_0^{\frac{R-a}{m}} \\
 &= \pi \left[\frac{-m^2}{3} \left(\frac{R-a}{m} \right)^3 - am \left(\frac{R-a}{m} \right)^2 - a^2 \left(\frac{R-a}{m} \right) + R^2 \left(\frac{R-a}{m} \right) \right]
 \end{aligned}$$

This expression was simplified to yield Equation 4.10.

$$V_{outer} = \pi \left[\frac{1}{3m} a^3 - \frac{R^2}{m} a + \frac{2R^3}{3m} \right] \quad (4.10)$$

The volume of the inner revolution was determined similarly.

$$\begin{aligned} V_{inner} &= \pi \int_{\frac{r-a}{m}}^0 ((mx + a)^2 - r^2) dx \\ &= \pi \left[\frac{m^2}{3} x^3 + amx^2 + a^2x - r^2x \right]_{\frac{r-a}{m}}^0 \\ &= -\pi \left[\frac{m^2}{3} \left(\frac{r-a}{m} \right)^3 + am \left(\frac{r-a}{m} \right)^2 + a^2 \left(\frac{r-a}{m} \right) + r^2 \left(\frac{r-a}{m} \right) \right] \end{aligned}$$

As before, this expression was simplified to yield Equation 4.11.

$$V_{inner} = \pi \left[\frac{1}{3m} a^3 - \frac{r^2}{m} a + \frac{2r^3}{3m} \right] \quad (4.11)$$

Next, in order to construct a general formula for the y-axis intercept, a , the following equality was defined, and rearranged for a . Note that the volume of the buffer studs was included in this expression so that the reduction in compression tube volume due to the presence of the buffer studs could be accounted for directly in L1D4.

$$\begin{aligned} V_{outer} &= V_{inner} - V_{buffer} \\ \pi \left[\frac{1}{3m} a^3 - \frac{R^2}{m} a + \frac{2R^3}{3m} \right] &= \pi \left[\frac{1}{3m} a^3 - \frac{r^2}{m} a + \frac{2r^3}{3m} \right] - V_{buffer} \end{aligned}$$

This expression was simplified to yield Equation 4.12.

$$a = \frac{2 \left(R^3 - r^3 + \frac{3mV_{buffer}}{2\pi} \right)}{3(R^2 - r^2)} \quad (4.12)$$

Where, for the standard X2 six-stud buffer configuration, the total buffer volume was approximated as the sum of six cylindrical volumes.

$$V_{buffer} = 6 \cdot \frac{\pi L_b D_b^2}{4} \quad (4.13)$$

For calculations without the inclusion of buffer studs, the volume of the buffers was simply set to zero. With a known, the x-locations of the start and end of the profile change were solved using Equations 4.14 and 4.15.

$$x_{outer} = \frac{R - a}{m} \quad (4.14)$$

$$x_{inner} = \frac{r - a}{m} \quad (4.15)$$

These points were then added to the x-location of each discontinuous profile change listed in Table 4.4. Note that this analysis assumed an expanding cross-section, thus for a reduction of cross-sectional area, the sign the aspect ratio, m , must be reversed. At last, with the volume conservation model established, the L1D4 representation of the X2 driver was finalised. The full details of the finalised geometry are presented in Table 4.5 and Figure 4.5.

Table 4.5: Basic L1D4 breakpoint representation of the X2 driver.

ID	x-location (m)	Diameter (m)	Comment	Volume (m ³)	
1	-8.7188	0.3160	Start of Reservoir	Segment	Total
2	-5.8093	0.3160	-	0.2282	-
3	-5.7913	0.2440	-	0.0011	-
4	-5.1893	0.2440	-	0.0281	-
5	-5.1733	0.1800	-	0.0006	-
6	-5.1436	0.1800	-	0.0008	-
7	-5.1376	0.1561	-	0.0001	-
8	-4.8332	0.1561	-	0.0058	Reservoir:
9	-4.8080	0.2568	End of Reservoir / Piston Rear	0.0009	$\Sigma = 0.2656$
10	-4.5870	0.2568	Piston Front / Start of Compression Tube	-	-
11	-0.1299	0.2568	-	0.2309	-
12	-0.0869	0.0850	Buffer Plate	0.0011	Compression Tube:
13	0.0000	0.0850	Primary Diaphragm	0.0005	$\Sigma = 0.2325$

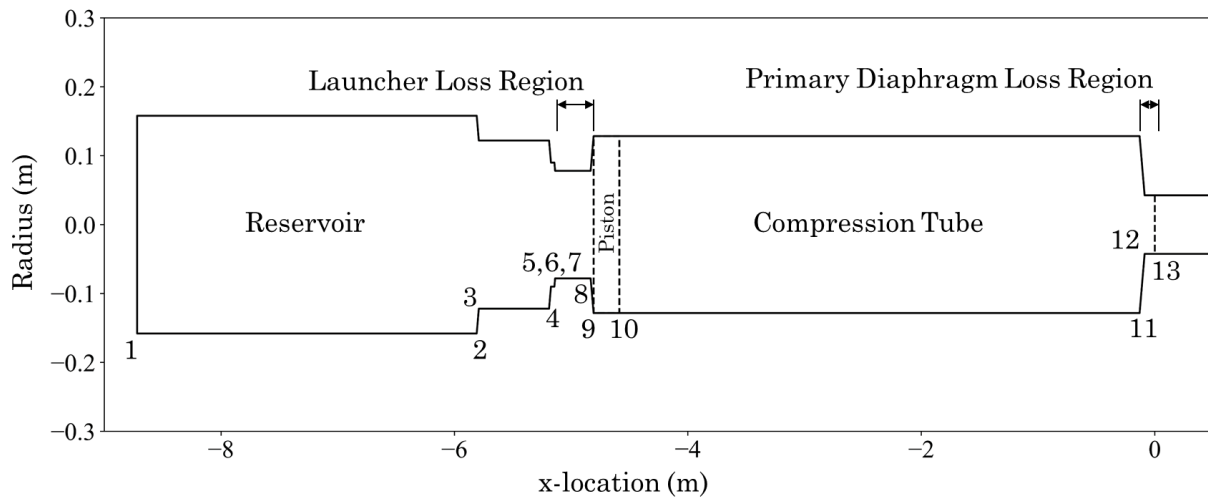


Figure 4.5: Basic L1D4 X2 driver configuration – without buffers or an orifice plate.

Finally, note that the breakpoint representations for the shock tube, the acceleration tube, and the Mach 10 nozzle were simply adopted from the X2 script provided in the GDTK L1D4 examples folder.

4.3.2 L1D4 Simulation Parameters

While key driver parameters such as reservoir fill pressure, driver composition, rupture pressure etc. were altered between different driver conditions, there were also several L1D4 simulation parameters which remained constant across all conditions. For convenience, these have been categorised into general parameters and gas slug specific parameters, as listed in Table 4.6.

Table 4.6: L1D4 simulation parameters.

General Parameters	Details
Piston Mass (kg)	10.524
Ambient Temperature (K)	295.15
Viscous Effects (-)	Enabled for the reservoir & driver gas slugs
Launcher Loss Factor	0.1
Primary Diaphragm Loss Factor	0.7
Secondary Diaphragm Loss Factor	0.2
Gas Slug Details	
Reservoir	Gas model: Ideal air Cell count: 39 * Mesh-scale-factor
Driver	Gas model: Thermally perfect He-Ar mixture Cell count: 46 * Mesh-scale-factor
Test	Gas model: CEA LUT ionising air Cell count: 34 * Mesh-scale-factor Initial pressure: 10 kPa
Accelerator	Gas model: CEA LUT ionising air Cell count: 66 * Mesh-scale-factor Initial pressure: 10 Pa

The general parameters were broadly adopted from the X2 script provided in the GDTK L1D4 examples folder. In slight contrast to the example script, viscous effects were only enabled within the reservoir and driver gas slugs. This decision was made to support the timely development of preliminary driver conditions; the validity of this decision was reinforced by the project scope, where it was deemed that the detailed optimisation of the test flow conditions (through manipulation of the driven-tube configuration) was

beyond the scope of the project. It should be noted that while the launcher loss factor was initially set to 0.1, as featured in the example script, it was later calibrated experimentally (see Section 5).

Regarding the gas slug parameters, first, the fill pressures for the shock tube and acceleration tube were set in accordance to the project scope. Next, the cell count formula for each slug was defined as the product of a mesh-scale-factor (for ease of mesh refinement) and an integer which approximately correlated to ten times the initial length of the given gas slug. Finally, each gas slug was assigned a gas model in accordance to the composition and chemistry each slug was required to model.

- **Reservoir Gas Slug:** An ideal air model was utilised as the temperatures in the reservoir remain low during X2 operation.
- **Driver Gas Slug:** A thermally perfect helium-argon model was utilised as the noble gases comprising the driver were assumed to be chemically inert at the temperatures encountered during low-enthalpy X2 operation.
- **Test & Accelerator Gas Slugs:** A CEA look-up-table air model was utilised as these gas slugs reach appreciable temperatures, at which air becomes highly chemically reactive.

4.3.3 Preliminary L1D4 Results

The initial driver condition design phase was brought to an end by transferring the infrastructure-minimising set of driver conditions from SPARK to L1D4 – the resulting set of driver conditions will be hereon referred to as the preliminary driver conditions. This transition also marked the switch from primary shock speed, v_{sl} , to flight equivalent velocity, U_e , as the main performance metric. The preliminary driver conditions and their associated performance characteristics are summarised in Table 4.7.

Table 4.7: Preliminary CR 15 driver conditions.

CR (-)	m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{di} (kPa)	He / Ar Mole Frac. (%)
15	10.5	85	0.72	32.853	0 / 100
		77			22 / 78
		72.1			48 / 52
		65			75 / 25
15	10.5	85	1.97	76.657	56 / 44
		77			68 / 32
		72.1			80 / 20
		65			94 / 6

Table 4.7: Continued.

L_{buffer} (mm)	v_{imp} (m/s)	p_{rup} (MPa)	p_{over} (%)	t_{hold} (ms)	$t_{req. hold}$ (ms)	U_e (m/s)
130	4.330	3.0	10.93	3.133	2.136	3300
	4.275		10.56	3.098	2.085	3620
	4.641		10.34	3.067	1.972	4150
	5.183		10.97	3.146	1.816	4400
130	2.901	7.0	10.71	2.031	1.556	4770
	7.284		9.97	1.903	1.420	5100
	5.966		10.41	1.982	1.342	5690
	8.283		10.88	2.045	1.235	6230

A brief comparison between these preliminary driver conditions and their SPARK counterparts shows that the two sets of conditions are very similar. However, two minor discrepancies were observed: the L1D4 conditions required greater reservoir fill pressures and generally higher helium content. The preliminary conditions generated test flows with flight equivalent velocity in the range of 3.3-6.2 km/s, which was deemed satisfactory given the project target range of 3-6 km/s. Within this range, each condition was separated by approximately 400-600 m/s, which effectively ensured that a continuous performance envelope could be designed through future manipulation of the driven tube fill parameters.

5 Coarse Refinement of Driver Conditions

With a promising set of preliminary driver conditions established, the next step was refinement of the driver conditions via blanked-off calibration. Blanked-off testing involves the study of driver pressure and piston displacement using a non-rupturing diaphragm. This section begins with an overview of the initial blanked-off simulations conducted in L1D4, before detailing the launcher loss-factor calibration process. Subsequently, the coarsely-refined set of driver conditions is presented. Finally, the remaining experimental work is outlined for the convenience of future project contributors.

5.1 Preliminary L1D4 Blanked-Off Simulations

Before X2 blanked-off tests could be conducted, L1D4 blanked-off simulations were first required to establish predictions for the maximum driver pressure and maximum piston displacement; these initial predictions inform decisions regarding PCB selection and the welding rod length used in the real experiments. If the implementation of orifice plates is assumed to enable perfect translation of free-piston mechanics between varied driver gas compositions, then the eight preliminary driver conditions can effectively be reduced to two primary conditions – the ‘low-enthalpy’ 3 MPa-rupturing condition and the, relatively, ‘high-enthalpy’ 7 MPa-rupturing condition.

These two primary conditions were tested in blanked-off configuration using L1D4. This involved making three changes to the standard L1D4 model: the primary diaphragm rupture pressure was increased by three orders of magnitude (to ensure it would not burst), the buffer length was set to 100 mm (indicative of the buffers used in the real blanked-off experiments), and the driver gas composition was set to pure helium. For these initial simulations the launcher loss-factor was maintained at 0.1 as per the GDTK example script. The results of the preliminary L1D4 blanked-off simulations are presented in Table 5.1.

Table 5.1: Preliminary L1D4 blanked-off results.

Condition	p_{res} (MPa)	p_{4i} (kPa)	$p_{4,max}$ (MPa)	x_{max} (m)
CR 15 (Low-Enthalpy)	0.72	32.853	6.922	4.292
CR 15 (High-Enthalpy)	1.97	76.657	17.134	4.296

5.2 X2 Blanked-Off Experiments

An X2 blanked-off driver experiment is broadly similar to standard free-piston driver operation, except for the inclusion of a non-rupturing primary diaphragm, commonly referred to as a blanked-off plate, which effectively isolates the driver from the remainder of the facility downstream. A PCB pressure transducer is housed within the blanked-off plate to record the driver pressure trace.

During a blanked-off experiment, the free-piston does not come to rest at end of the compression tube; instead, it rebounds back and forth before eventually settling at an equilibrium point where the reservoir and driver pressure forces are equal. In order to determine the maximum piston displacement, aluminium welding rods are installed into the buffer studs at the end of the compression tube. These rods extend upstream in a staggered array and are deformed by the piston as it inflects for the first time during its trajectory. As shown in Figure 5.1, inspection of the welding rods post-experiment yields the location of the piston's inflection point with respect to the primary diaphragm, ΔL . In turn, this metric can be related to piston displacement using the known geometry of the compression tube.

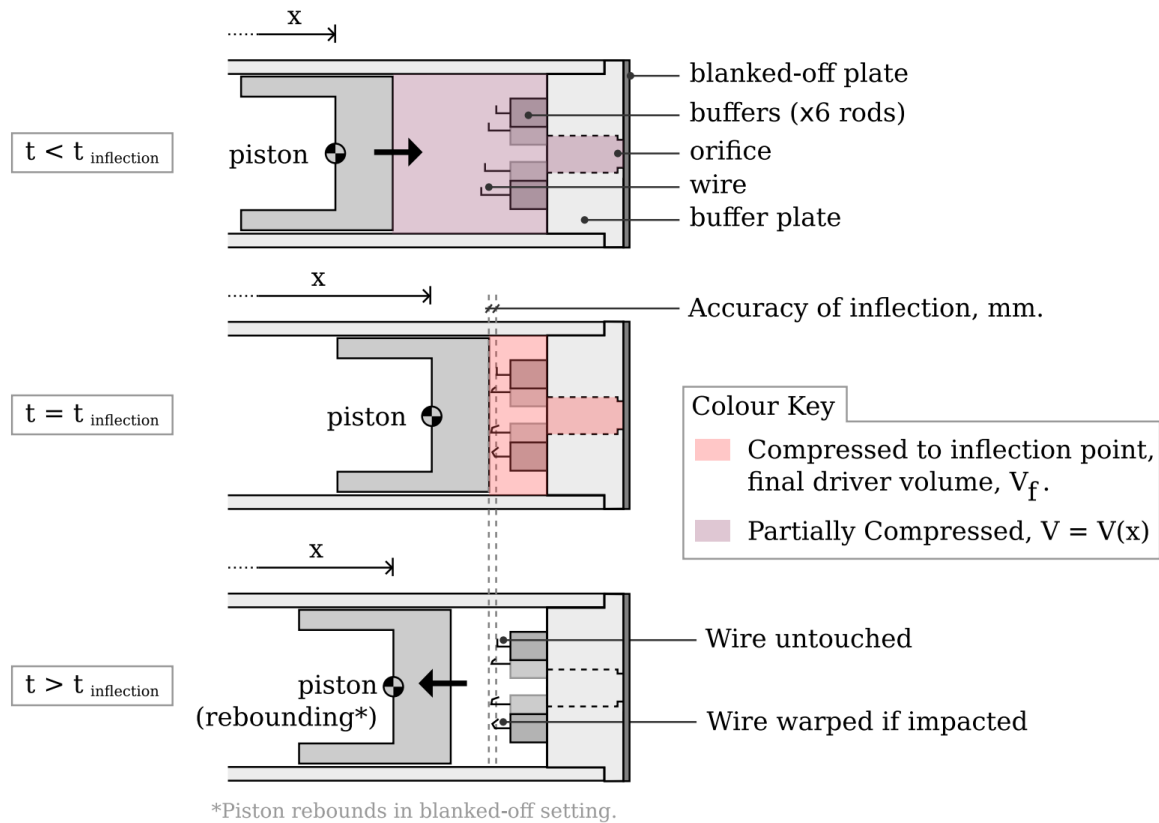


Figure 5.1: The stages of a blanked-off experiment [10].

Following the standard operating procedure presented in Appendix I.1, four successful, pure-helium blanked-off experiments were conducted – consisting of two trials of the conditions previously presented in Section 5.1. The results of these experiments can be found in Table 5.2.

Table 5.2: X2 blanked-off results.

Condition	Shot ID	p_{res} (MPa)	p_{4i} (kPa)	$p_{4,max}$ (MPa)	ΔL_I (m)
CR 15 (Low-Enthalpy)	x2s5983	0.7	32.85	5.517	0.302
	x2s5984	0.7	32.85	5.574	0.302
CR 15 (High-Enthalpy)	x2s5987	2.0	76.66	12.046	0.325
	x2s5988	2.0	76.66	12.152	0.325

Table 5.2: Continued.

x_{max} (m)	V_{final} (m ³)	CR (-)	$\gamma_{4,polytropic}$ (-)
4.285	9.298×10^{-3}	24.869	1.59
4.285	9.298×10^{-3}	24.869	1.60
4.262	9.298×10^{-3}	22.044	1.64
4.262	9.298×10^{-3}	22.044	1.64

Comparing the experimental results to the preliminary L1D4 simulations, it is clear that the L1D4 model severely underpredicted the losses incurred across the piston launcher resulting in overestimation of peak driver pressure – approximately +25% for the low enthalpy condition and +40% for the high enthalpy condition; it was anticipated that the error would be greater for the high-enthalpy condition as the reservoir pressure is higher, which leads to greater reservoir gas velocities on average, and consequently greater losses through the launcher geometry. The maximum piston displacement predicted by the L1D4 model was fairly accurate, though with such a significant disparity between the pressure results, the need for calibration of the L1D4 model remained evident.

$$\gamma_{4,polytropic} = \frac{\ln \left| \frac{p_{max,observed}}{p_{4i}} \right|}{\ln |CR_{observed}|} \quad (5.1)$$

Also included in Table 5.2, a polytropic compression index was calculated for each blanked-off trial using Equation 5.1 (the initial volume of the driver, V_i , is 0.2312 m^3). Polytropic compression index is a useful metric to consider as it provides an indication of heat loss in the driver during the compression process. For isentropic compression with no heat loss, $\gamma = 5/3 = 1.667$ for helium and argon, for isothermal compression, $\gamma = 1$, and for real, 'polytropic' compression, as observed in X2's free-piston driver, γ is expected to fall within these bounds. In fact, the polytropic compression indexes determined from the blanked-off experiments were approximately 1.59 for the low-enthalpy condition and 1.64 for the high-enthalpy condition. At first glance, it may seem counter-intuitive that the high-enthalpy condition would exhibit less heat loss, but there is a logical reason behind this result. As can be seen in Figures 5.2 & 5.3, the low-enthalpy condition maintains the highly-compressed state for significantly longer than the high-enthalpy condition. Considering that the two conditions exhibited similar compression ratios (24.869 & 22.044), it follows that maximum driver temperatures would also be comparable. Thus, if the heat flux out of the driver gas is assumed to be similar, one would expect the low-enthalpy condition to incur greater heat loss due to the extended duration of the highly-compressed, high-temperature state. It is interesting to note that the polytropic compression index for the low-enthalpy blanked-off tests imply a similar degree of total heat loss to that of very high compression ratio conditions tested previously in X2 [10].

It is also important to note that a single pure-argon blanked-off experiment was conducted for the low-enthalpy condition (as the corresponding 85 mm orifice plate driver condition features a pure-argon driver). Unfortunately, this trial was inconclusive as the piston inflected so far upstream that it did not impact the welding rods at all. While the exact location of the piston's inflection point remained ambiguous, the fact that it did not reach the welding rods implies that driver gas composition has a significant impact on the free-piston mechanics at low compression ratios.

This phenomenon is predicted by basic kinetic gas theory. At the same temperature, argon atoms move slower, on average, than helium atoms because argon atoms are heavier and $\text{KE} = \frac{1}{2}mv^2$. As a result, during driver operation, argon atoms collide less frequently with the walls of the compression tube, leading to a lower conductive heat

loss. Consequently, for a given piston displacement, an argon driver will exhibit a greater driver pressure than a helium driver. Therefore, the driver pressure begins to dominate the force balance across the piston sooner, which results in an inflection point further upstream. Ultimately, it was not foreseen that this discrepancy in gas properties would compound with the high heat loss observed in the low-enthalpy trials to yield such pronounced driver composition sensitivity.

Furthermore, Gildfind [3] suggests that because L1D4 uses a pipe flow model for calculating heat loss, it struggles to accurately predict heat loss for compression processes in which the gas is hot, but is moving with a relatively low velocity (as is particularly true for the slow moving piston featured in the low-enthalpy condition). This limitation, paired with the teachings of the failed pure-argon trial, highlighted the need for a secondary round of blanked-off experiments where the specific composition of each driver condition could be tested. As such, the following L1D4 calibration was deemed ‘coarse’.

5.3 Coarse L1D4 Calibration

The next step was the ‘coarse’ calibration of the L1D4 launcher loss-factors for the new driver conditions. This was an iterative process in which the magnitude of each loss-factor was tuned until the simulated maximum driver pressure reflected the experimental maximum. Note that the effects of secondary wave processes on the maximum pressure readings were not considered for this initial calibration pass. The results of the calibration are presented in Table 5.3 and Figures 5.2 & 5.3.

Table 5.3: L1D4 calibrated blanked-off results.

Condition	Loss Factor	$p_{4,max}$ (MPa)	x_{max} (m)
Experimental Avg. CR 15 (Low-Enthalpy)	1.8	5.546	4.285
L1D4 CR 15 (Low-Enthalpy)		5.571	4.267
Experimental Avg. CR 15 (High-Enthalpy)	2.2	12.099	4.262
L1D4 CR 15 (High-Enthalpy)		12.107	4.256

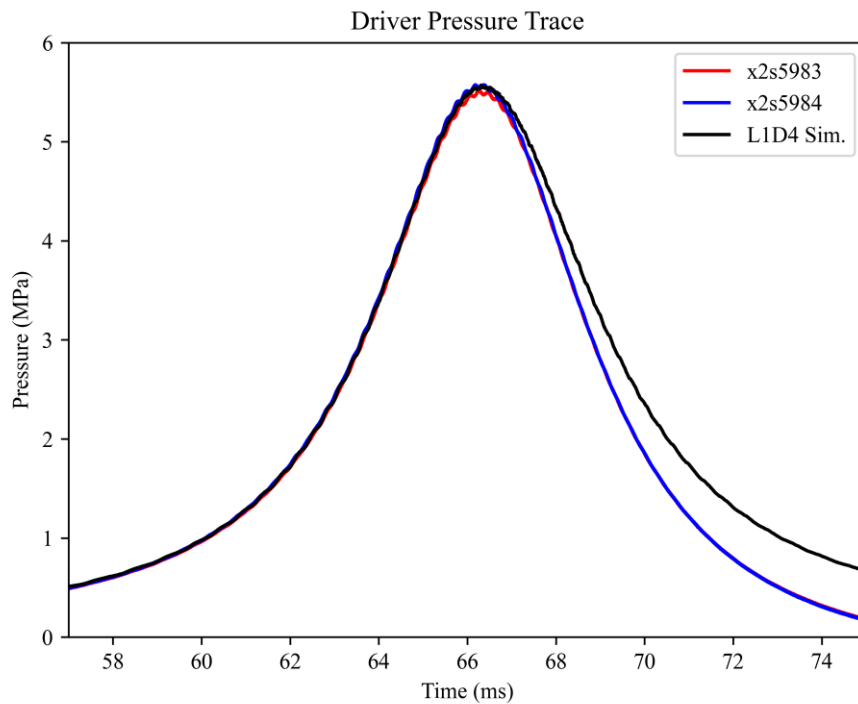


Figure 5.2: Time-aligned blanked-off pressure traces comparing shots x2s5983 & x2s5984, and the calibrated 'low-enthalpy' L1D4 model.

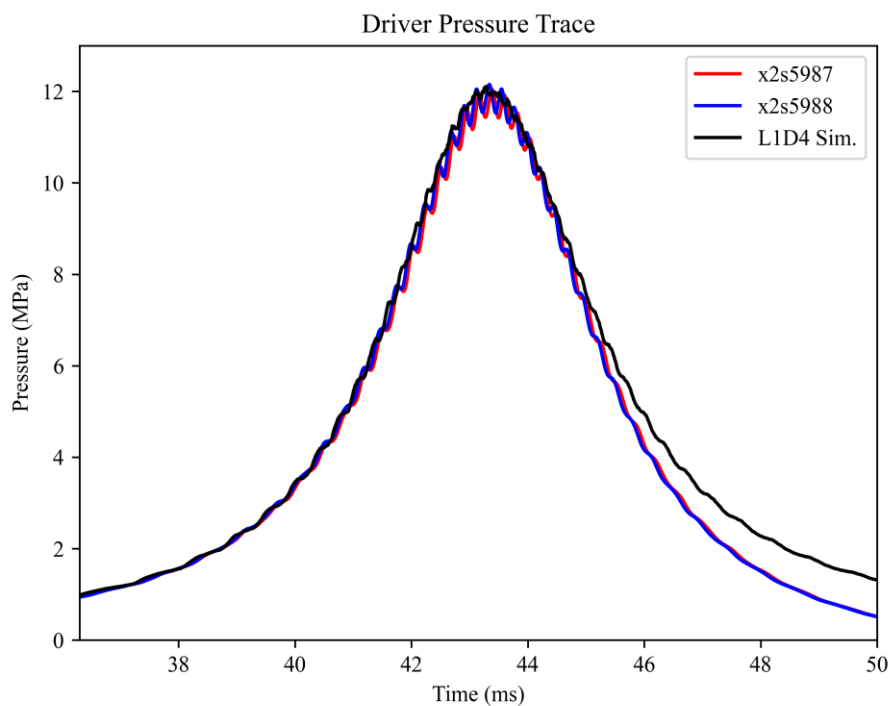


Figure 5.3: Time-aligned blanked-off pressure traces comparing shots x2s5987 & x2s5988, and the calibrated 'high-enthalpy' L1D4 model.

After calibration, the average experimental and L1D4 peak pressures exhibit a strong alignment; errors of just 0.45 % and 0.07 % were observed for the low and high-enthalpy conditions respectively. Likewise, the maximum piston displacements show good correlation, particularly for the high-enthalpy trial which had a deviation of 0.14 %. The displacement error was slightly larger for the low-enthalpy trial, at 0.42 %, which follows from L1D4's underprediction of this driver condition's heat loss as explained previously. The required launcher loss-factors are significantly lower than those featured in Gildfind's L1D4 simulations, which were in the range of 3.1-3.5 [3]; this is to be expected as very low reservoir pressures are utilised for the low compression ratio driver conditions.

5.4 Summary of Coarsely-Refined Driver Conditions

Finally, with the L1D4 model calibrated, work could begin towards a semi-refined set of driver conditions in preparation for the secondary round of detailed blanked-off tests. All eight, updated low compression ratio driver conditions are presented in Table 5.4 along with a variety of performance metrics. Furthermore, several associated L1D4 plots, including driver pressure traces and piston velocity profiles, can be found in Appendix F. Please note that each condition has been assigned an ID in accordance to its compression ratio, its rupture pressure and the internal diameter of the orifice plate used. This identification code is intended to be employed throughout the remainder of the driver condition design process before eventually transitioning to the established X2 driver condition naming convention once the new primary diaphragms have been developed.

Table 5.4: Coarsely-refined CR 15 driver conditions.

Condition ID	CR (-)	m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{di} (kPa)
CR15-1.8MPa-85mm	15	10.5	85	0.45	19.71
CR15-1.8MPa-77mm			77		
CR15-1.8MPa-72.1mm			72.1		
CR15-1.8MPa-65mm			65		
CR15-5.5MPa-85mm	15	10.5	85	1.6	54.76
CR15-5.5MPa-77mm			77		
CR15-5.5MPa-72.1mm			72.1		
CR15-5.5MPa-65mm			65		

Table 5.4: Continued.

He / Ar Mole Frac. (%)	L_{buffer} (mm)	v_{imp} (m/s)	p_{rup} (MPa)	p_{over} (%)	t_{hold} (ms)
0 / 100	100	9.057	1.8	9.191	3.687
20 / 80		8.497		8.769	3.750
44 / 56		9.455		9.552	3.675
65 / 35		9.195		9.247	3.947
70 / 30	100	15.022	5.0	11.342	2.515
78 / 22		14.626		10.236	2.404
87 / 13		16.141		11.031	2.513
95 / 5		17.641		9.363	2.461

Table 5.4: Concluded.

U_e (m/s)	M_s (-)	p_s (Pa)	ρ_s (kg/m ³)
3090	12.4	110	0.0026
3200	12.5	118	0.0026
3420	12.7	127	0.0025
3630	12.8	130	0.0023
4820	12.9	250	0.0026
5020	12.8	270	0.0025
5530	12.8	330	0.0025
5800	12.6	350	0.0024

As a direct result of the loss-factor calibration, some features of the preliminary driver conditions had to be altered in order to ensure an appropriate U_e range was maintained. The most notable change was the reduction of the two primary diaphragm rupture pressures from 3 & 7 MPa to 1.8 & 5 MPa. As a convenient consequence of this change, in combination with some small model tweaks, the buffer length required for soft piston-landing was reduced from 130 mm to 100 mm, which is equal to X2's existing 'long' buffers. Therefore, the proposed driver conditions will only require the development of two new primary diaphragms to be fully implemented. The coarsely-refined driver conditions can generate test flows with flight equivalent velocities in the range of 3.1-5.8 km/s. However, this new envelope comes with a caveat in the form of a performance

gap which spans 3.7-4.7 km/s. Bridging this gap should be as simple as designing four additional ‘mid-enthalpy’ CR 15 driver conditions with a U_e target of approximately 3.8-4.6 km/s. Unfortunately, this performance gap was discovered too late in the placement for such conditions to be included in this thesis. To touch further on the coarse nature of these driver conditions, future refinements should endeavour to achieve lower piston impact speeds and pressure overshoot values closer to 10 %.

Referring to the plots in Appendix F, a number of additional observations can be made. First, in the driver pressure traces, secondary wave processes can be observed in the form of periodic perturbations about the main pressure trace curve. It is critical to note that these perturbations became more aggressive as the orifice plates were increased up to a diameter of 85 mm. Meanwhile, for equivalent orifice plates, the relative magnitudes of the waves were approximately equal, irrespective of the driver composition. This is interesting as a prevailing X2 sentiment suggests that high argon content in the driver gas is the cause of these perturbations; however, this is believed to be an oversimplification.

The driver pressure perturbations are the combined result of two longitudinal wave processes, the compression waves generated by the free-piston as it accelerates the driver gas in front of it, and the expansion waves from the rupture of the primary diaphragm. The severity of the compression waves contribution to the net perturbations is primarily dictated by the velocity of the piston relative to the sound speed of the driver gas – a higher relative piston velocity produces stronger compression waves. Therefore, because the piston’s velocity profile is approximately constant between different orifice plates, when the size of the orifice is increased, and in turn more argon is added to the driver gas to maintain the mass flow rate into the shock tube, the sound speed of the driver gas decreases, thereby increasing the strength of the compression waves produced by the piston. In this sense, increased argon content is more of a correlation than a cause. This accounts for how the perturbations observed for the ‘CR15-1.8MPa-85mm’ and ‘CR15-5MPa-85mm’ conditions can be of a similar relative magnitude despite having vastly different argon contents – 100 % vs. 30 % respectively.

Generally, it is advised to limit these driver pressure perturbations when possible, as they increase the volatility of a driver condition because the sudden spikes in pressure

can lead to premature rupturing of the primary diaphragm. In a bid to remedy this disadvantage, the definition of constant-pressure driver operation was relaxed slightly to allow +10 % overshoot of the primary pressure trace curve, ignoring all secondary wave processes; this was achieved by median filtering the driver pressure trace before taking the maximum value for the overshoot calculation. This decision allowed the piston to be overdriven further than before, which contributed to increased driver hold time and reduced buffer length. While a highly overdriven piston could be considered dangerous for driver conditions with high rupture pressures, where maximum driver pressures can approach the compression tube limit, the relatively low rupture pressures featured in the low compression ratio driver conditions ensure that a high degree of overdrive can be implemented safely.

Further observations can be drawn from the test flow plots, where predictions of both test time and test flow steadiness are illustrated. Overall, the new driver conditions are shown produce very stable test times in the order of 80-200 microseconds (test time increases as U_e decreases). Some conditions, particularly ‘CR15-5MPa-85mm’ and ‘CR15-5MPa-77mm’, did exhibit signs of test flow disturbances, however detailed optimisation of the test flow steadiness was not within the scope of this project.

Finally, the tail-end of many of the test flow plots feature a series of step-like numerical artefacts which correspond to the arrival of cell boundaries at the test-flow history point location in L1D4. These are simply a result of the low grid resolution that was used in order to complete the full set of driver conditions in time for the conclusion of the ENGG7291 placement. It is strongly recommended that further development of the driver conditions should use a higher grid resolution, preferably per the insights of a grid sensitivity analysis.

5.5 Proposal for Future Experimental Campaigns

At the conclusion of the author's involvement with the project, there remained three experimental campaigns required to complete the development of the low compression ratio driver conditions.

- **Detailed Blanked-Off Tests:** Based on the findings of Section 5.2, a series of driver-composition-specific blanked-off experiments should be conducted for each of the coarsely-refined driver conditions presented in Table 5.4. This will enable more confident refinement of the final driver conditions.
- **Static Rupture Tests:** Once the driver conditions are finalised in L1D4, new primary diaphragms need to be developed in order to conduct diaphragm-rupturing experiments in X2. The process of making new diaphragms includes static rupture testing using the X2 rupture rig (refer to Section 3.1 and Appendix H.4 for details).
- **Diaphragm-Rupturing Validation Tests:** Once the new primary diaphragms have been developed, the final step is to conduct a series of pitot rake experiments in X2 for each of the candidate driver conditions. These experiments should focus on determining key properties of the test flows generated by the CR 15 driver conditions; X2 officials will assess the efficacy of the new driver conditions based on the results of this experimental campaign.

6 Conclusions

The contents of this thesis have detailed the early development of eight low compression ratio driver conditions for the X2 expansion tube. The test flows generated by these conditions exhibited freestream flight equivalent velocities in the range of 3.1-5.8 km/s, thereby satisfying the goal of enabling standard X2 operation in the 3-6 km/s test flight regime. As a result of the novel, infrastructure-minimising philosophy applied throughout the design process, the new driver conditions require just two new diaphragms (with rupture pressures of 1.8 & 4.5 MPa) to be fully integrated into X2's existing 'long buffer' driver configuration. If the new driver conditions are as successful in practice as they are in simulation, the extension of X2's performance envelope will enable X2 researchers to collect low-enthalpy, low-density validation data for the Modified Marrone-Treanor thermochemical nonequilibrium air chemistry model.

Additionally, a series of X2 facility upgrades were successfully implemented, specifically: a more accurate pressure monitoring system for the reservoir, an improved wear pad material and rear O-ring gland for the piston, and recommissioning of the X2 static rupture rig. Appreciable progress was also made towards a new laser-centric diaphragm scoring process and driver-length instrumentation.

7 Recommendations for Future Work

The future work established throughout the project is extensive, however it is hoped that this document may provide some guidance for future contributors. With respect to the low compression ratio driver conditions, future work includes: design of additional ‘mid-enthalpy’ CR 15 driver conditions, detailed blanked-off testing, detailed L1D4 calibration, finalisation of the driver conditions, development of the required primary diaphragms, and finally diaphragm-rupturing validation tests.

Further work is also required to complete some of the facility upgrades initiated during the project. Future work towards the goal of laser-scored primary diaphragms should include: finalisation of the laser-scoring rig design, manufacturing of the laser rig, and iterative tuning of the laser-scoring process such that it can accurately capture the static rupture characteristics of X2’s 1.2 mm, 2.0 mm and 2.5 mm steel diaphragms. Throughout the future development of this new scoring process, consideration of the operator’s user-experience is imperative. Finally, there is a great deal of work required to complete the driver-length instrumentation, including: finalisation of the PCB housing design, FEA of the housing/tunnel wall assembly, development of a custom magnetic drill mount (with an associated drilling procedure), and final installation.

References

- [1] Doolan, C., Morgan, R., *Test-time Optimization of a Large Scale Hypervelocity Expansion Tunnel*. In Eighth National Space Engineering Symposium, Barton, ACT, 1993.
- [2] Gibbons, N. N., Damm, K. A., Jacobs, P. A., and Gollan, R. J., *Eilmer: an Open-Source Multi-Physics Hypersonic Flow Solver*. Computer Physics Communications. Volume 282, 108551, 2023.
- [3] Gildfind, D. E., *Development of High Total Pressure Scramjet Flow Conditions using the X2 Expansion Tube*, Ph.D., School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2012.
- [4] Hornung, H. G., *The Piston Motion in a Free-piston Driver for Shock Tubes and Tunnels*, Graduate Aeronautical Laboratories, California Institute of Technology, California, 1988
- [5] Itoh, K., Ueda, S., Komuro, T., Sato, K., Takahashi, M., Myajima, H., Tanno, H., Muramoto, H., *Improvement of a free piston driver for a high-enthalpy shock tunnel*. Shock Waves, 8(4):215–233, 1998.
- [6] Jacobs, P. A., *Shock tube modelling with L1d*. Report 13/98, School of Mechanical and Mining Engineering., University of Queensland, Queensland, 1998.
- [7] James, C. M., *A Yarn About Free-Piston Drivers*. CfH Talk, School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2024.
- [8] James, C. M., *An Introduction to PITOT3*. CfH Talk, School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2021.
- [9] James, C. M., Gildfind, D. E., Lewis, S. W. et al. *Implementation of a State-to-state Analytical Framework for the Calculation of Expansion Tube Flow Properties*. Shock Waves 28, 349–377, 2018.

- [10] James, C. M., van den Herik, T. J., Thompson, M., *Free-Piston Driver Performance at High Compression Ratios in the X2 Expansion Tube*. School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2022.
- [11] McGilvray, M., *The Use of Expansion Tube Facilities for Scramjet Testing*. Ph.D., School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2008.
- [12] Paull, A., Stalker, R. J., *Test flow disturbances in an expansion tube*. J. Fluid Mech. 245(1), 493–521 1992.
- [13] Scott, M. P., *Development and Modelling of Expansion Tubes*. Ph.D., School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2006.
- [14] Stennett, S. J., *Development of an Extended Test Time Operating Mode for a Large Reflected Shock Tunnel Facility*. Ph.D., School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2020.
- [15] Watson, A., *X2 Driver Pressure Plate Instrumentation*. Master's Thesis., School of Mechanical and Mining Engineering., University of Queensland, Queensland, 2024.
- [16] Whitside, R. W., Chan, W. Y. K., Gildfind, D. E., Smart, M. K. et al. *Nitrogen Driver for Low-Enthalpy Testing in Free-Piston-Driven Shock Tunnels*. Shock Waves 31, 541–550, 2021.
- [17] Chaudhry, R. S., Boyd, I. D., and Candler, G. V., “Vehicle-Scale Simulations of Hypersonic Flows using the MMT Chemical Kinetics Model,” *AIAA AVIATION 2020 FORUM*, Virtual Event, June 15-19, 2020. doi:10.2514/6.2020-3272.
- [18] Torres, E., Gross, T., and Schwartzentruber, T. E., “Implementation of new multi-temperature nonequilibrium air chemistry model for CFD based on first-principles calculations,” *AIAA AVIATION 2023 Forum*, San Diego, C.A., U.S.A. & Online June 12-16, 2023. doi:10.2514/6.2023-3489.

-
- [19] Finch, P. M., Girard, J. J., Schwartz, T., Strand, C. L., Hanson, R. K., Yu, W. M., Austin, J. M., and Hornung, H. G., “Measurements of T5 Shock Tunnel Freestream Temperature, Velocity, and Composition,” *AIAA Journal*, Vol. 61, No. 4, 2023, pp. 1555–1578. doi:10.2514/1.j062383.
- [20] Finch, P. M., Schwartz, T., Girard, J., Yu, W. M., Strand, C., Austin, J. M., Hornung, H. G., and Hanson, R. K., “Near-Body Measurements in Hypersonic Cylinder Flows in the T5 Reflected Shock Tunnel by Tunable Diode Laser Absorption Spectroscopy,” *AIAA SCITECH 2022 Forum*, San Diego, C.A., U.S.A. & Virtual, January 3-7, 2022. doi:10.2514/6.2022-1705.
- [21] Finch, P. M., Schwartz, T., Granowitz, Z. N., Yu, W. M., Gross, T., Strand, C. L., Schwartzentruber, T. E., Austin, J. M., Hornung, H. G., and Hanson, R. K., “Near-Body Measurements in Hypersonic Wedge Flows in the T5 Reflected Shock Tunnel by Tunable Diode Laser Absorption Spectroscopy,” *AIAA SCITECH 2023 Forum*, National Harbor, M.D., U.S.A. & Online January 23-27, 2023. doi:10.2514/6.2023-2518.

Appendices

A Documentation of Generative AI Usage in Research

AI tools were NOT utilised for the completion of any aspect of this project or any of the associated documentation.

B Project Gantt Chart



C Project Risk Assessment

At the beginning of this thesis placement, it was critical to consider the risks involved with the project and discuss the strategies by which these risks would be managed. Project risks were categorised into safety, scheduling and technical risks, each of which are presented below as they were presented in the project proposal. The severity of each risk was evaluated in accordance to the risk matrix defined in Table C.1. The project opportunities originally identified are also listed briefly at the end of this section.

Table C.1: Risk matrix.

		Consequences				
		Negligible	Minor	Moderate	Major	Catastrophic
Definitions	Health & Safety	Near miss	First aid	Medical treatment	Prolonged Hospitalisation	Permanent injury, Fatality
	Scheduling	Insignificant delays - no effect on project schedule	Minor delays - no long term consequences	Delays - project is recoverable with great effort	Severe delays - reduction of project scope	Project completion infeasible
	Technical	Insignificant effect on project deliverables	Minor effect on project deliverables	Moderate effect on project deliverables	Severe effect - reduction of project deliverables	Project completion infeasible
Likelihood	Almost Certain	Medium	Medium	High	Extreme	Extreme
	Likely	Low	Medium	High	High	Extreme
	Possible	Low	Low	Medium	High	Extreme
	Unlikely	Low	Low	Medium	Medium	High
	Very Unlikely	Low	Low	Low	Medium	High

C.1 Health & Safety Risks

Health and safety risks refer to hazards that may cause physical harm to an individual. This project combines office work with physical tasks such as manufacturing and X2 experimentation; associated health and safety risks are presented in Table C.2.

Table C.2: Health & safety risk management.

Risk	Consequence	Likelihood	Risk Level	Prevention	Mitigation
X2 lab incident	Moderate injuries such as eyesight/hearing damage, finger pinches, etc.	Possible	Medium	Complete X2 lab induction. Adhere to in-place X2 safety regulations	X2 has a comprehensive system of mitigation strategies in place - call 000 if necessary
UQ Innovate laser cutter incident	Minor injuries such as cuts from burrs, and finger pinches	Unlikely	Low	Complete laser cutter induction. Adhere to in-place Innovate safety regulations	UQ Innovate has a comprehensive system of mitigation strategies in place - call 000 if necessary
Office related fatigue	Minor, temporary strain injuries - eyes and back	Almost Certain	Medium	Take small, frequent breaks from office tasks	Consult a doctor if necessary

C.2 Scheduling Risks

Scheduling risks threaten to delay progress and derail the proposed project timeline. In extreme cases, such risks can result in reduction of project scope or even failure to meet entire deliverables. Key scheduling risks are presented briefly in Table C.3.

Table C.3: Scheduling risk management.

Risk	Consequence	Likelihood	Risk Level	Prevention	Mitigation
Supply chain delays for new pressure gauges	Could delay/limit high-fill-pressure driver condition experiments	Unlikely	Medium	Order the gauges early. Discuss lead times with supplier	Borrow similar equipment temporarily from within the Cfh faculty
Delay of X2 machinist manufacturing schedule – PCB housings, drill mount, new buffers, etc.	Might delay the experimental campaigns which rely on the new components	Possible	Low	Submit requests early. Check capabilities of manufacturer to reduce complications	Attempt to submit emergency jobs to the APW
Delay of diaphragm manufacturing due to student demand for laser cutter	Might delay final validation campaign	Possible	Low	Book timeslots in advance via Innovate website	Gain access to the machine outside of student hours
Delay of experimental campaigns due to X2 schedule changes/maintenance	Substantial delays late in project could mean no final validation of new driver conditions	Possible	Medium	Negotiate campaigns early in fortnightly XLabs meetings	Attempt to swap experimental blocks with a fellow researcher who may have looser time constraints
File loss	Progress setback while work is repeated	Very Unlikely	Low	Routinely backup work to a USB and cloud storage	Recover previous versions where possible and redo work

Supervisor's unavailable	Progress may halt briefly while waiting for responses	Very Unlikely	Low	Schedule critical meetings in advance	Supervisors are almost always readily available on Slack
Illness/unexpected life events	Progress delays due to inability to work	Unlikely	Low	Maintain healthy work-life balance	Commit extra hours upon return

C.3 Technical Risks

Technical risks are a broad category of risks which can directly impact the deliverables of the project. Such risks may stem from the likes of gaps in technical understanding, impractical project aims, or hardware failure. Key technical risks are listed in Table C.4.

Table C.4: Technical risk management.

Risk	Consequence	Likelihood	Risk Level	Prevention	Mitigation
Failure to accurately replicate driver dynamics in L1D4	Corrections during the experimental validation phase may become significant – major time sink	Unlikely	Low	Experience suggests calibration step should ensure that the L1D4 model is accurate enough. Grid independence studies will confirm numerical reliability	Consult experienced XLabs members for assistance with experimental tuning
X2 driver machining accidents	Highly damaging incidents might lead to an extended facility repair period which could severely limit project deliverables	Unlikely	Medium	Produce a comprehensive procedure for all tunnel machining; X2 officials should be included in this process	Initiate facility repairs as soon as feasible i.e. plugging failed holes etc.
Failure to calibrate the laser cutter for engraving	Major delays to project timeline as new diaphragm manufacturing method will be required	Unlikely	Low	Start task early. Consult experienced XLabs members for assistance. Assign ample time in project schedule	Pivot to CNC scoring of primary diaphragms
Failure to recommission static rupture rig	Unable to test/commission new primary diaphragms, thus infeasible to complete development of new driver conditions	Very Unlikely	Medium	Start task early. Consult experienced XLabs members for assistance. Assign ample time in project schedule	Attempt to borrow similar equipment from other UQ shock facilities. Commission new rig if time permits
Poor experimental data	Might require additional tests – minor progress delay	Possible	Low	Careful setup of the experiment – good design	Repeat poor trials if practical

C.4 Project Opportunities

Key project opportunities will arise as a result of the project's success, however other opportunities may arise throughout the course of the project. Major opportunities include:

- Potential for NASA & Defence contracts if X2 performance envelope is successfully expanded to include 3-6km/s 'low performance' test conditions.
- Potential for detailed validation of existing and future numerical X2 facility models if the driver instrumentation is successful implemented.
- Potential for shared success with other UQ hypersonic testing facilities – X3R has expressed shared interest in a new piston wear pad material, T4 has expressed a shared interest in recommissioning the static rupture rig for testing of their own primary diaphragms.
- Potential for the author to gain valuable industry experience and form relations with leading hypersonics researchers.

D SPARK Control Driver Conditions

Table D.1: CR 17.5 control driver conditions.

m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{4i} (kPa)	He / Ar Mole Frac. (%)
10.5	85	3.01	131.277	78.5 / 21.5
	77			89.2 / 10.8
	72.1			94.3 / 5.7
	65			100 / 0
20	85	2.56	131.277	49.6 / 50.4
	77			69.6 / 30.4
	72.1			79.2 / 20.8
	65			90.1 / 9.9
35	85	2.31	131.277	4 / 96
	77			39 / 61
	72.1			55.7 / 44.3
	65			74.5 / 25.5

Table D.1: Continued.

L_{buffer} (mm)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{sI} (m/s)
126	1.176	1.113	3071
	1.172	1.014	3371
	1.168	0.955	3578
	1.169	0.875	3905
114	1.683	1.375	2485
	1.685	1.246	2743
	1.685	1.170	2921
	1.678	1.064	3212
83	2.269	1.671	2046
	2.269	1.502	2276
	2.269	1.403	2436
	2.269	1.268	2695

Table D.2: CR 15 control driver conditions.

m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{4i} (kPa)	He / Ar Mole Frac. (%)
10.5	85	3.60	169.741	83.2 / 16.8
	77			92.3 / 7.7
	72.1			96.7 / 3.3
	65			-
20	85	3.02	169.741	59.6 / 40.4
	77			76.4 / 23.6
	72.1			84.5 / 15.5
	65			93.5 / 6.5
35	85	2.71	169.741	22.5 / 77.5
	77			51.5 / 48.5
	72.1			65.2 / 34.8
	65			80.8 / 19.2

Table D.2: Continued.

L_{buffer} (mm)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{sI} (m/s)
152	1.328	1.094	3124
	1.330	0.998	3426
	1.324	0.940	3635
	-	-	-
141	1.875	1.341	2549
	1.876	1.215	2813
	1.865	1.141	2996
	1.877	1.040	3286
125	2.509	1.619	2111
	2.502	1.457	2346
	2.517	1.363	2507
	2.518	1.233	2771

Table D.3: CR 12.5 control driver conditions.

m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{4i} (kPa)	He / Ar Mole Frac. (%)
10.5	85	4.41	230.029	87.6 / 12.4
	77			95.3 / 4.7
	72.1			99 / 1
	65			-
20	85	3.67	230.029	68.8 / 31.2
	77			82.6 / 17.4
	72.1			89.3 / 10.7
	65			96.7 / 3.3
35	85	3.26	230.029	39.5 / 60.5
	77			62.9 / 37.1
	72.1			74.1 / 25.9
	65			86.7 / 13.3

Table D.3: Continued.

L_{buffer} (mm)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{sI} (m/s)
190	1.531	1.077	3175
	1.526	0.982	3482
	1.513	0.925	3695
	-	-	-
178	2.137	1.307	2615
	2.144	1.186	2883
	2.124	1.113	3070
	2.126	1.015	3367
171	2.835	1.568	2180
	2.836	1.412	2420
	2.823	1.321	2587
	2.820	1.197	2856

Table D.4: CR 10 control driver conditions.

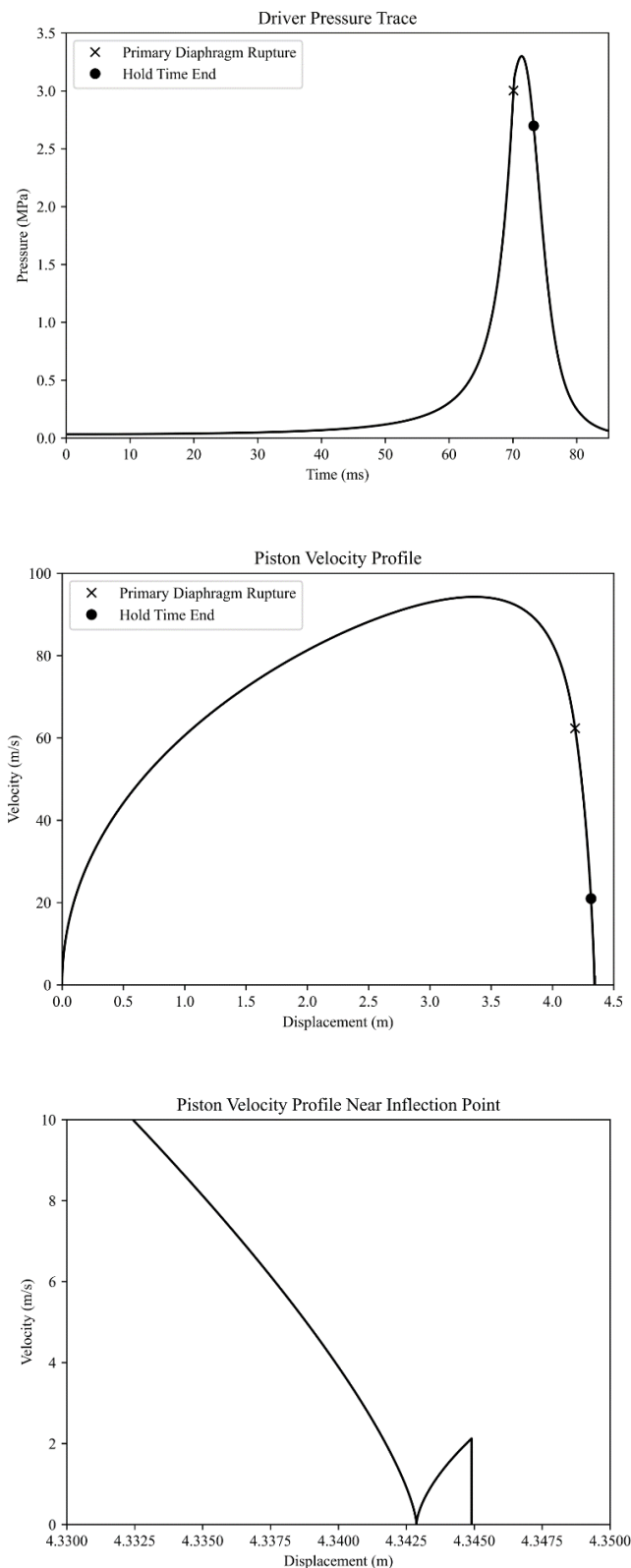
m_{piston} (kg)	$\phi_{orifice}$ (mm)	p_{res} (MPa)	p_{4i} (kPa)	He / Ar Mole Frac. (%)
10.5	85	5.61	333.681	91.4 / 8.6
	77			97.8 / 2.2
	72.1			-
	65			-
20	85	4.60	333.681	77.2 / 22.8
	77			88.3 / 11.7
	72.1			93.6 / 6.4
	65			99.5 / 0.5
35	85	4.06	333.681	54.8 / 45.2
	77			73.2 / 26.8
	72.1			82 / 18
	65			91.9 / 8.1

Table D.4: Continued.

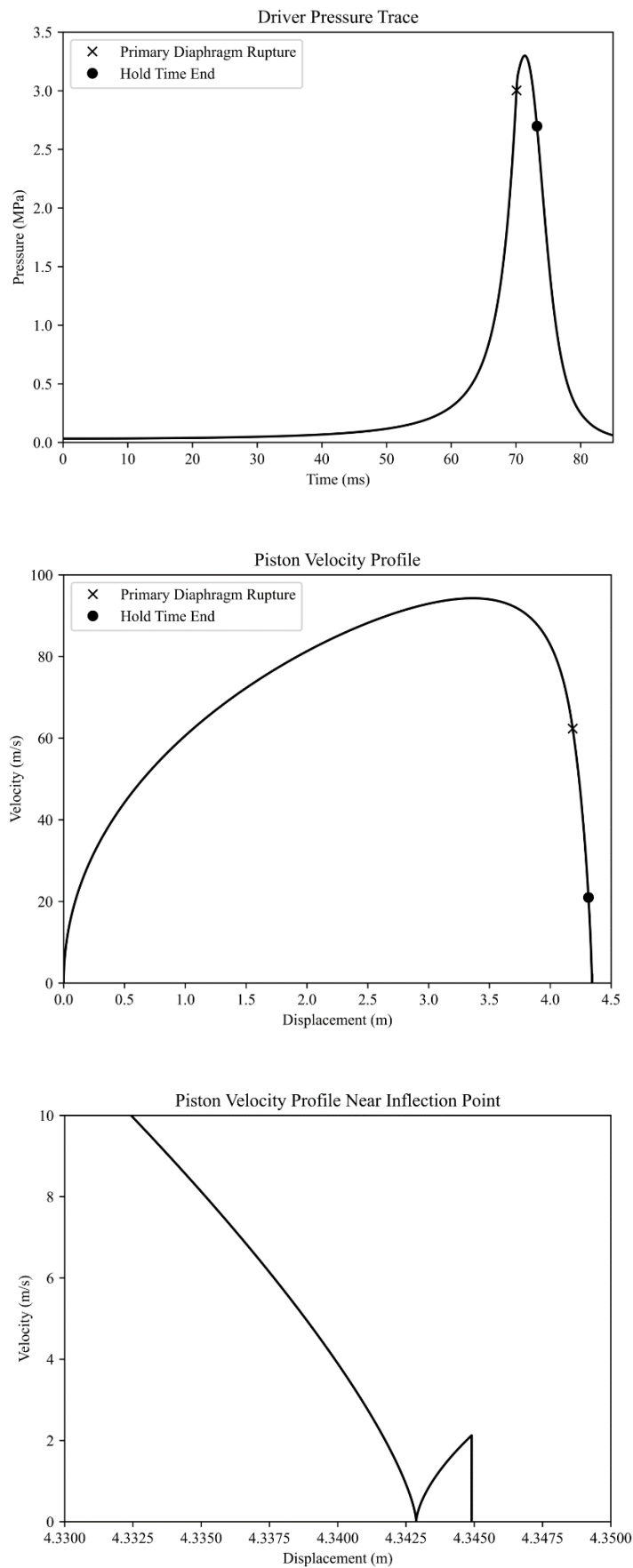
L_{buffer} (mm)	t_{hold} (ms)	$t_{req. hold}$ (ms)	v_{sI} (m/s)
251	1.845	1.067	3204
	1.852	0.974	3509
	-	-	-
	-	-	-
236	2.506	1.276	2679
	2.501	1.157	2953
	2.492	1.089	3140
	2.518	0.994	3438
227	3.302	1.518	2251
	3.303	1.369	2497
	3.298	1.283	2665
	3.291	1.163	2939

E SPARK Plots for Preliminary Driver Conditions

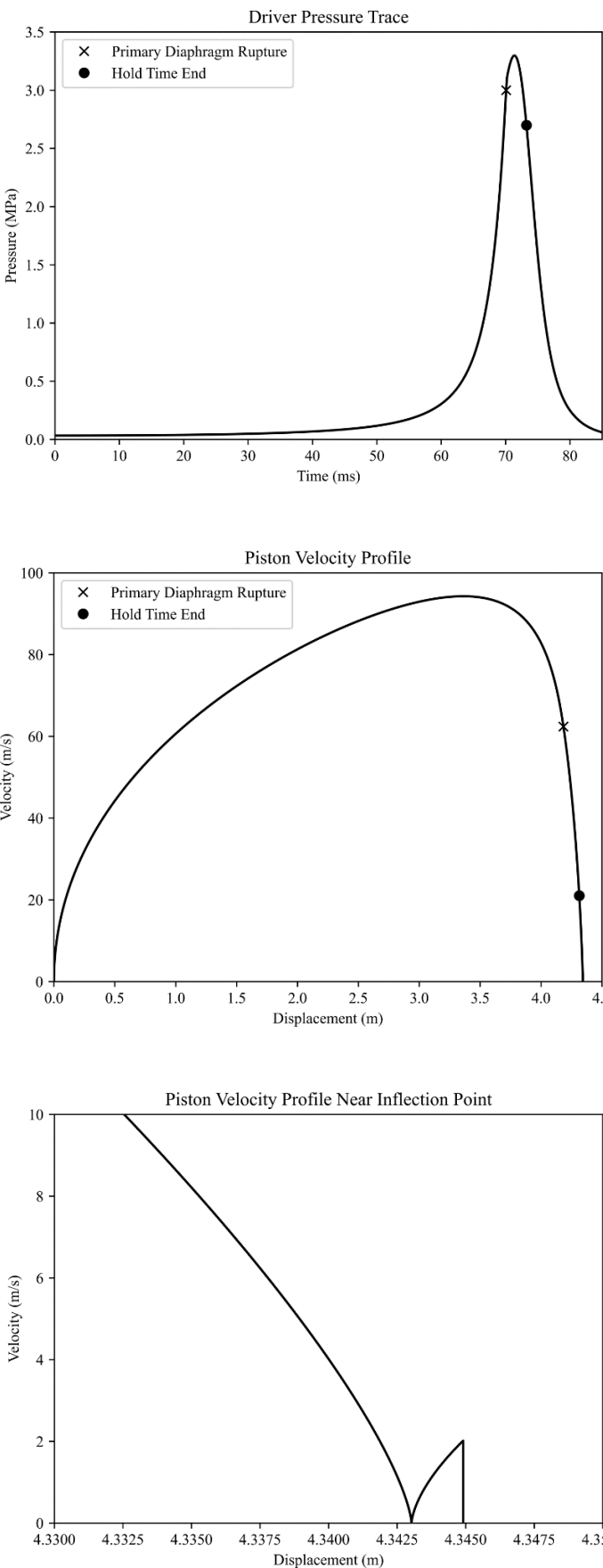
E.1 3 MPa Rupture - 85 mm Orifice Plate



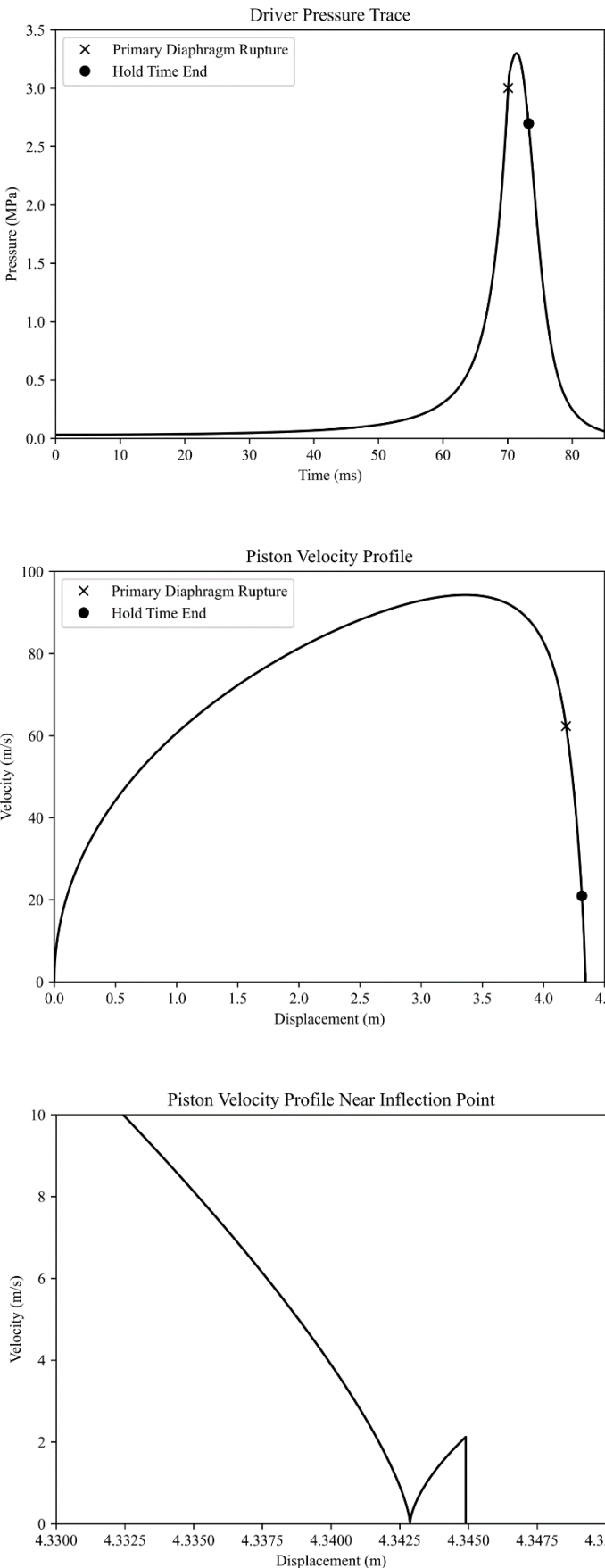
E.2 3 MPa Rupture - 77 mm Orifice Plate



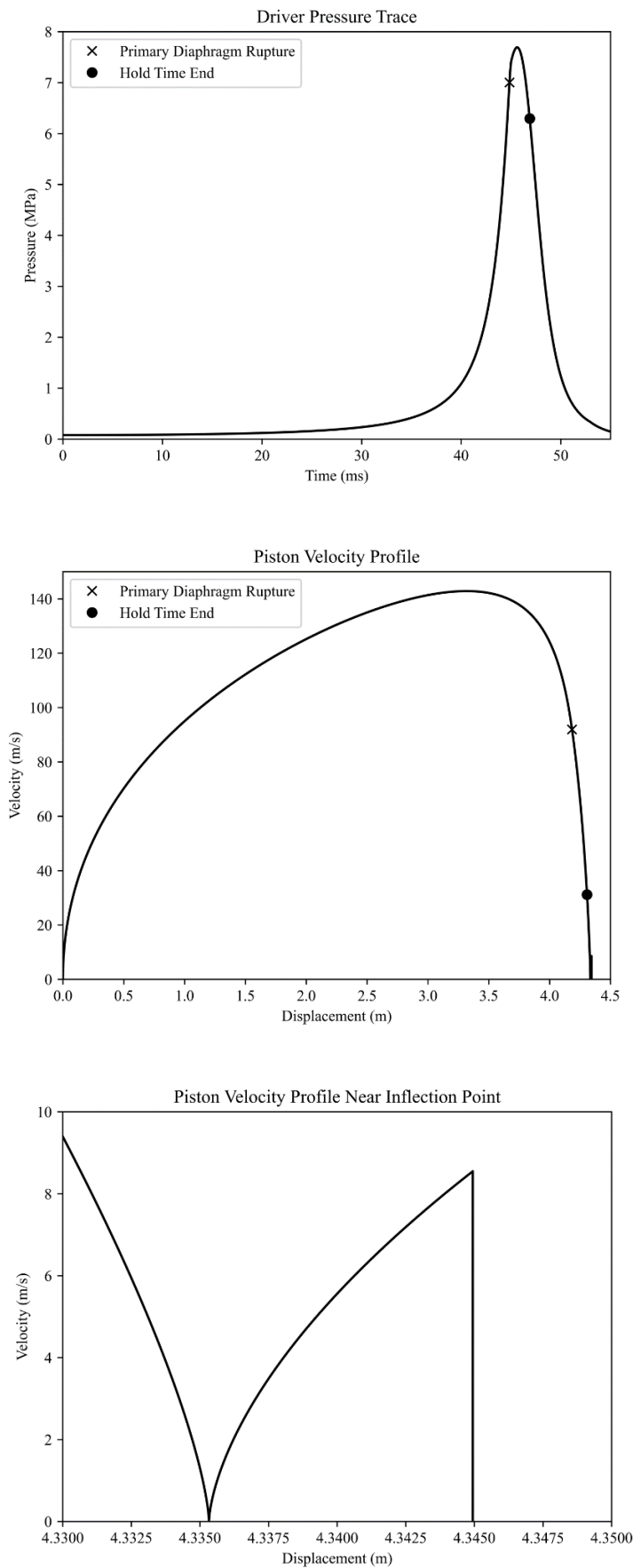
E.3 3 MPa Rupture - 72.1 mm Orifice Plate



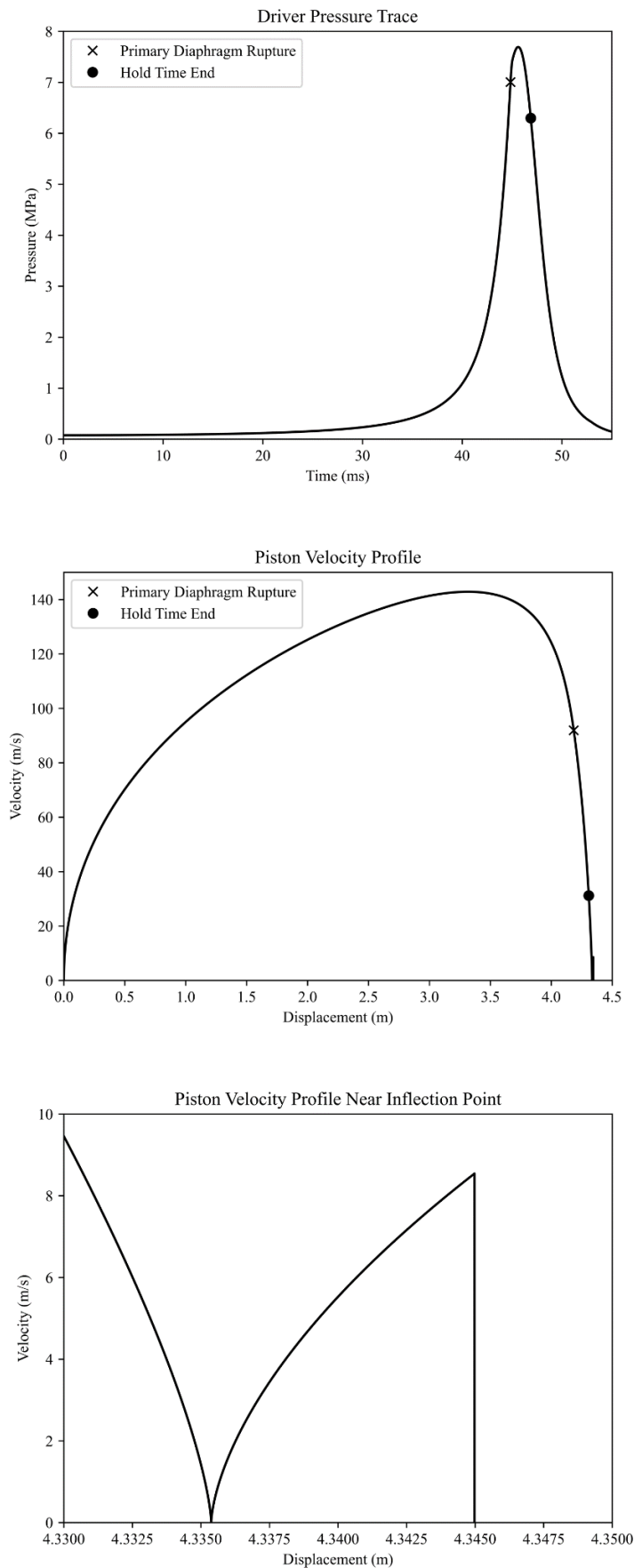
E.4 3 MPa Rupture - 65 mm Orifice Plate



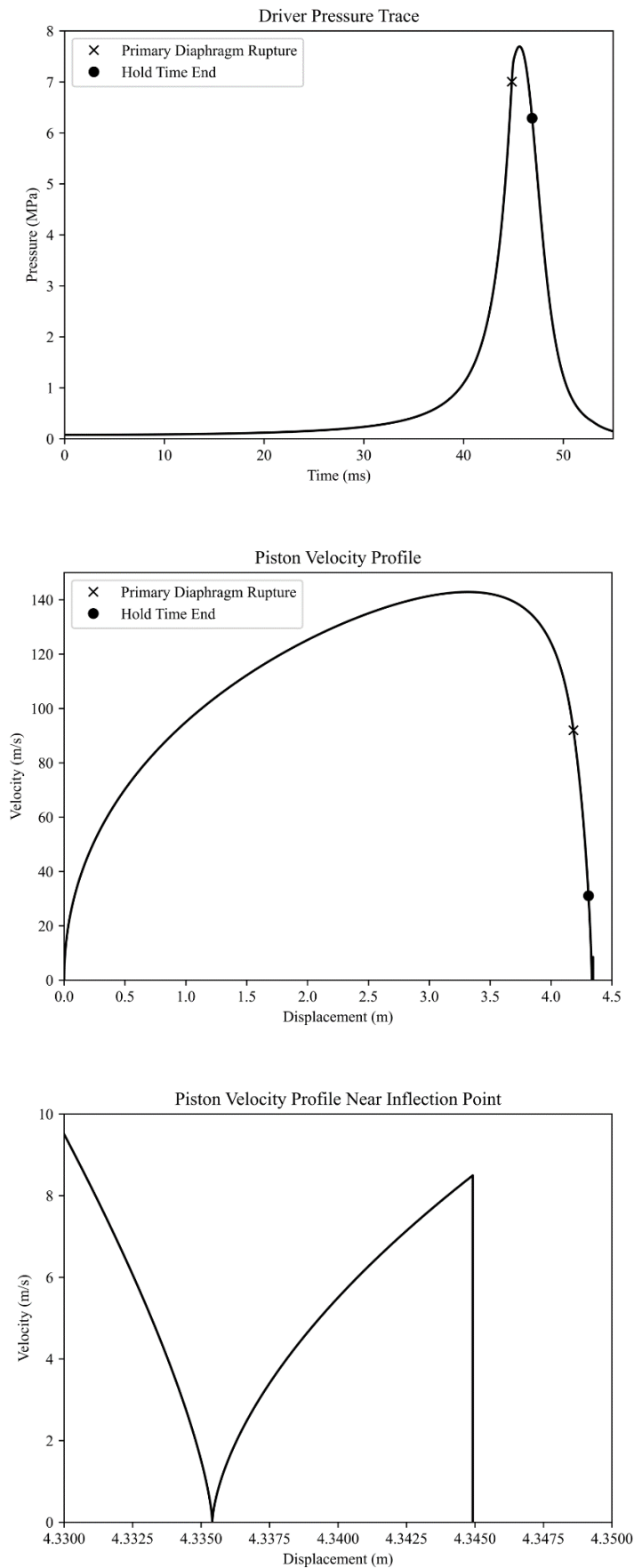
E.5 7 MPa Rupture - 85 mm Orifice Plate



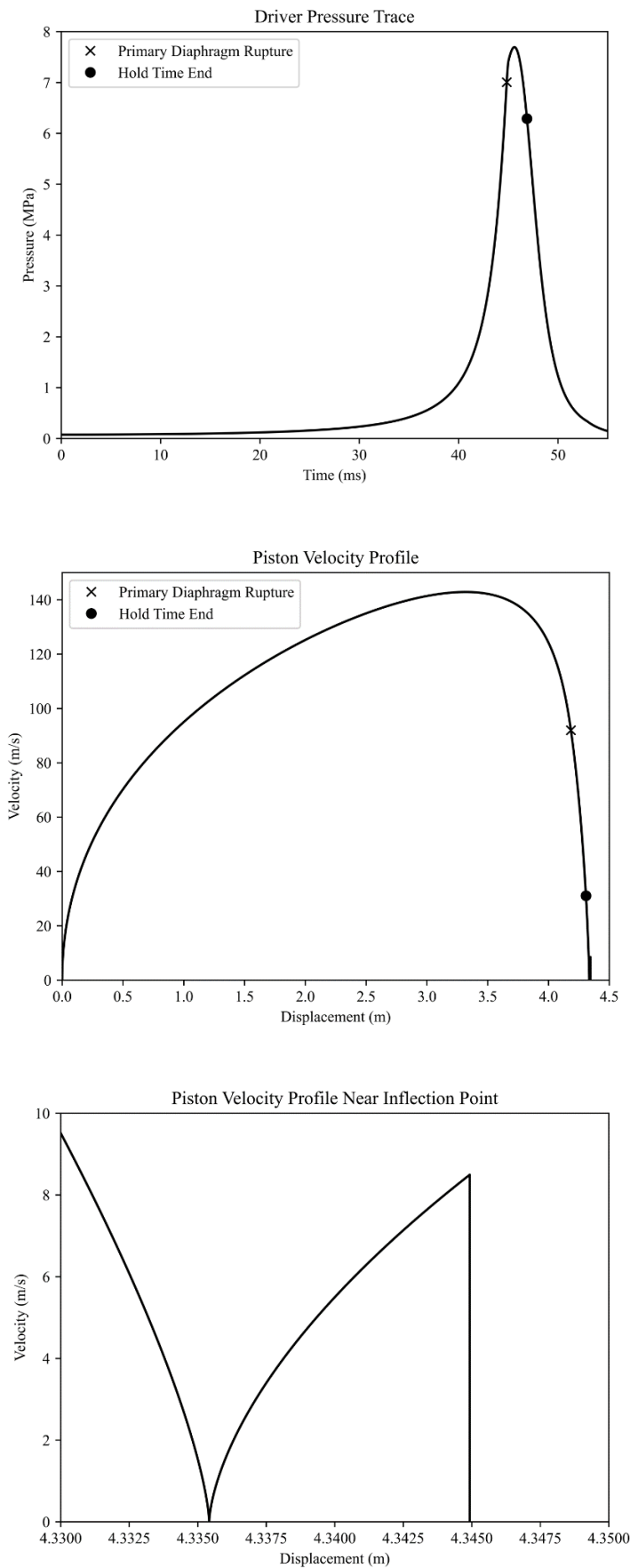
E.6 7 MPa Rupture - 77 mm Orifice Plate



E.7 7 MPa Rupture - 72.1 mm Orifice Plate

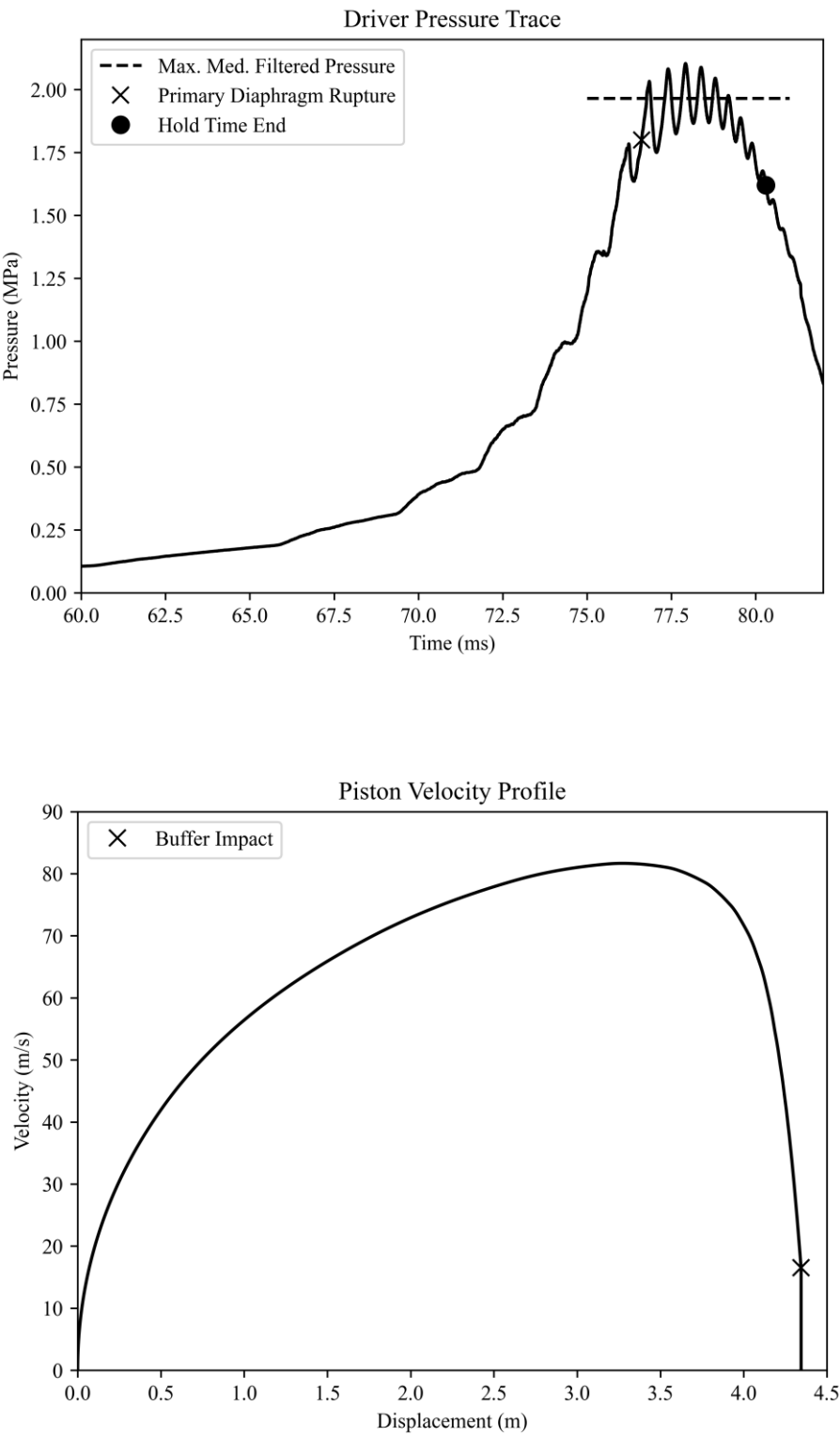


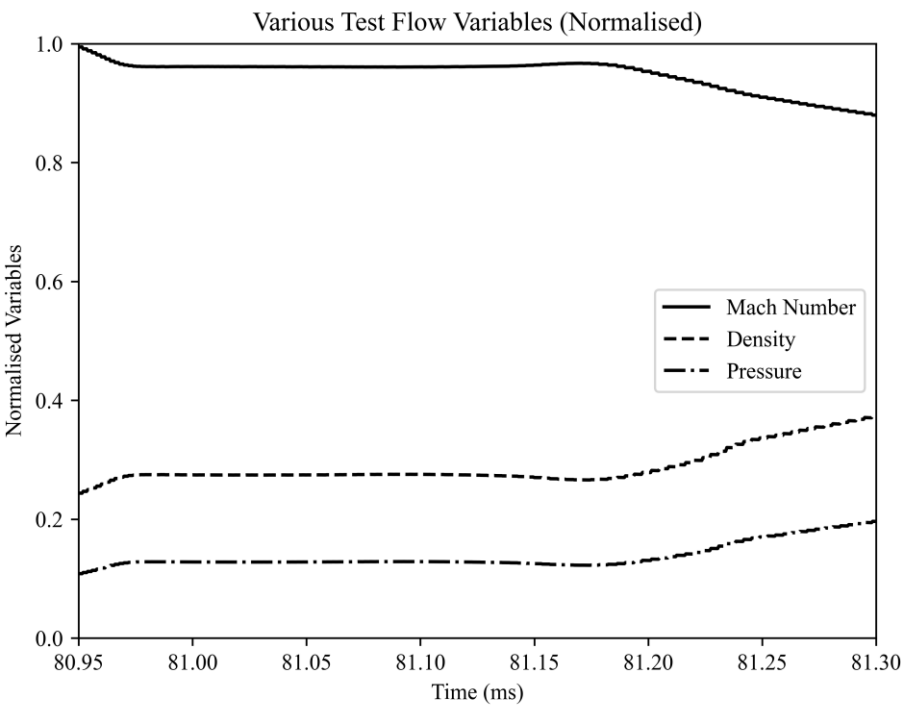
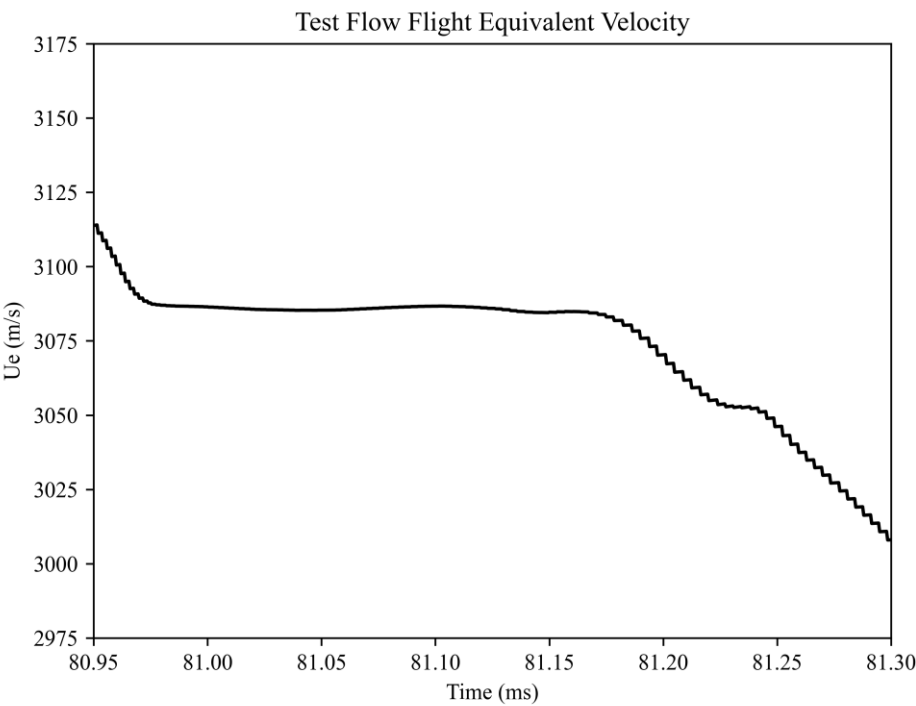
E.8 7 MPa Rupture - 65 mm Orifice Plate



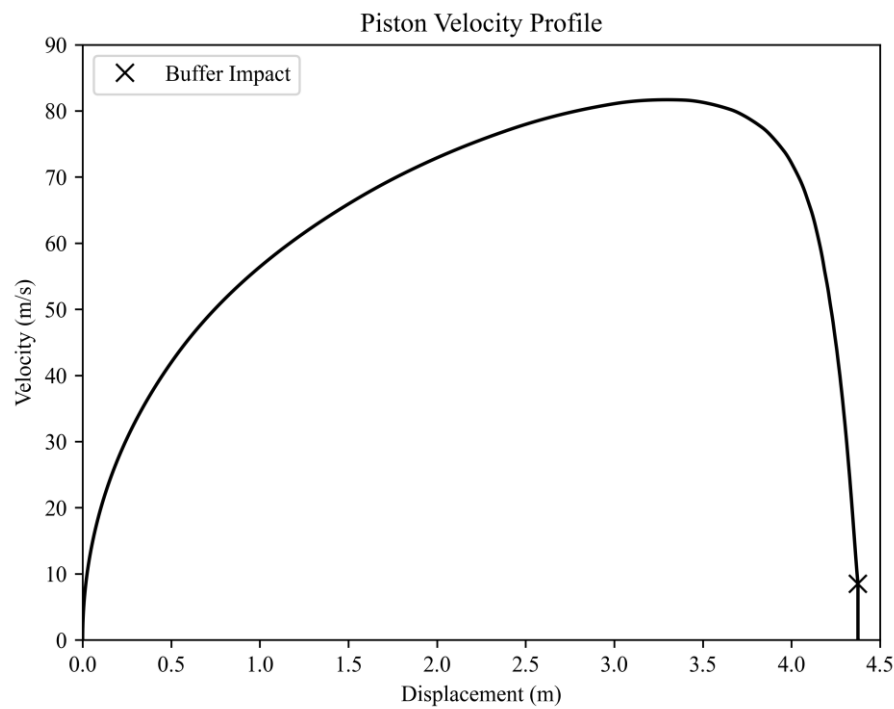
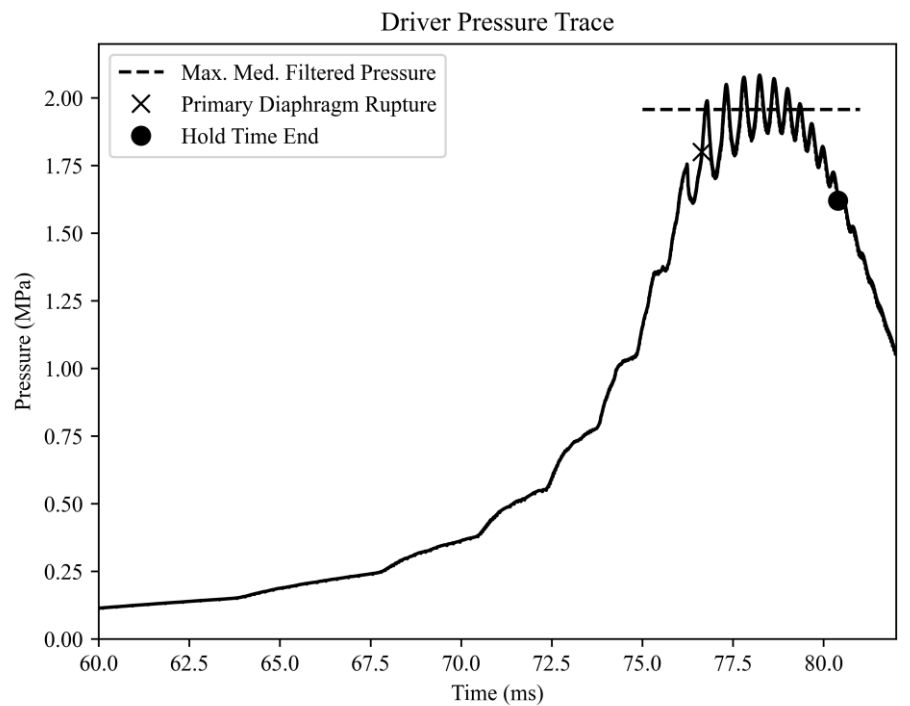
F L1D4 Plots for Coarsely-Refined Driver Conditions

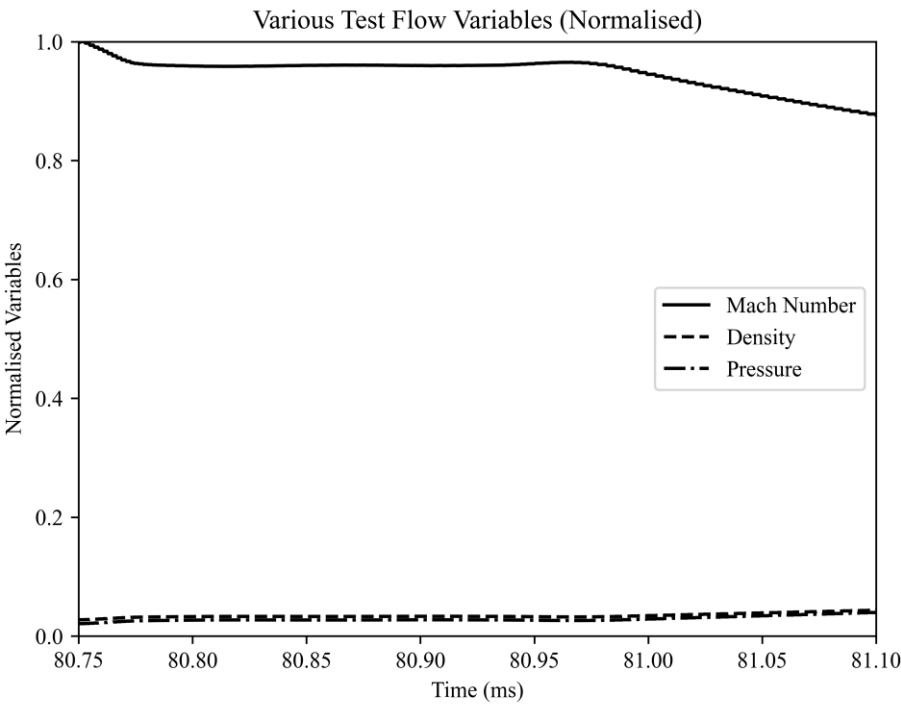
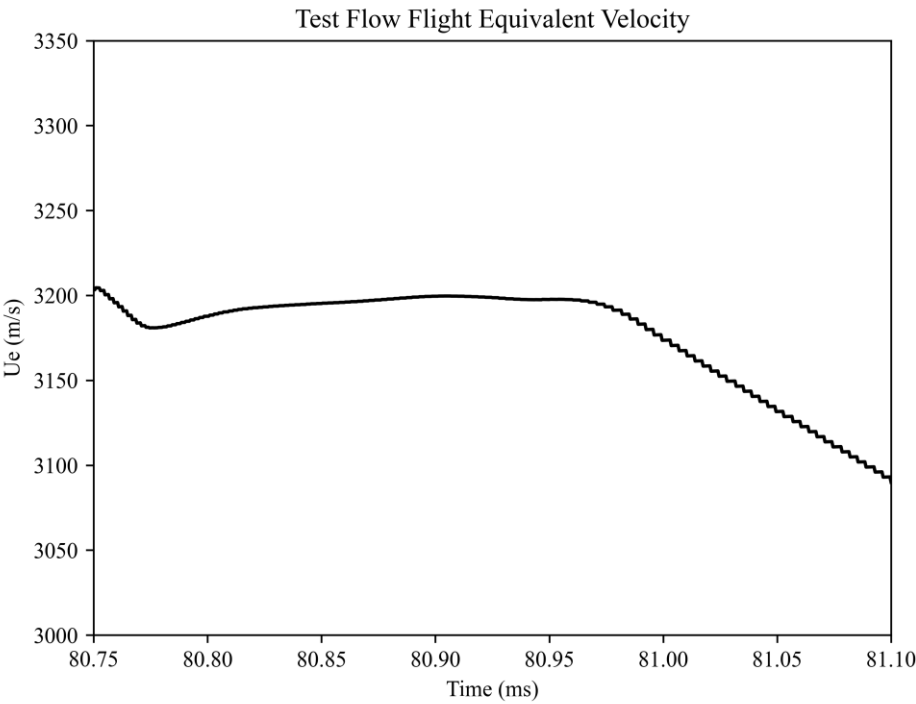
F.1 CR15-1.8MPa-85mm



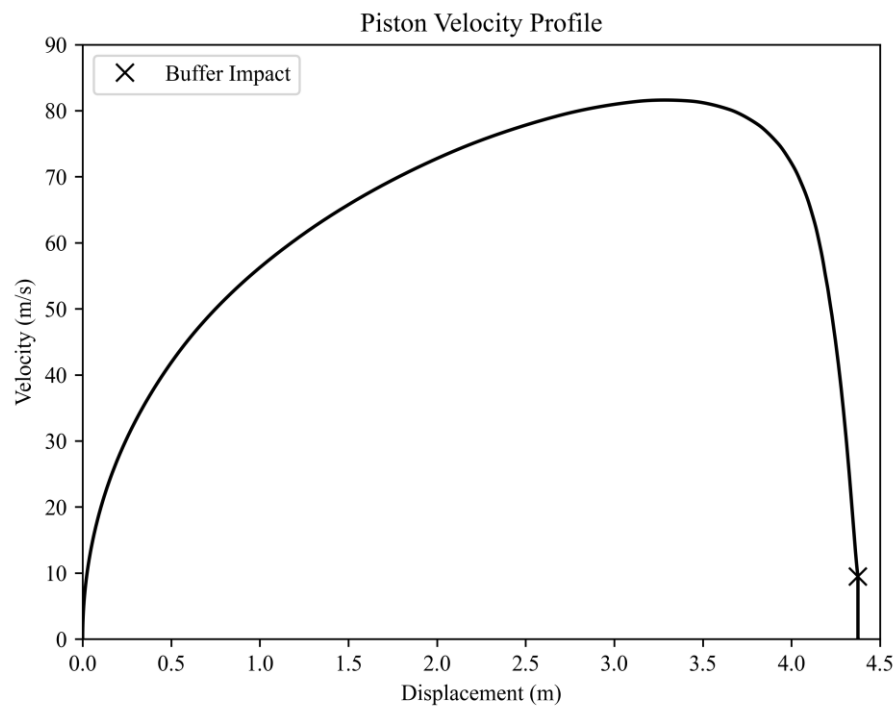
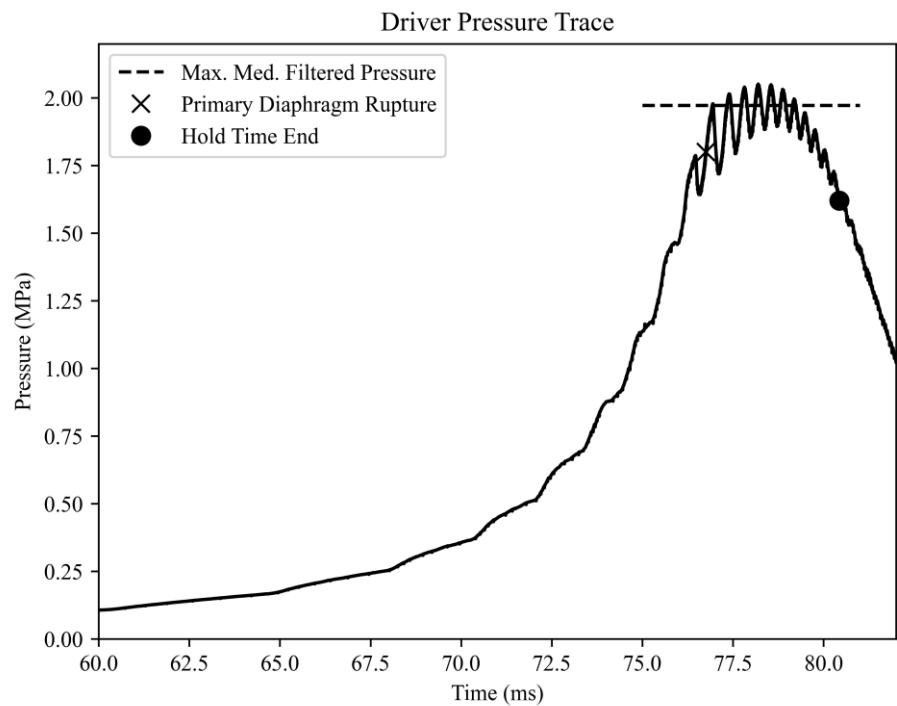


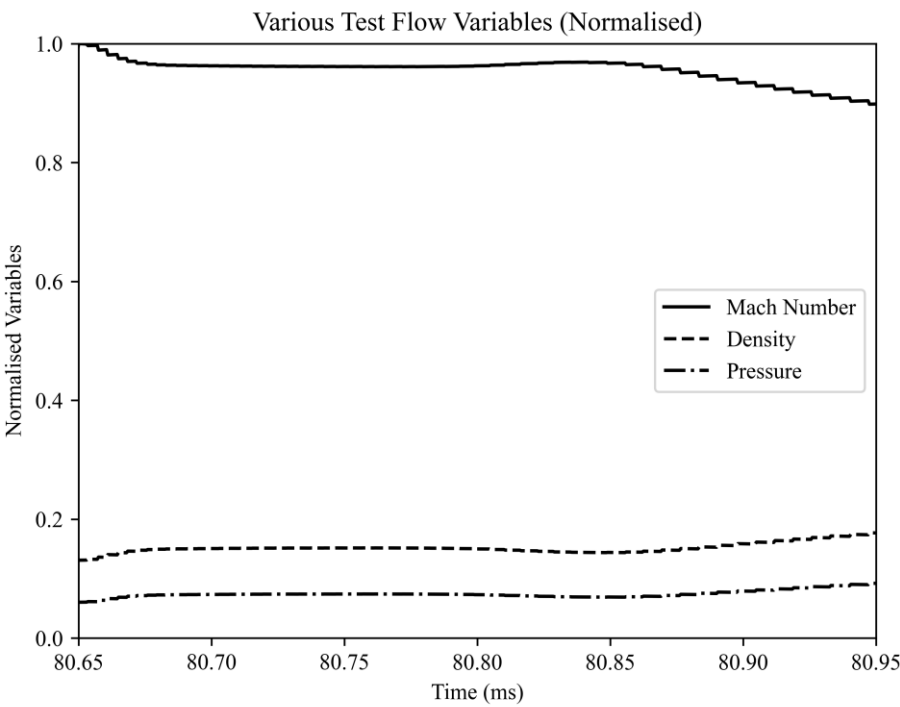
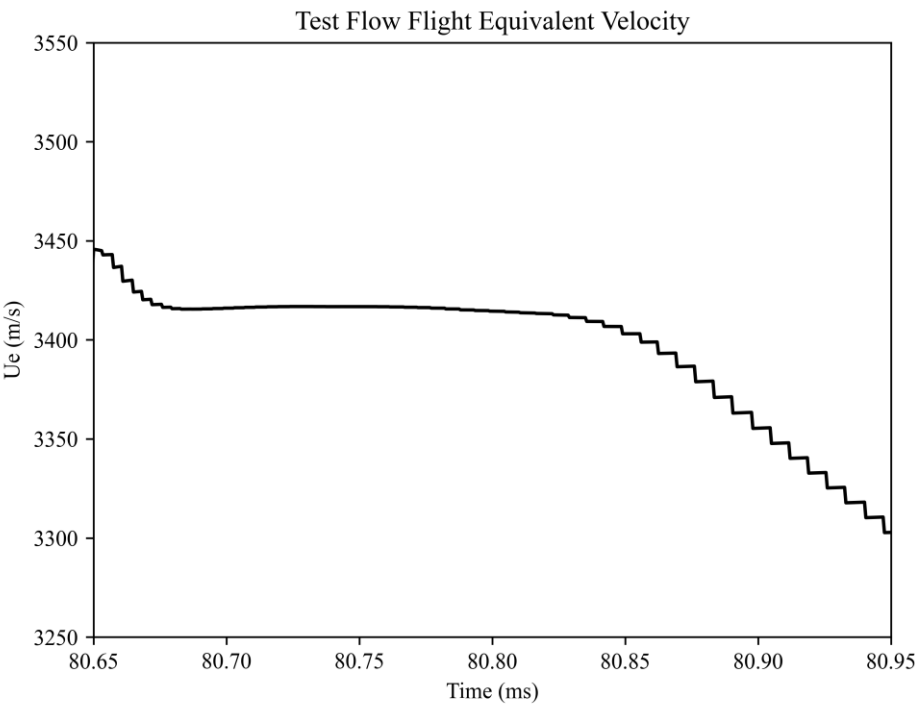
F.2 CR15-1.8MPa-77mm



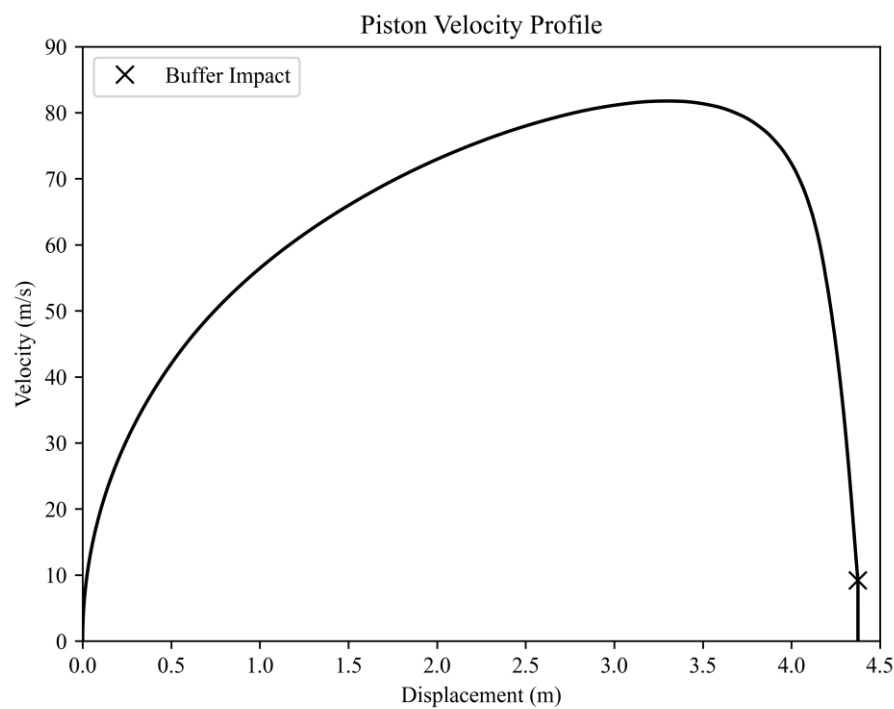
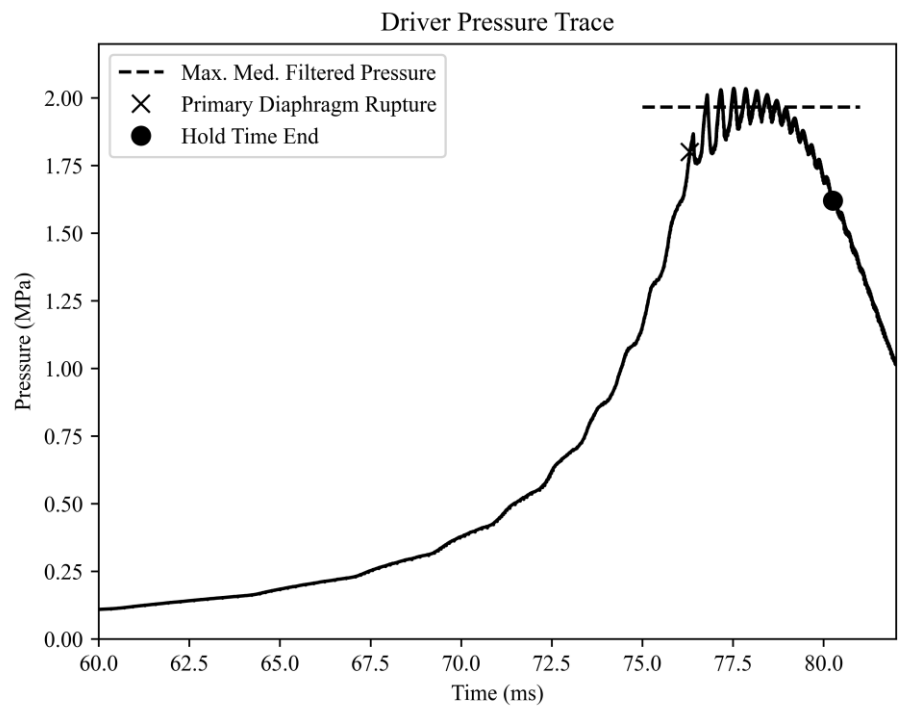


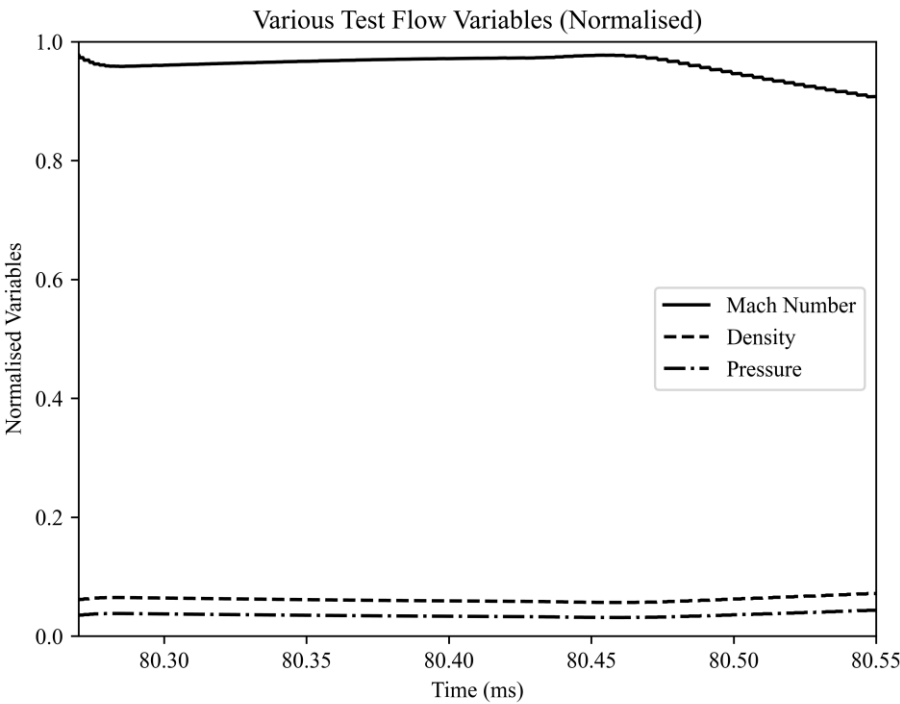
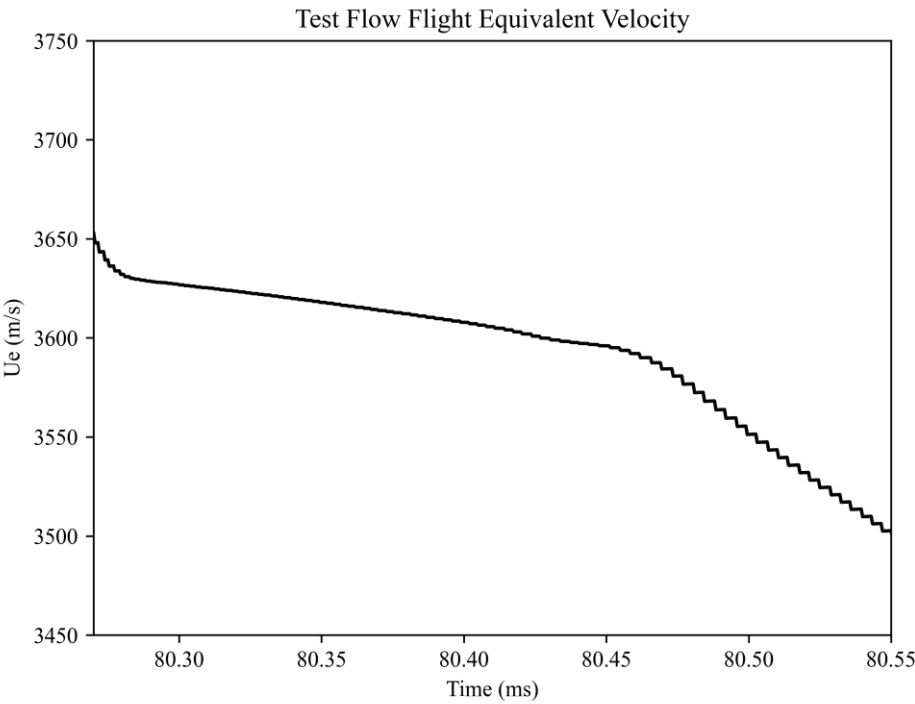
F.3 CR15-1.8MPa-72.1mm



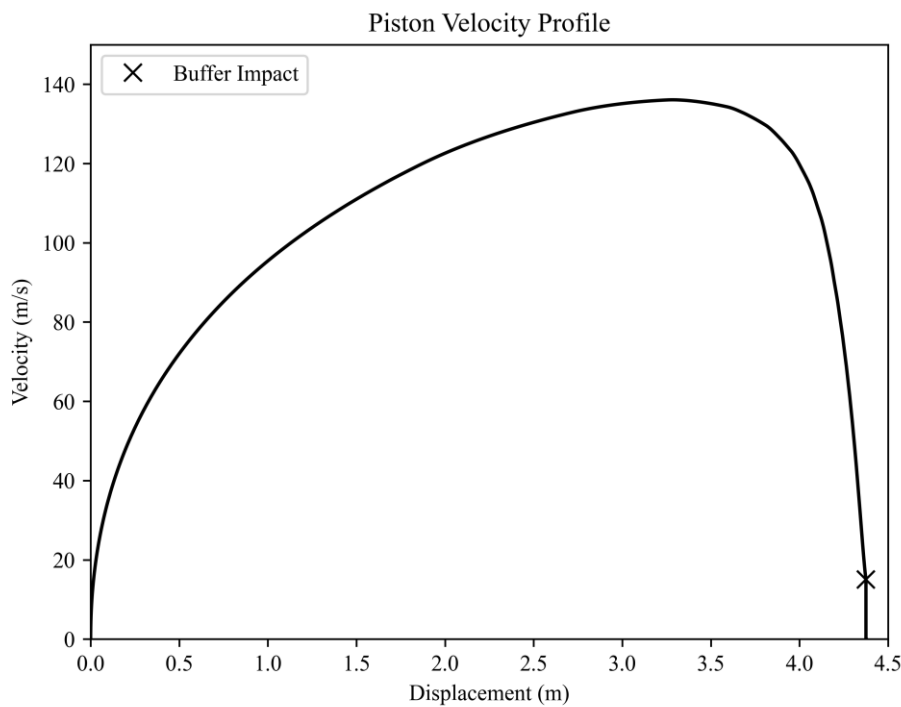
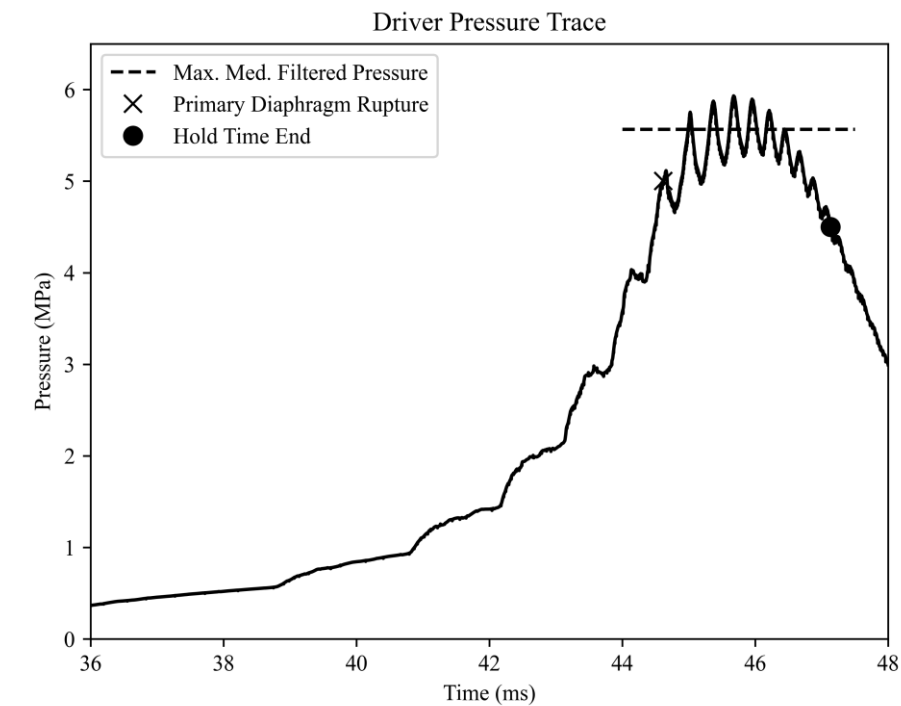


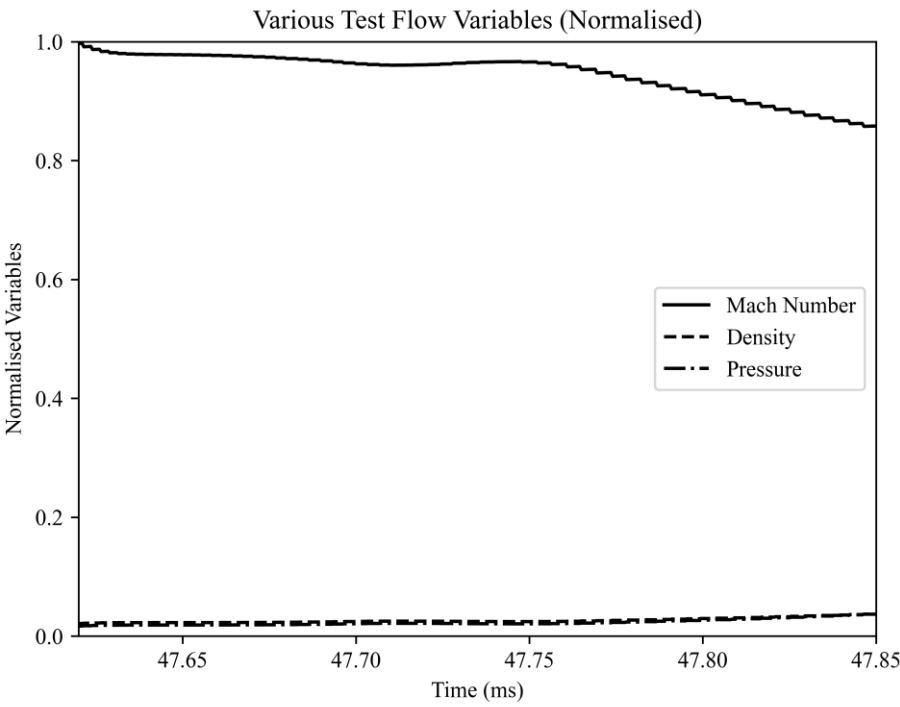
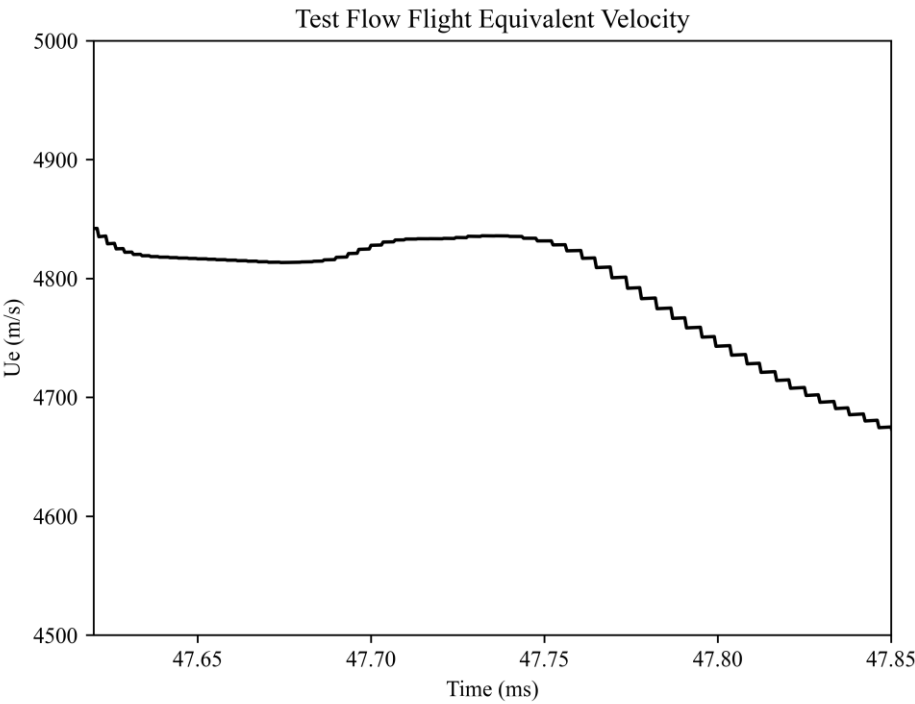
F.4 CR15-1.8MPa-65mm



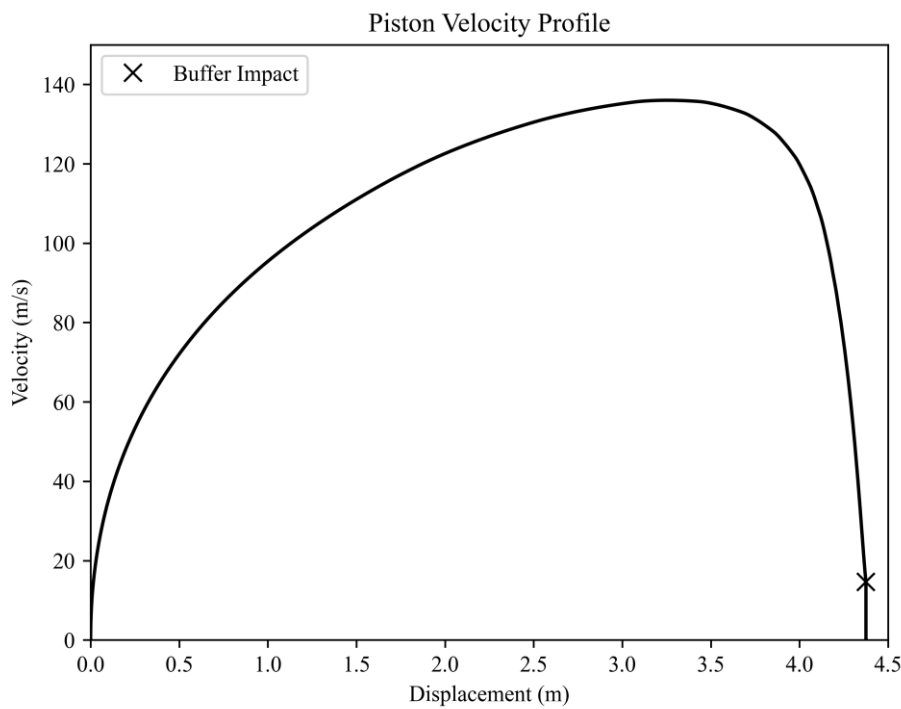
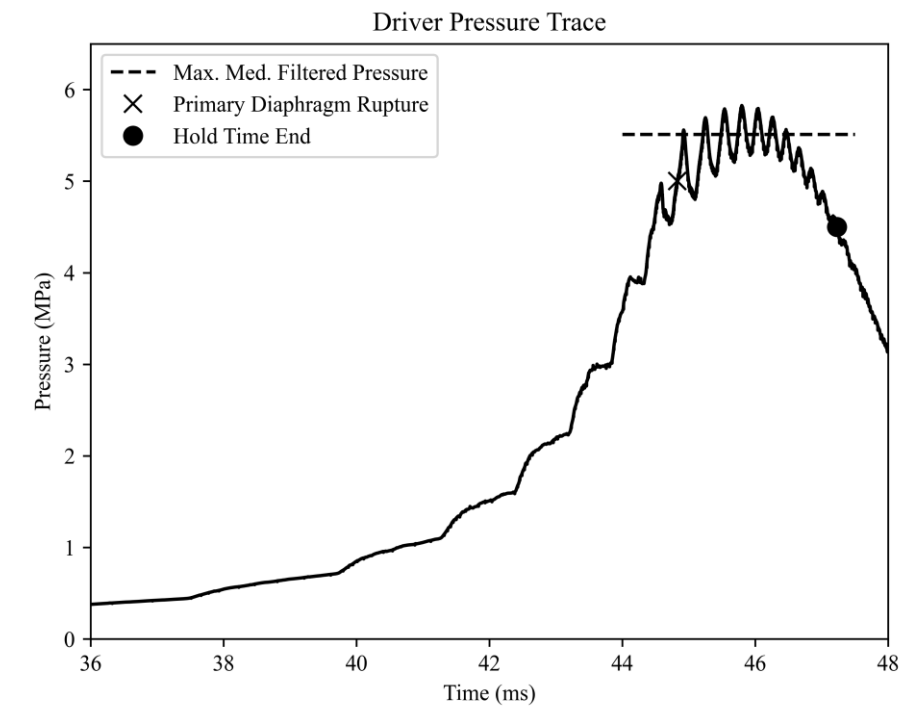


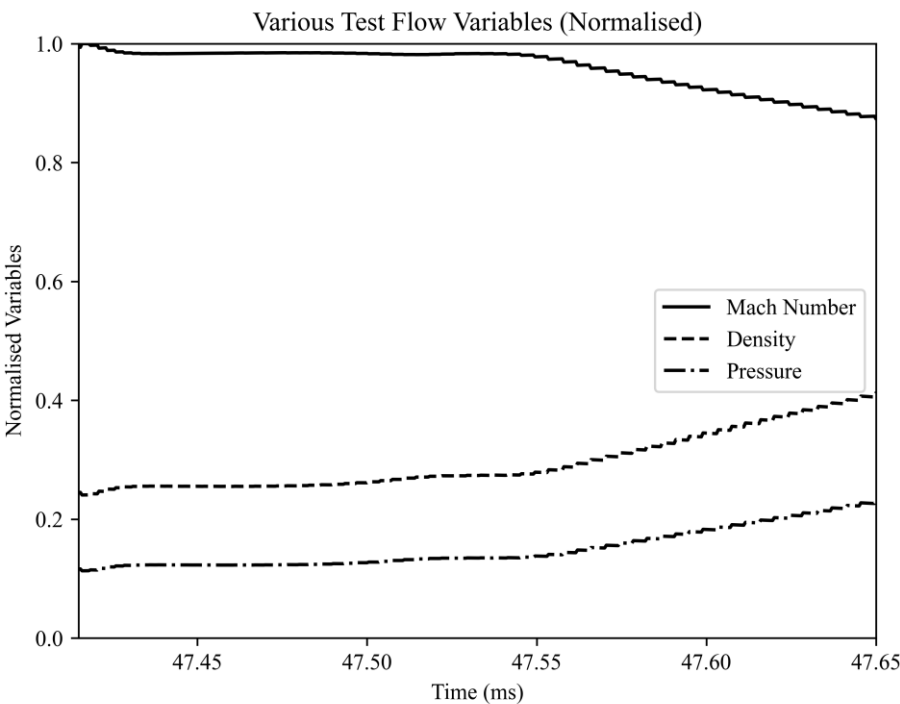
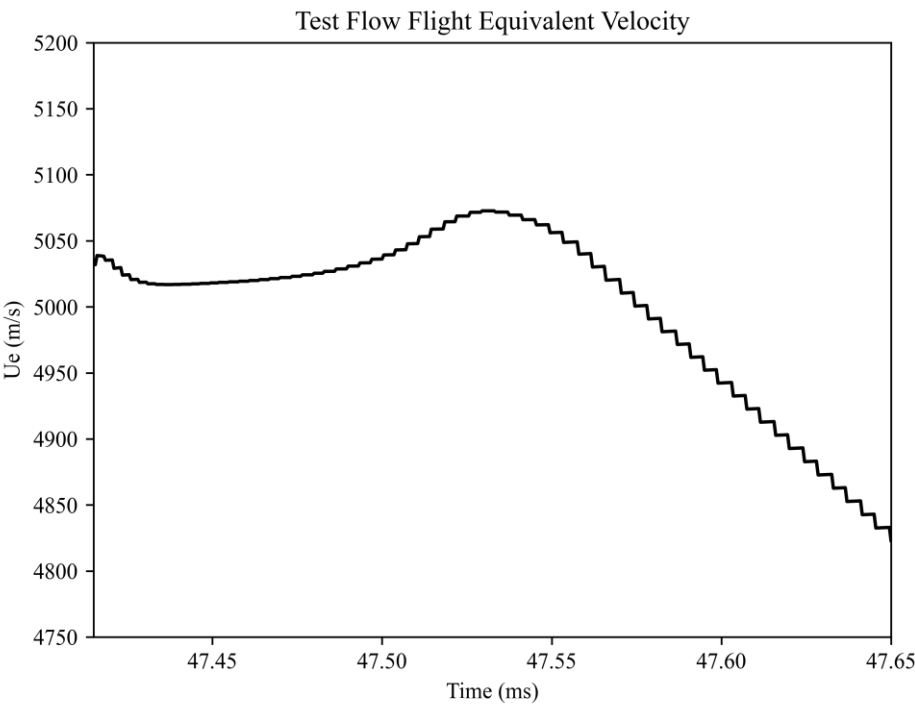
F.5 CR15-5MPa-85mm



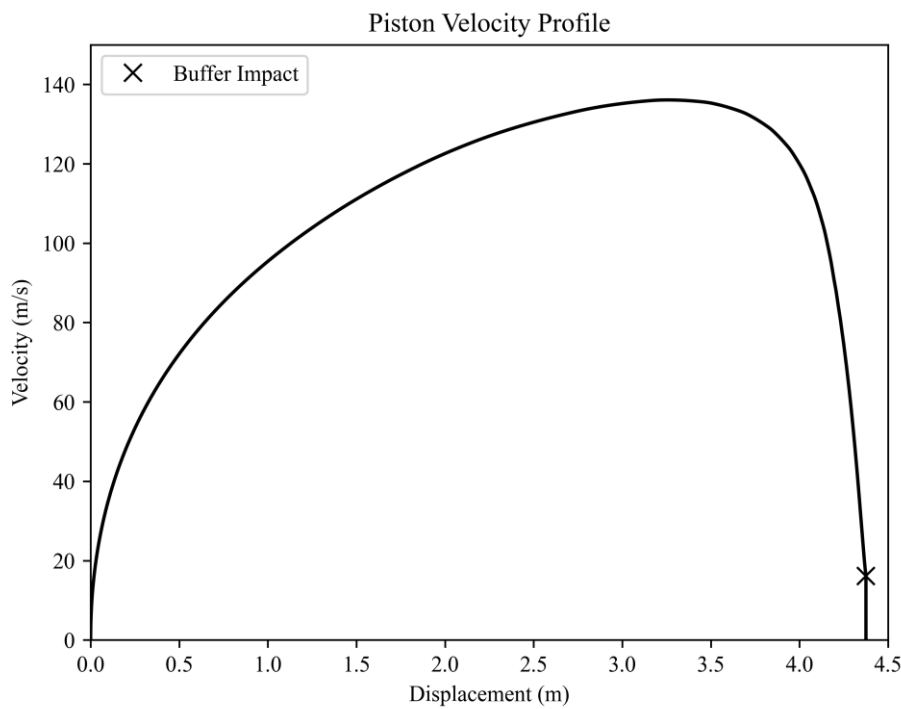
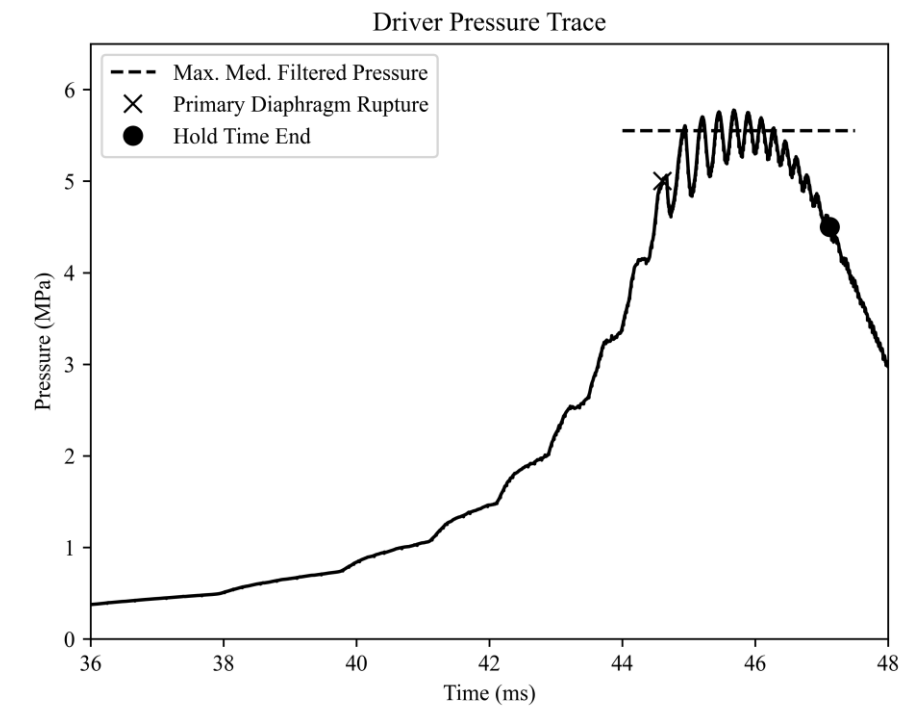


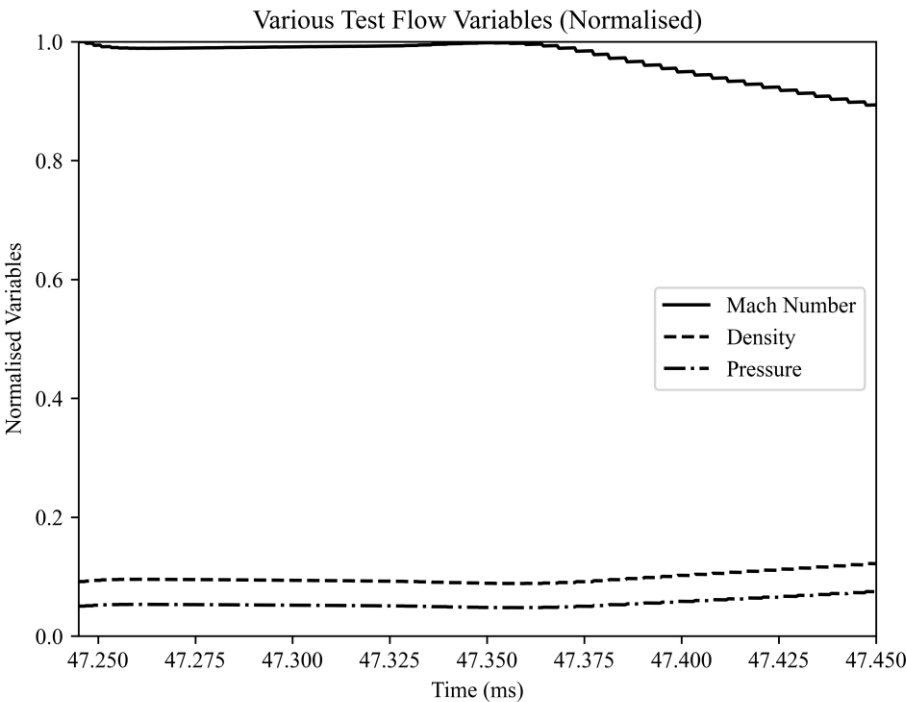
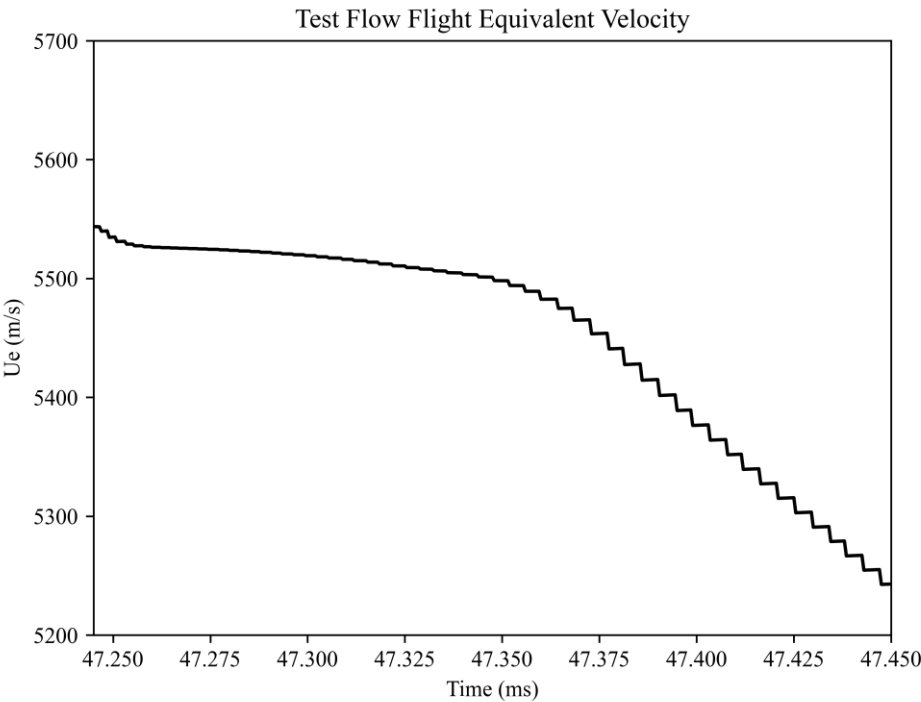
F.6 CR15-5MPa-77mm



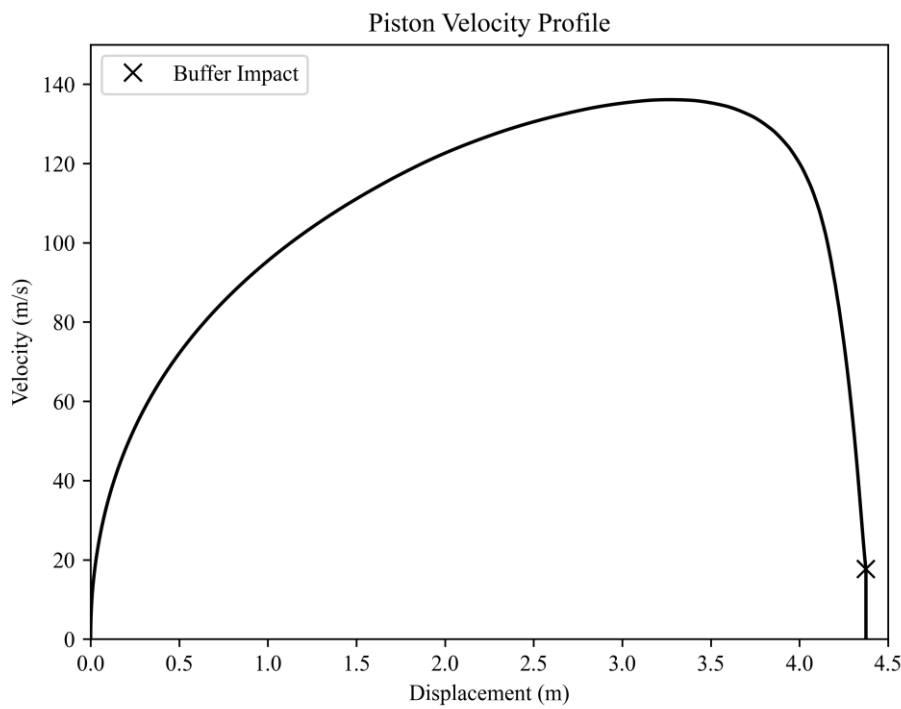
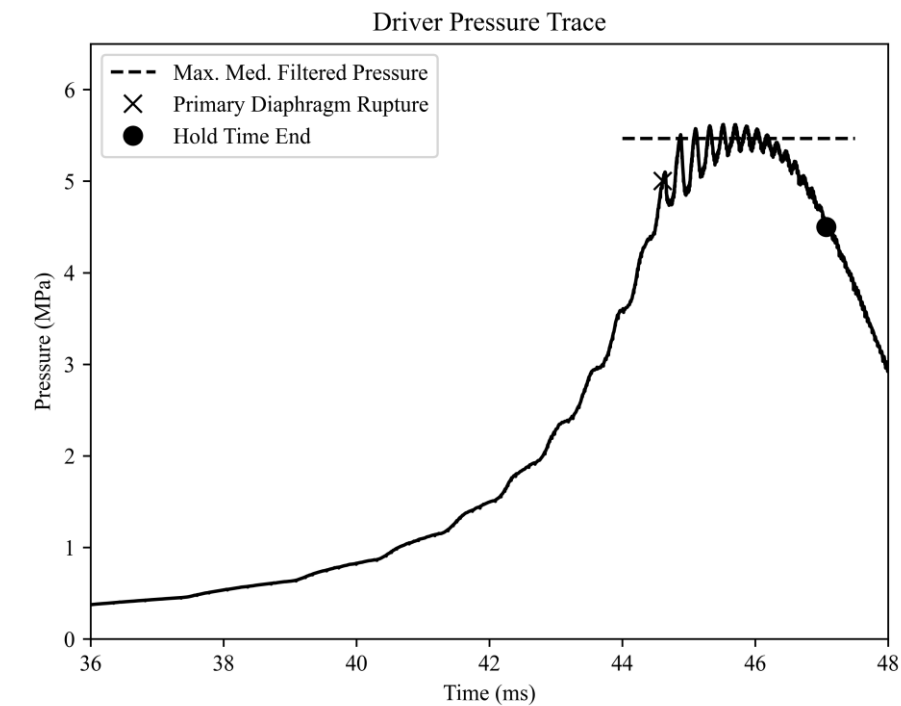


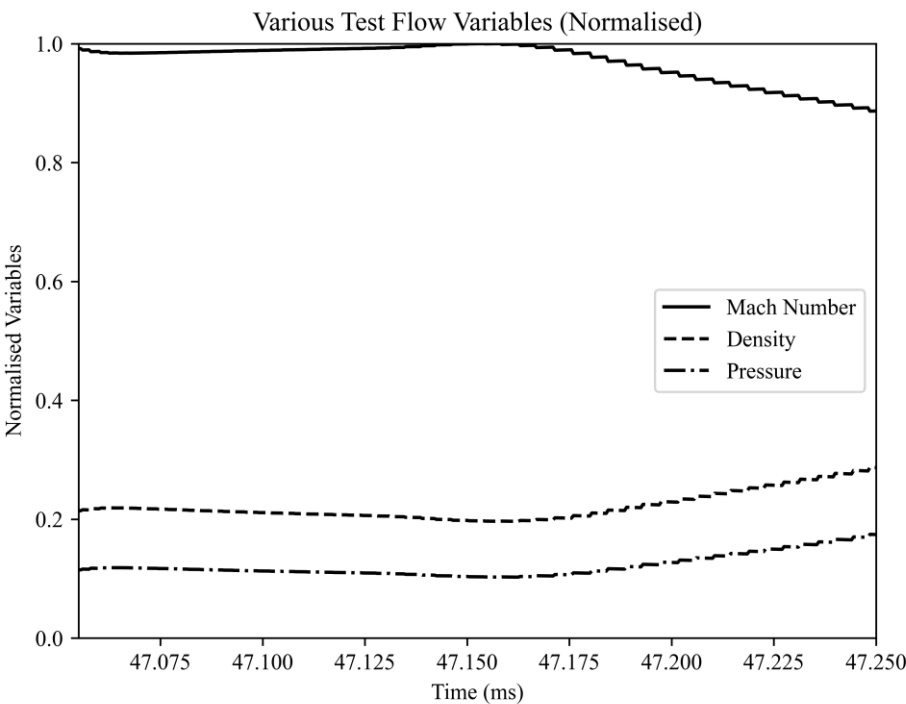
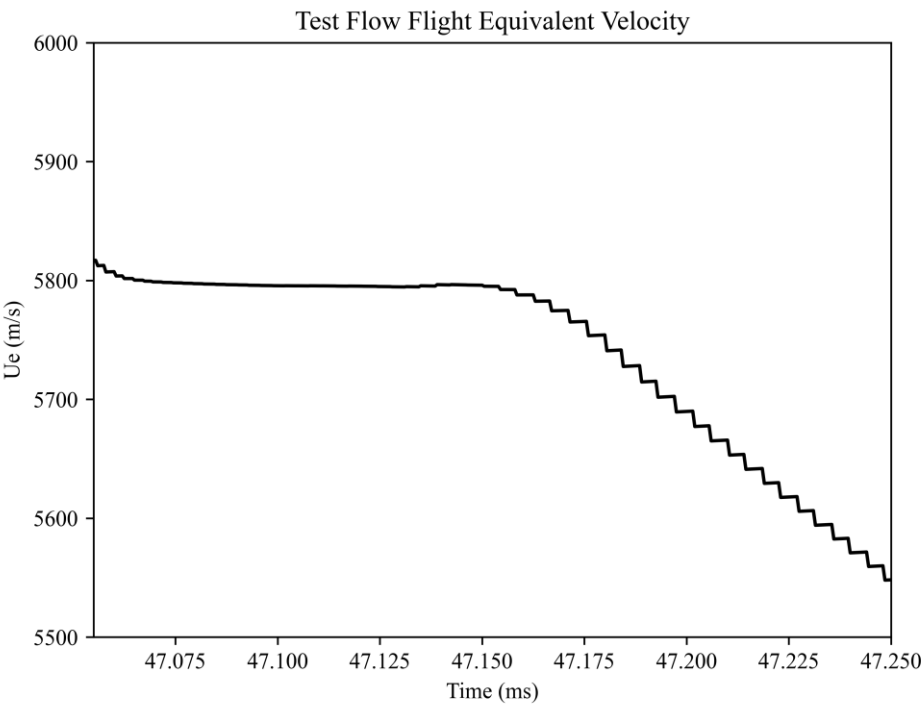
F.7 CR15-5MPa-72.1mm





F.8 CR15-5MPa-65mm





G Buffer Stud Failure Analysis

It was originally thought that a new set of nylon buffer studs would need to be designed to accompany the new low compression ratio driver conditions. While this ultimately was not the case (X2's existing 100 mm 'long' buffers can be utilised), a simple buffer stud failure analysis was derived early in the project which could still prove useful if, for any reason, the buffer studs need to be extended in the future.

First, the energy absorbing capacity of the buffer system was determined using Gildfind's [3] constant-volume, constant-stress plastic deformation model. For a single buffer stud of initial length L_s , and area A_s , the cross-sectional area with respect to a change in stud length x , is defined as,

$$A_{s,x} = \frac{A_s L_s}{L_s - x} \quad (\text{G.1})$$

The work W_s , required to deform the stud by length x , at constant stress (the yield strength of nylon in this case) is as follows,

$$W_s = \int_0^x \sigma A_{s,x} dx = \int_0^x \sigma \left(\frac{A_s L_s}{L_s - x} \right) dx \quad (\text{G.2})$$

Evaluating this integral and equating the total work to deform n studs to the kinetic energy of the free-piston at impact yields the following expression,

$$nW_s = -n\sigma A_s L_s \ln \left| \frac{L_s - x}{L_s} \right| = \frac{1}{2} m_p v_{imp}^2 \quad (\text{G.3})$$

Rearranging this equation gives the maximum absorbable piston impact speed for a given buffer stud configuration,

$$v_{imp} = \sqrt{-n\sigma \frac{\pi D_s^2}{2m_p} L_s \ln \left| \frac{L_s - x}{L_s} \right|} \quad (\text{G.4})$$

Next, buffer deformation constraints were determined using a process adapted from Stennett [14]. Provided the 256.8 mm internal diameter of the X2 compression tube and the buffer mount pitch circle diameter of 170.5 mm, two geometric limits exist:

- Stud-Stud Interference: The constant volume deformation of the nylon studs results in stud-stud surface collision once $r_s > r_{PCD} \sin\left(\frac{\pi}{n}\right)$.
- Stud-Wall Interference: The constant volume deformation of the nylon studs results in stud-wall surface collision once $r_s > (r_{driver} - r_{PCD})$.

When either of these limits are reached, the radial deformation of the studs becomes constrained and the interfacing stresses increase by orders of magnitude as further deformation requires volumetric reduction as opposed to plastic redistribution [3]. Exceeding these limits would result in buffer stud fracture, thus the maximum absorbable piston impact speed is dictated by the length reduction x , at which either of the limits first become applicable.

As an example, for buffer studs which are 140 mm in length and 50 mm in diameter, the maximum absorbable impact speed for the 10.5 kg piston is 115.8 m/s – a catastrophic driver accident if it were to occur. In turn, fracture is predicted to occur after a 65.6% length change (equivalent to 92 mm). As expected, elongated studs have a higher impact capacity than X2's existing studs (which are limited to a 97.9 m/s impact for the 100 mm buffers).

Finally, it is critical to ensure that an increased buffer length does not introduce a buckling failure mode, as this would invalidate the previous stress analysis. To evaluate the risk of buckling failure, the slenderness ratio of a stud design can be calculated and used to assess the column type within the Euler buckling model. First, the radius of gyration r_G , must be calculated,

$$r_G = \sqrt{\frac{I}{A}} = \sqrt{\frac{\pi D_s^4 / 64}{\pi D_s^2 / 4}} = \frac{D_s}{4} \quad (\text{G.5})$$

Then, assuming a fixed-free support configuration for a conservative assessment, the effective length L_e of the stud becomes double its actual length. The slenderness ratio follows,

$$SR = \frac{L_e}{r_G} \quad (\text{G.6})$$

Provided the slenderness ratio does not exceed a value of 40, a stud is considered to be a ‘short column’ for which the failure mechanism is crushing (yielding), as opposed to buckling. The previous stress analysis and its conclusions remain valid as long as buckling failure modes are not probable.

H X2 Facility Upgrades

A significant portion of the project was dedicated to the development of several upgrades for the broader X2 facility. The following appendix details the progress made towards these upgrades.

H.1 Commissioning of Improved Reservoir Pressure Monitoring System

With the development of new, low compression ratio driver conditions, which utilise very low reservoir fill pressures, the accuracy of the existing reservoir pressure monitoring system rendered it unfit for purpose. The previous system centred around an 8 MPa analogue pressure gauge with an effective accuracy of ± 0.05 MPa. This pressure gauge was replaced with a 10 MPa UNIK 5000 series digital pressure gauge (PMP5076-TA-A3-CA-H0-PE) with an effective accuracy of ± 0.004 MPa. To integrate the UNIK gauge into the reservoir manifold, a Red Lion PAX2A000 digital meter was also purchased.

H.2 Commissioning of Improved Piston Wear Pad Material

Numerous reports from X2 operators suggested that the existing material used for the wear rings and chevron on the X2 piston was leaving a 'burned-plastic' residue along the length of the compression tube. The laborious process associated with the removal of this residue was significantly extending the duration of tunnel turn-arounds, which was limiting X2's effective up-time. A wide range of candidate products were evaluated with respect to a set of key material properties including: wear rate, coefficient of friction, coefficient of thermal expansion, melting temperature, machineability and cost/m². Ultimately, the existing material, Nylatron GSM PA6, was replaced with a more advanced nylon compound, Nylatron 703 XL PA6 (see Figure H.1 in the following subsection). While the 703 XL variant was almost double the price of the old material, its wear rate is significantly lower, marketed at 2.5 microns/km, compared to the 11 microns/km offered by the GSM variant.

H.3 Correction of The X2 Piston's Rear O-Ring Gland

Further reports from X2 operators suggested that the X2 piston was failing to maintain a strong vacuum on the launcher – this increases the risk of pre-firing which can ruin X2 experiments. X2 officials suspected that this issue was caused by an incorrectly specified rear O-ring gland depth. After consulting O-ring handbooks, it was confirmed that the piston technical drawings did, in fact, feature a sub-optimal gland geometry. An excerpt from the amended technical drawing set is provided in Appendix J.1, where the new rear O-ring gland depth of 245.4 mm is shown. Subsequently, X2 officials commissioned a new piston, presented in Figure H.1, which featured both the new wear pad material and the amended O-ring gland.



Figure H.1: Newly manufactured X2 piston (before final lathing of the wear pads).

H.4 Recommissioning of The X2 Static Rupture Rig

The static rupture rig is an important piece of equipment used by both X2 and T4 for static hydraulic testing of new primary diaphragms. Unfortunately, the rig had fallen into disrepair after years of standing idle. In preparation for future testing of the new diaphragms required for the new driver conditions, the static rupture rig was recommissioned – this was a three-step process. First, a 100 MPa WIKA digital pressure gauge was purchased; this gauge features a full-scale accuracy of $\pm 0.05\%$ and data logging capabilities which enable it to be used independently from the main X2 data acquisition system. Next, a new set of hydraulic fittings were purchased to connect the Enerpac pump to the WIKA gauge and rupture station. Quick-release fittings were selected for convenient disconnection and isolation of the hydraulic system from the rupture station (which must be completed after each static rupture test). It is important to note that the hydraulic system is uniformly rated to a maximum pressure of 70 MPa. A snapshot of the reassembled rig is provided in Figure H.2 below. Finally, a standard operating procedure was drafted (see Appendix I.2) and a static rupture test was successfully conducted (the results of which were beyond the scope of this project).



Figure H.2: The X2 static rupture rig (operational).

H.5 Development of Laser-Scoring Techniques for X2 Primary Diaphragms

Prior to experimentation, X2's primary diaphragms are mechanically scored with an 'X' pattern to control the dynamic deformation which occurs during the rupture event (see Figure H.3). Counterproductively, the method for scoring the diaphragms is believed to suffer from poor repeatability. In a bid to yield primary diaphragms with more reliable rupture pressures, X2 officials have expressed interest in exploring laser-engraving as a new scoring technique.

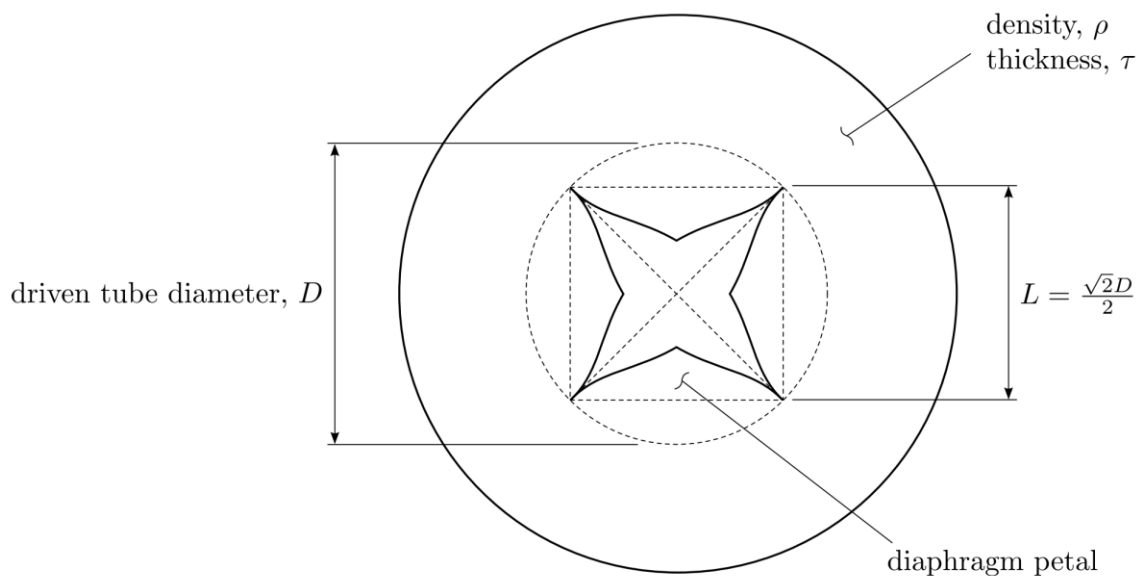


Figure H.3: Schematic of a rupturing diaphragm [3].

After liaising with UQ Innovate staff for some time, the development process began with iterative calibration of Innovate's Gweike 1500 W fibre laser machine to enable engraving of X2's 2.5 mm steel diaphragms (the machine had only been utilised for laser-cutting in the past). This was achieved through trialling of various combinations of duty cycle and scan speed; an example of this process is provided in Figure H.4. However, before much progress could be made, it became obvious that the laser machine's peak bed was too uneven to engrave consistently (see Figure H.5). In fact, it was virtually impossible to ensure each diaphragm was level, thus the inevitable deviation from the optimal laser focal length resulted in non-uniform application of power.

One possible solution to this challenge is a laser-scoring rig, capable of steadily holding an X2 diaphragm above the peak bed of the laser machine. A preliminary schematic for such a design is presented Appendix J.2. The design features a stiff chassis milled from 40 mm mild steel plate, a self-locating diaphragm station, and a magnetic base to ensure the rig will not move during operation. Unfortunately, the rig could not be manufactured before the conclusion of the ENGG7291 placement.

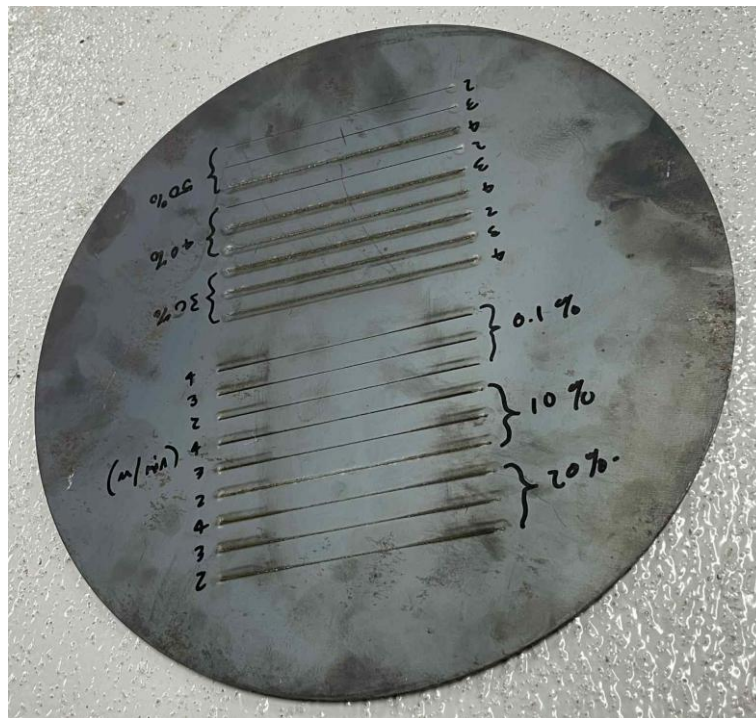


Figure H.4: Example of iterative laser engraving calibration.

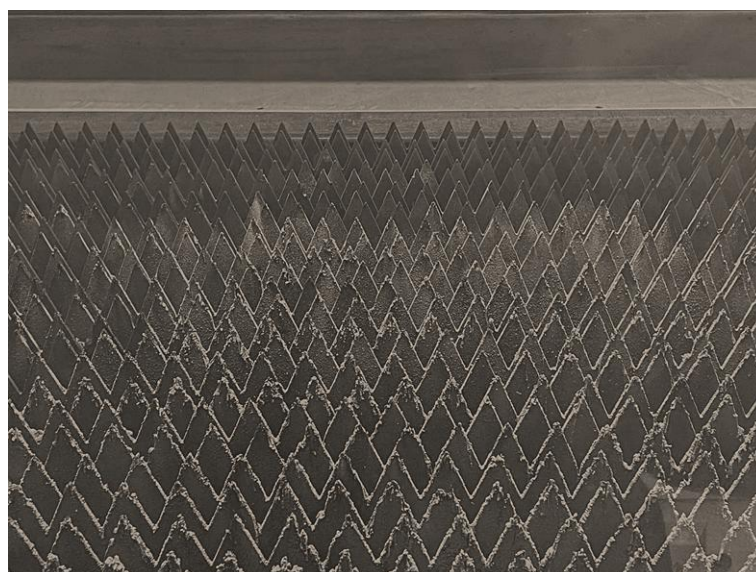


Figure H.5: UQ Innovate laser machine peak bed.

H.6 Development of Driver-Length Instrumentation

Finally, to assist in the design of future driver conditions, development of new driver instrumentation was commissioned by X2 officials. This system is intended to facilitate measurement of previously unobtainable data such as detailed driver pressure distributions and piston trajectory. In the short term, these new insights will enable more accurate calibration of current numerical models of the X2 facility. In the long term, the same instrumentation will enable robust validation of emerging facility models which aim to capture more detailed driver phenomena.

The first step in the design of the driver-length instrumentation was optimisation of the PCB distribution along the compression tube. A python script was devised to run SPARK simulations of Gildfind's [3] three primary driver conditions whilst attempting to recreate the piston trajectories using a set of seven theoretical PCB locations. This model assumed that the wall-mounted PCB's have zero thickness in the longitudinal tunnel coordinate, and that the arrival of both the piston's front and rear edges can be accurately determined from the PCB pressure traces; the latter leads to the concept of 'inferred' points which arise one piston length downstream of each real PCB location. As such, discrete samples of the piston's trajectory were used to construct a spline-fit estimation of the 'real' piston trajectory. The spline function was subsequently derived to yield the piston's velocity profile.

The outputs of the python script are provided in Figures H.6-H.8, which also include plots of the maximum driver pressure exerted on each PCB and the location of each PCB mapped to the external geometry of the X2 driver. Note the accuracy with which the 'real' SPARK data (plotted in black) is recreated by the spline-fit model (plotted in red). Interestingly, the peak driver pressure experienced by any of the PCB's is predicted to occur during the 15.5 MPa driver condition. It is also important to mention that the final three PCB's require installation into the thickest region of the compression tube wall – this will require special considerations.

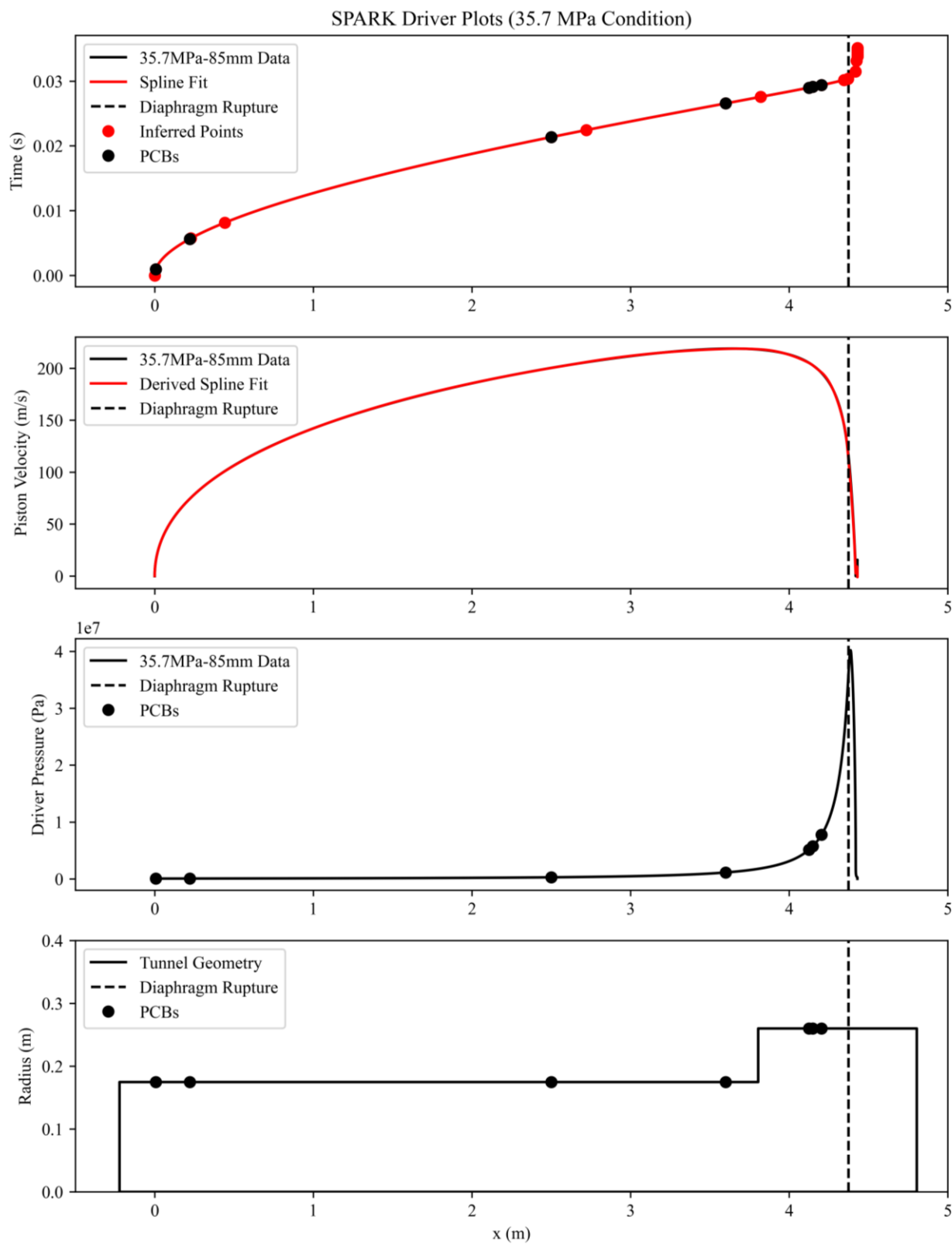


Figure H.6: Driver-length instrumentation plots (35.7 MPa condition).

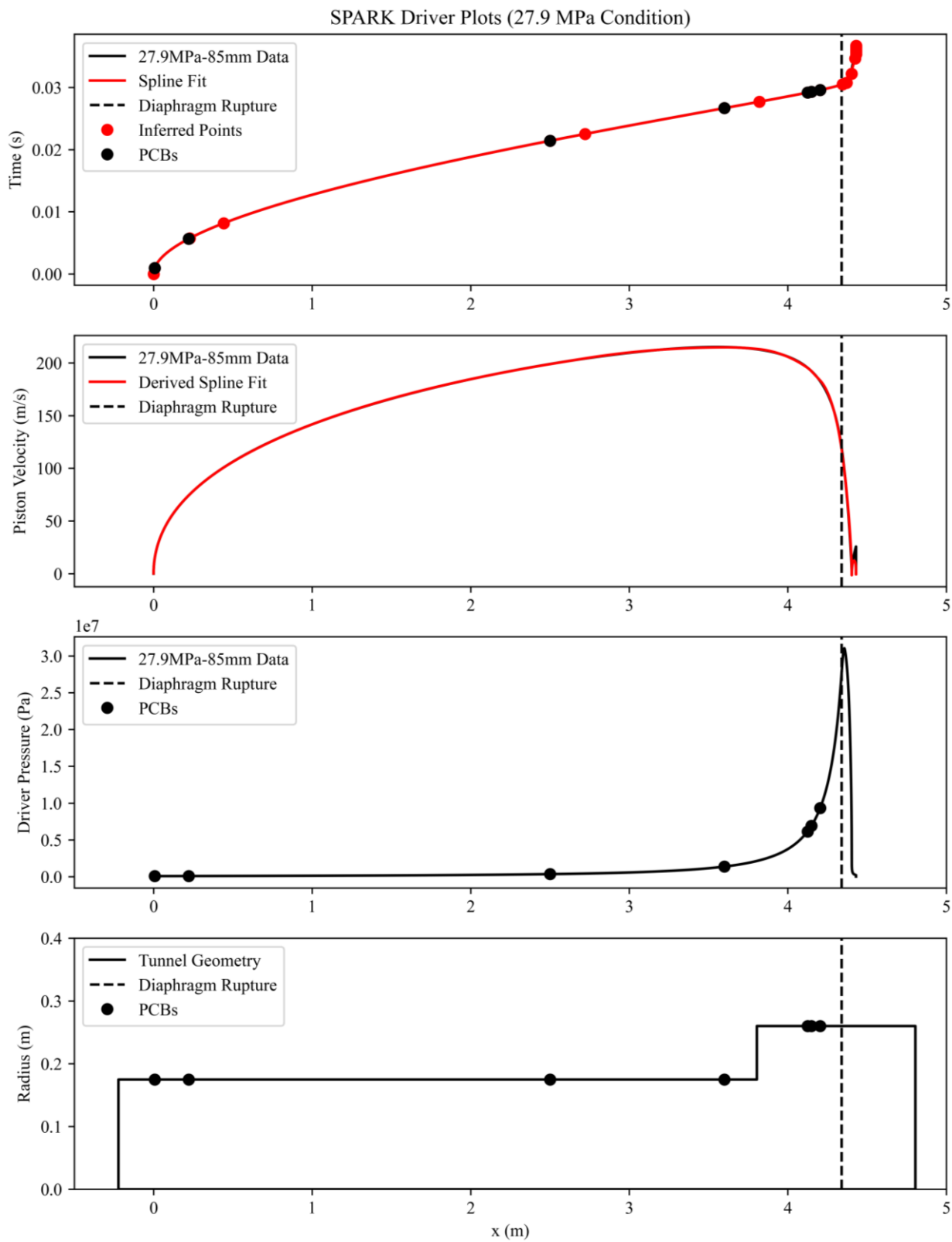


Figure H.7: Driver-length instrumentation plots (27.9 MPa condition).

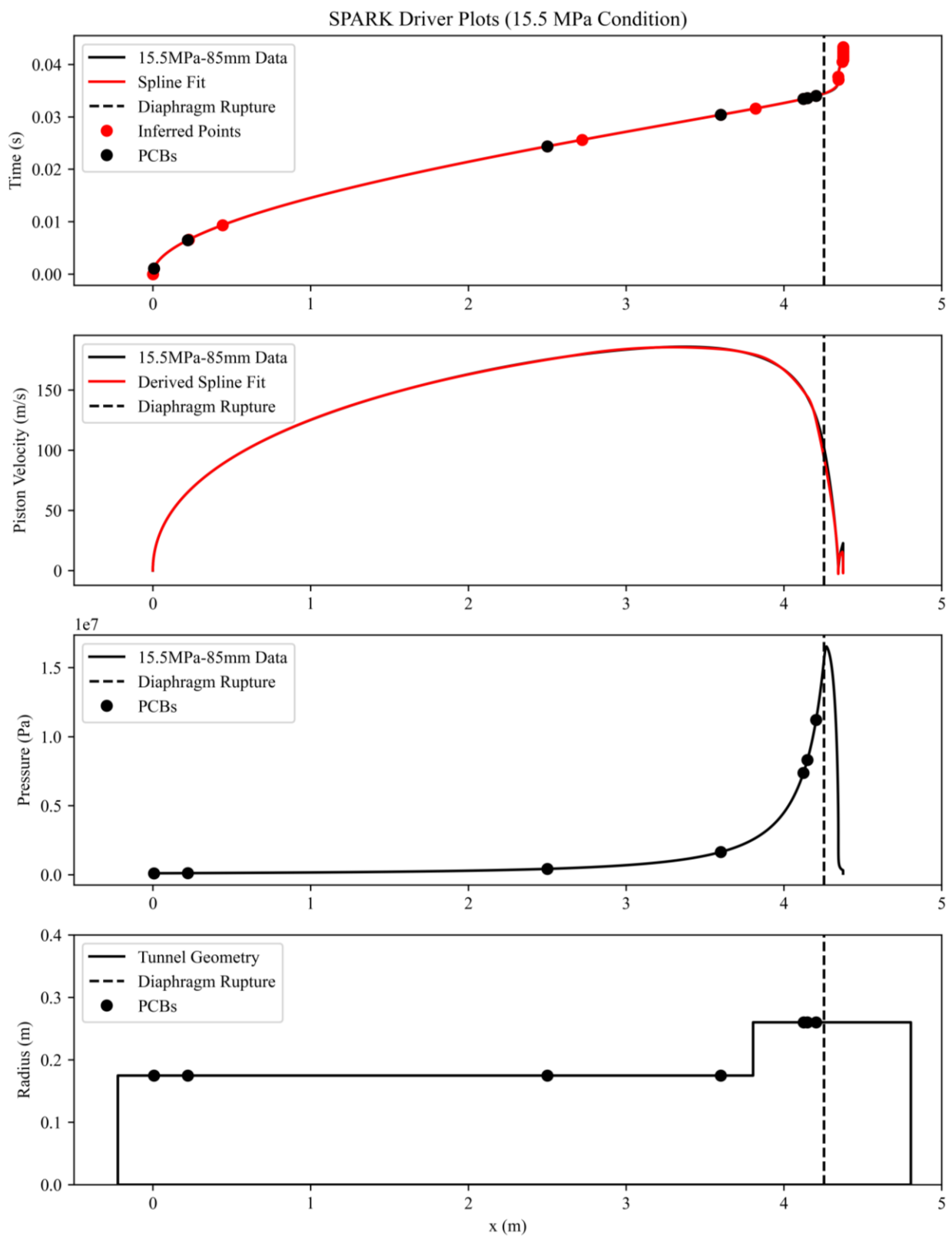
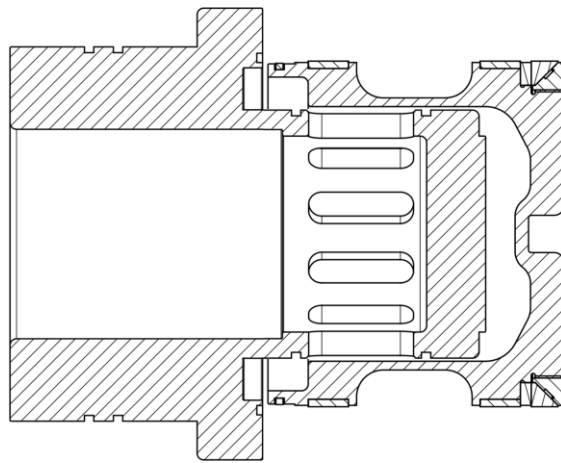
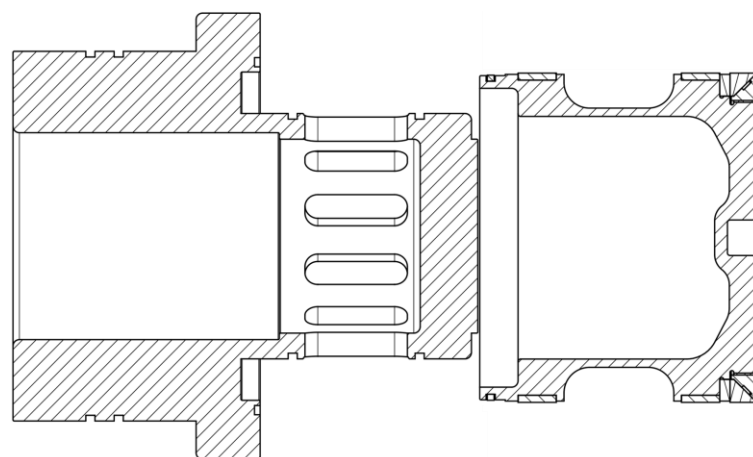


Figure H.8: Driver-length instrumentation plots (15.5 MPa condition).

The proposed PCB distribution, showcased in the plots above, is $x = [0.015, 0.221, 2.5, 3.6, 4.119, 4.149, 4.204]$, in metres, measured from the front face of the piston when seated on the launcher. The first two points in the distribution were strategically placed to precisely capture the time at which the launcher slots are first exposed and the time at which the piston fully clears the launcher. Illustrated in Figure H.9, these events are critical for characterising the piston's early acceleration profile as it departs from the launcher. Moving downstream, the next three PCB's were arbitrarily placed to improve the accuracy of the spline fit. The final two points were placed a piston length plus 5 mm upstream of the 100 mm and 45 mm buffer studs respectively; these PCB's capture the piston mechanics at the extreme end of the compression tube.



Exact time launcher slots exposed



Exact time launcher cleared

Figure H.9: Key events during X2 piston launch.

With the optimal PCB distribution established, PCB housings were designed so that the pressure transducers could be fixed securely in the wall of the compression tube. A schematic for a preliminary PCB housing design is presented in Appendix J.3. Shown also in Figure H.10 below, the brass housing is designed to flush-mount the PCB (shown in silver) to the inner surface of the compression tube. The housing's interior geometry was adopted from PCB Piezotronics' technical schematics, while the exterior was designed to incorporate a face-sealing O-ring and threaded rear ports for ease of removal from the compression tube holes. X2 officials have expressed assembly concerns regarding the face-sealing O-ring, so it is recommended that future design iterations implement a radial O-ring configuration instead. The housing is intended to be fixed into the tunnel wall by a short M20 bolt, hollowed along its central axis to allow routing of a microdot cable to the PCB. Further recommendations for future work include FEA of the final design to ensure safe operation up to a maximum driver pressure of 50 MPa, and the development of a custom magnetic drill mount (plus an associated procedure) for installation of the final assembly.

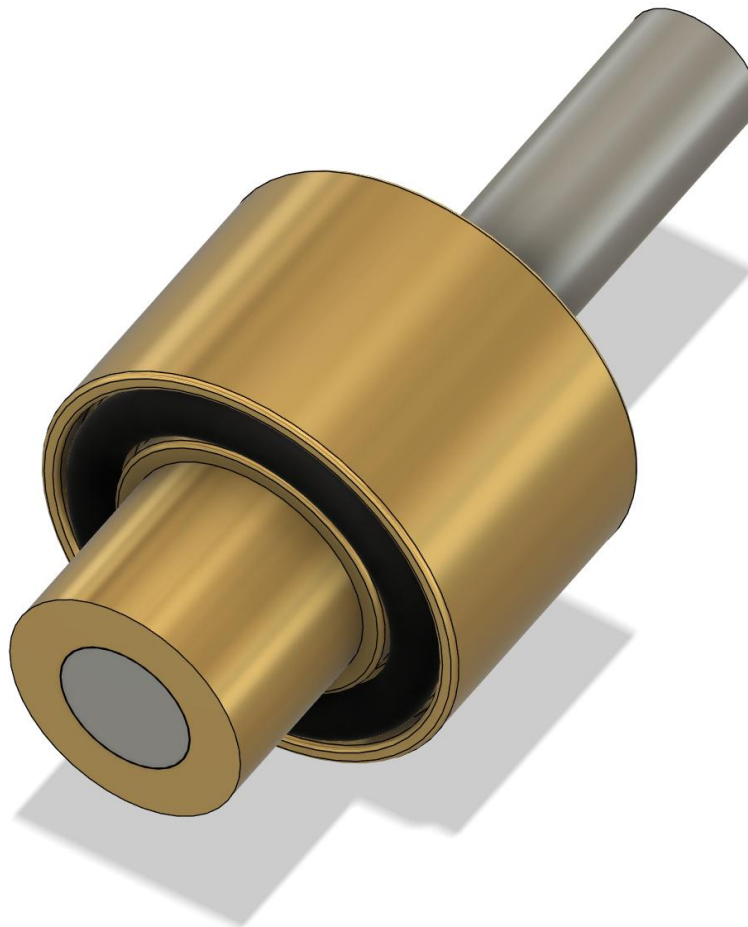


Figure H.10: Preliminary PCB housing design.

I Standard Operating Procedures

I.1 Updated X2 Blanked-Off Test SOP

Standard Operating Procedure – X2 Blanked-Off Tests

DO NOT conduct blanked-off tests unless you have been trained and deemed competent.

Risk Assessment: #2706

Potential Hazards & Injuries:

- Injuries if pressure remains on the back of the piston while opening the tunnel and the piston is fired towards the operators.
- Injuries to eyes if pressures in front of the piston have not vented to atmosphere before opening the tunnel.
- Damage to equipment if this procedure is not strictly followed (X2 expansion tube, pressure gauges).

Required Training:

- XLABS Laboratory Induction
- X2 Expansion Tube Training Program & Manual
- OHSB07 – Compressed Gases Safety Assessment
- OHSB08 – General Workplace Safety Assessment
- OHSB09 – Fire Safety Assessment

Note: An approved X2 operator **MUST** be present to conduct the experiment.

Required PPE for Facility Operators & Bystanders:

- Safety Glasses
- Hearing Protection
- Enclosed Footwear

Pre-Experiment Safety Checks:

1. Ensure experiments have been discussed and authorised by laboratory manager.
2. Verify computationally that maximum blanked-off driver pressure will not exceed 50MPa. Note: Blanked-off tests can produce peak pressures several times higher than nominal diaphragm-rupturing conditions.
3. Computationally estimate maximum piston displacement and ensure appropriate buffers are installed. Note: Piston will impact buffers directly if too long.

Experiment Setup:

1. Calibrate PCB's sensitivity.
2. Install PCB's into the blanked-off plate – finger tight with an additional $\frac{1}{4}$ turn with a small spanner. Careful not to overtighten.
3. Apply RTV to the exposed face of each PCB. This ensures that elevated driver gas temperatures do not impact the accuracy of the transducers.
4. Route microdot cables through the st1 port to the blanked-off plate thereby connecting the PCB's to the DAQ. Note: it may be useful to tape multiple cables together or feed through a length of conduit to reduce the risk of severance when the tunnel is closed.
5. Check PCB's are transmitting signals to the DAQ.
6. Install welding rods into nylon buffer studs as required by the driver condition being tested.

Pre-Shot Procedure:

1. Update manifold boards with the specified fill pressures.
2. Fully vent facility and open balance lines.
3. Reload piston into the driver and ensure piston is correctly mounted on launcher and cavity is pumping down to vacuum.
4. Ensure buffer plate is securely bolted to the pressure plate and welding rods are fully seated in the nylon buffers.
5. Install buffer/pressure plate assembly. Be careful not to damage the welding rods.
6. Install the blanked-off plate, ensuring the PCB cables are not damaged.

7. Slide the shock tube into firing alignment.
8. Remove as much PCB cable slack as possible whilst ensuring that the blanked-off plate remains seated in the driver.
9. Confirm that the cables are taut and unlikely to become pinched when reassembling the tunnel.
10. Reconnect the acceleration tube to the shock tube and tighten the secondary capstan. Stop if there is any unusual resistance – the PCB cables can be easily sheared during this step.
11. Ensuring that the cables remain taut, slowly reconnect the driver and tighten the primary capstan. Stop if there is any unusual resistance – the PCB cables can be easily sheared during this step. The capstan should align with the 1.2 mm diaphragm indicator.
12. Check that the PCB's are still transmitting signal after tunnel assembly.
13. Move dump-tank into firing position.
14. Raise dump-tank motor-drive pinion from the linear rack.
15. Ensure driven tubes and dump tank are at atmospheric pressure. Note: for blanked-off tests, the driven tube and test section are not evacuated and are not sealed due to PCB pressure transducer cables running through shock tube from the st1 port.
16. Evacuate the driver using the corresponding vacuum pump.
17. Arm the DAQ.
18. Conduct trigger test.
19. Rearm the DAQ.
20. Apply eye protection.
21. Fill driver with specified driver gas mixture and isolate from manifold.
22. Apply ear protection.
23. Isolate reservoir and gauges.
24. Fill reservoir with compressed air to specified pressure.

Shot Procedure:

1. Check the data acquisition system is still armed.
2. Fire facility in accordance with X2 standard operating procedure.

Post-Shot Procedure:

1. Slowly vent reservoir to atmospheric pressure ($\frac{1}{4}$ turn of the vent valve, $\frac{1}{2}$ turn once reservoir pressure is < 0.5 MPa, fully open once reservoir pressure is < 0.3 MPa), as to ensure the piston does not slam back onto the launcher. Ensure reservoir is fully vented before proceeding.
2. Open manifold to driver and check pressure.
3. Vent driver to atmosphere.
4. Vent entire facility. Turn off all vacuum pumps.
5. Disconnect the dump tank.
6. Feed ample PCB cable slack into st1 port.
7. Disassemble secondary capstan and move the acceleration tube downstream.
8. Ensuring there is ample cable slack, disassemble primary capstan and move the driver upstream.
9. Feed additional slack and slide the shock tube to the side.
10. Insert a table under the downstream end of the driver tube.
11. Remove blanked-off plate from the driver and place on the table.
12. Remove the orifice plate.
13. Remove the pressure plate in accordance with X2 standard operating procedure.
Do not let the welding rods hit any part of the tunnel during removal.
14. Inspect the welding rods, determining the maximum excursion of the free piston by measuring the distance between the base of the pressure plate and the deformed rods.
15. Remove and clean the piston in accordance with X2 standard operating procedure.

Emergency Response Procedures:

A. In the case of pressure rise in driven tube and dump tank following experiment (evidenced by dump tank pressure increase):

1. Ensure reservoir vent is fully open.
2. Turn on vacuum pump to test section and open valve.
3. Stand clear until fully vented.

B. In the case of piston pre-launch (evidenced by facility recoil and a noise during reservoir filling):

1. Immediately discontinue filling reservoir.
2. Vent reservoir to atmospheric pressure. Ensure reservoir is fully vented before proceeding.
3. Open manifold to driver and check pressure.
4. Vent driver to atmosphere.
5. Vent entire facility. Turn off all vacuum pumps.

C. In the case of an experiment abort (for any other reason):

1. Vent reservoir to atmospheric pressure. Ensure reservoir is fully vented before proceeding.
2. Open manifold to driver and check pressure.
3. Vent driver to atmosphere.
4. Vent entire facility. Turn off all vacuum pumps.

I.2 Static Rupture Test SOP

Standard Operating Procedure – Static Rupture Rig

DO NOT operate the static rupture rig unless you have been trained and deemed competent.

Risk Assessment: #1206

Potential Hazards & Injuries:

- High pressure fluids can become dangerous if the hydraulic tubing is punctured. Hydraulic injection injuries can be severe.
- Incorrect venting of excess air in the hydraulic system can lead to rapid decompression upon diaphragm rupture.

Required Training:

- XLABS Laboratory Induction
- OHSB08 – General Workplace Safety Assessment

Required PPE for Trained Operators & Bystanders:

- Safety Glasses
- Hearing Protection
- Enclosed Footwear

Experiment Procedure:

1. Check the hydraulic tubing and fittings for any signs of deterioration.
2. Check the oil capture tray and clean if necessary (a clean tray ensures the oil can be recycled for future tests).
3. Insert a new diaphragm into the rig, ensuring the diaphragm is concentric with the rig, and the score pattern is facing downwards.
4. Install the hydraulic injection plate, ensuring the O-ring is fitted correctly.
5. Place the clamping plate on top of the injection plate and tighten the bolts using a 36mm spanner.
6. Connect the hydraulic tubing to the rig.

7. Fill the hydraulic system with oil, ensuring the fill point (Enerpac) is the highest point in the assembly (so that air does not become trapped (trapped air can result in dangerous, rapid decompression upon diaphragm rupture).
8. Vent trapped air out of the Enerpac.
9. Seal the Enerpac.
10. Check the pressure gauge is receiving signal and the maximum pressure log has been reset.
11. Slowly pump the Enerpac to raise the pressure in the hydraulic system.
12. Check consistently for any signs of leaks.
13. When the hydraulic pressure reaches 1 MPa, tighten the bolts once again to ensure maximum diaphragm clamping.
14. Continue to pump the Enerpac until the diaphragm ruptures.

Post-Experiment Procedure:

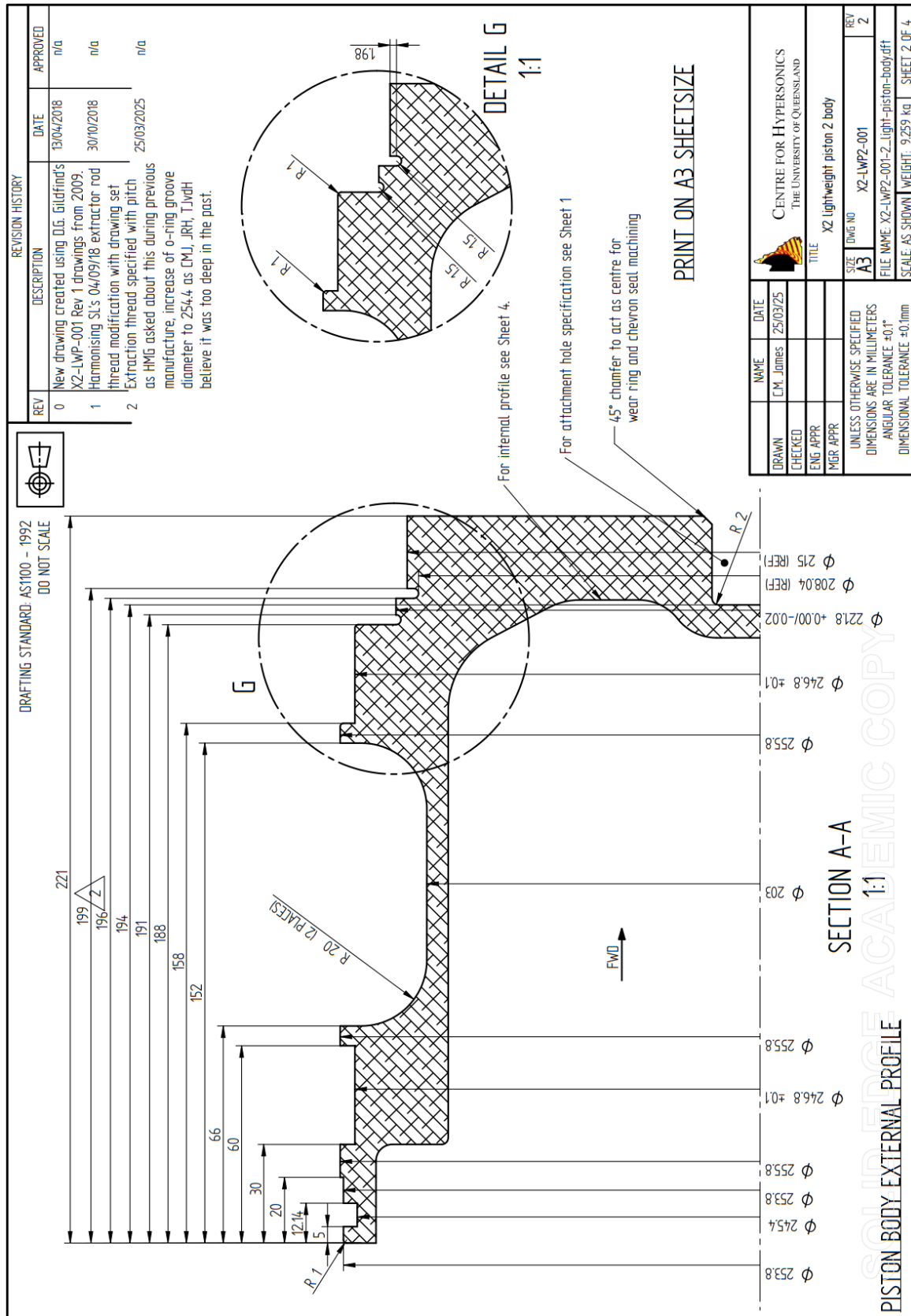
1. Allow the hydraulic oil to drain into the capture tray.
2. Check the pressure gauge to ensure the system has returned to atmospheric pressure.
3. Disconnect the hydraulic tubing from the rig.
4. Disassemble the rig, ensuring excess oil drips into the capture tray.
5. Discard the broken diaphragm.
6. Recycle the captured hydraulic oil for future tests.

Emergency Response Procedure:

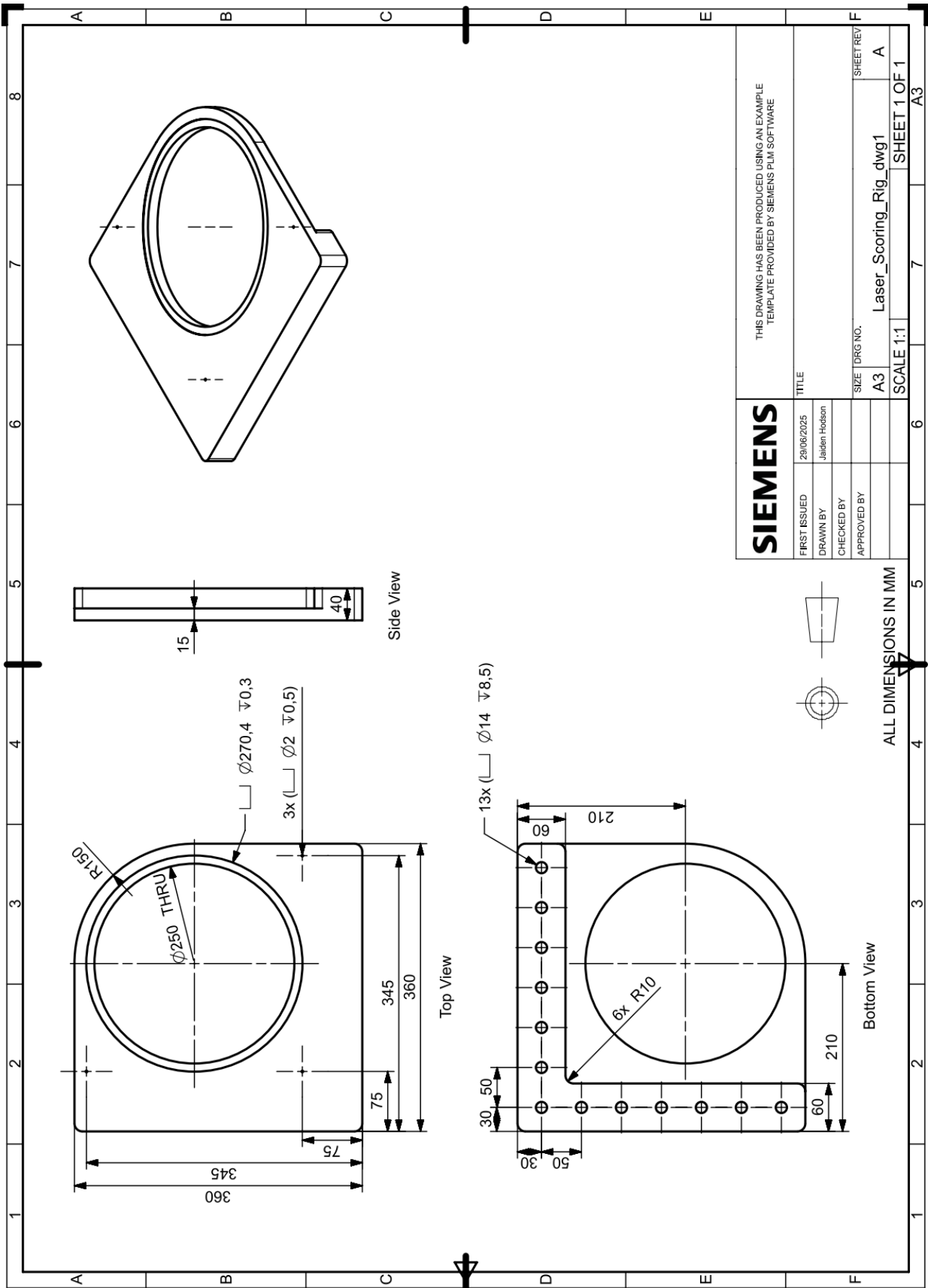
In the case of a leak in the hydraulics while the system is pressurised:

1. Immediately leave the area.
2. Notify other members of the laboratory to ensure the site is kept clear while the hydraulics return to a safe pressure.
3. Inform the laboratory supervisor and complete an incident report.

J.1 Excerpt from Corrected X2 Piston Drawing



J.2 Laser-Scoring Rig Drawing



J.3 PCB Housing Drawing

